

My Project

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Chapter 1

Data Structure Index

1.1 Data Structures

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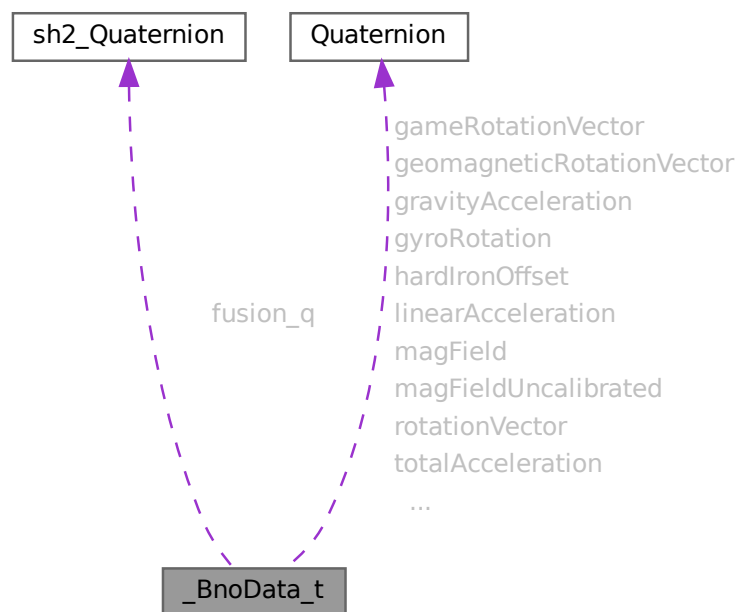
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Chapter 3

Data Structure Documentation

3.1 `_BnoData_t` Struct Reference

Collaboration diagram for `_BnoData_t`:



Data Fields

- [sh2_Quaternion_t](#) `fusion_q`
- [Quaternion](#) `rotationVector`
- [Quaternion](#) `totalAcceleration`
- [Quaternion](#) `linearAcceleration`
- [Quaternion](#) `gravityAcceleration`

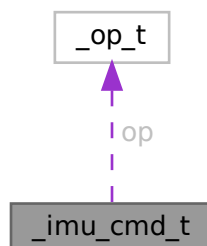
- [Quaternion](#) **gyroRotation**
- [Quaternion](#) **magField**
- [Quaternion](#) **magFieldUncalibrated**
- [Quaternion](#) **hardIronOffset**
- [Quaternion](#) **gameRotationVector**
- [Quaternion](#) **geomagneticRotationVector**
- bool **tapDetected**
- bool **doubleTap**
- enum Stability **stability**
- bool **stepDetected**
- uint16_t **stepCount**
- bool **significantMotionDetected**
- bool **shakeDetected**
- bool **circleDetected**
- bool **xAxisShake**
- bool **yAxisShake**
- bool **zAxisShake**
- uint8_t **majorSoftwareVersion**
- uint8_t **minorSoftwareVersion**
- uint16_t **patchSoftwareVersion**
- uint32_t **partNumber**
- uint32_t **buildNumber**
- float **rotationAccuracy**
- float **geomagneticRotationAccuracy**
- double **ambient**
- double **pressure**
- double **humidity**
- double **proximity**
- double **temperature**
- uint8_t **statusA**
- uint8_t **statusG**
- uint8_t **statusM**

The documentation for this struct was generated from the following file:

- `src/bno086.h`

3.2 `_imu_cmd_t` Struct Reference

Collaboration diagram for `_imu_cmd_t`:



Data Fields

- const uint16_t **id**
- uint32_t **board_serial_id**
- enum device_type **device**
- const uint8_t **cs**
- const uint8_t **spi_bytes**
- const uint8_t **acc_range_scl**
- uint8_t **acc_range**
- uint32_t **log_timeout**
- uint32_t **rs**
- uint32_t **ss**
- volatile bool **online**
- volatile bool **run**
- volatile bool **update**
- volatile bool **features**
- volatile bool **crc_error**
- volatile bool **angles**
- uint64_t **host_serial_id**
- bool **locked**
- bool **warn**
- bool **down**
- uint8_t **rbuf** [512]
- uint8_t **tbuf** [512]
- uint32_t **rbuf32** [2]
- uint32_t **tbuf32** [2]
- uint16_t **serial1**
- uint16_t **serial2**
- [op_t](#) **op**
- volatile bool **init_good**

The documentation for this struct was generated from the following file:

- `src/imu.h`

3.3 _imu_host_t Struct Reference

Data Fields

- uint16_t **id**
- uint64_t **host_serial_id**
- uint8_t **cmd**
- uint32_t **cmd_data** [4]
- uint64_t **secret**
- uint8_t **buf** [64]

The documentation for this struct was generated from the following file:

- `src/imu.h`

3.4 `_op_t` Struct Reference

Data Fields

- void(* **info_ptr**)(void)
- void(* **imu_set_spimode**)(void *)
- bool(* **imu_getid**)(void *)
- bool(* **imu_getdata**)(void *)

The documentation for this struct was generated from the following file:

- src/imu.h

3.5 `_SPid` Struct Reference

Data Fields

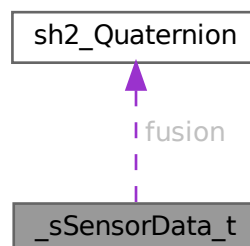
- double **dState**
- double **iState**
- double **iMax**
- double **iMin**
- double **iGain**
- double **pGain**
- double **dGain**

The documentation for this struct was generated from the following file:

- src/pid.h

3.6 `_sSensorData_t` Struct Reference

Collaboration diagram for `_sSensorData_t`:



Data Fields

- const uint16_t **id**
- [sh2_Quaternion_t](#) **fusion**
- double **x**
- double **y**
- double **z**
- float **xa**
- float **ya**
- float **za**
- float **xerr**
- float **yerr**
- float **zerr**
- uint32_t [sensortime](#)
- float **sensortemp**
- uint8_t **buffer** [64]

3.6.1 Field Documentation

3.6.1.1 sensortime

```
uint32_t sensortime
```

sensor time

3.6.1.2 x

```
double x
```

X-axis sensor data

3.6.1.3 xa

```
float xa
```

X-angle sensor data

3.6.1.4 y

```
double y
```

Y-axis sensor data

3.6.1.5 ya

```
float ya
```

Y-angle sensor data

3.6.1.6 z

```
double z
```

Z-axis sensor data

3.6.1.7 za

```
float za
```

Z-angle sensor data

The documentation for this struct was generated from the following file:

- src/imu.h

3.7 __SYS_RTCC_OBJ_STRUCT Struct Reference

Data Fields

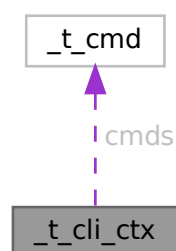
- RTCC_CALLBACK **callback**
- uintptr_t **context**

The documentation for this struct was generated from the following file:

- src/config/bma400/peripheral/rtcc/plib_rtcc.c

3.8 _t_cli_ctx Struct Reference

Collaboration diagram for _t_cli_ctx:



Data Fields

- `t_cmd` * **cmds**
- char **cmd** [CLI_CMD_BUFFER_SIZE]
- uint8_t **cpos**

The documentation for this struct was generated from the following file:

- src/cmd_scanner.h

3.9 `_t_cmd` Struct Reference

Data Fields

- const char * **cmd**
- void(* **fh**)(void *)

The documentation for this struct was generated from the following file:

- src/cmd_scanner.h

3.10 `ADCHS_CALLBACK_OBJECT` Struct Reference

Data Fields

- ADCHS_CALLBACK **callback_fn**
- uintptr_t **context**

The documentation for this struct was generated from the following file:

- src/config/bma400/peripheral/adchs/plib_adchs_common.h

3.11 `ADCHS_EOS_CALLBACK_OBJECT` Struct Reference

Data Fields

- ADCHS_EOS_CALLBACK **callback_fn**
- uintptr_t **context**

The documentation for this struct was generated from the following file:

- src/config/bma400/peripheral/adchs/plib_adchs_common.h

3.12 B_type Struct Reference

Data Fields

- volatile uint32_t **arate**
- volatile uint32_t **alimit**
- volatile uint32_t **srate**
- volatile uint32_t **slimit**
- volatile uint32_t **drate**
- volatile uint32_t **dlimit**

The documentation for this struct was generated from the following file:

- src/buzzer.h

3.13 CANFD_CALLBACK_OBJ Struct Reference

Data Fields

- CANFD_CALLBACK **callback**
- uintptr_t **context**

The documentation for this struct was generated from the following file:

- src/config/bma400/peripheral/canfd/plib_canfd_common.h

3.14 CANFD_OBJ Struct Reference

Data Fields

- uint32_t **errorStatus**

The documentation for this struct was generated from the following file:

- src/config/bma400/peripheral/canfd/plib_canfd_common.h

3.15 CANFD_RX_MSG Struct Reference

Data Fields

- uint32_t * **id**
- uint8_t * **buffer**
- uint8_t * **size**
- uint32_t * **timestamp**
- CANFD_MSG_RX_ATTRIBUTE * **msgAttr**

The documentation for this struct was generated from the following file:

- src/config/bma400/peripheral/canfd/plib_canfd_common.h

3.16 CANFD_RX_MSG_OBJECT Struct Reference

Data Fields

- uint32_t **r0**
- uint32_t **r1**
- uint8_t **data** []

The documentation for this struct was generated from the following file:

- src/config/bma400/peripheral/canfd/plib_canfd_common.h

3.17 CANFD_TX_EVENT_FIFO_ELEMENT Struct Reference

Data Fields

- uint32_t **te0**
- uint32_t **te1**
- uint8_t **timestamp** []

The documentation for this struct was generated from the following file:

- src/config/bma400/peripheral/canfd/plib_canfd_common.h

3.18 CANFD_TX_MSG_OBJECT Struct Reference

Data Fields

- uint32_t **t0**
- uint32_t **t1**
- uint8_t **data** []

The documentation for this struct was generated from the following file:

- src/config/bma400/peripheral/canfd/plib_canfd_common.h

3.19 CORETIMER_OBJECT Struct Reference

Data Fields

- CORETIMER_CALLBACK **callback**
- uintptr_t **context**
- volatile uint32_t **tickCounter**
- uint32_t **period**

The documentation for this struct was generated from the following file:

- src/config/bma400/peripheral/coretimer/plib_coretimer.h

3.20 DMAC_CHANNEL_OBJECT Struct Reference

Data Fields

- bool **inUse**
- DMAC_ERROR **errorInfo**
- DMAC_CHANNEL_CALLBACK **pEventCallBack**
- uintptr_t **hClientArg**

The documentation for this struct was generated from the following file:

- src/config/bma400/peripheral/dmac/plib_dmac.h

3.21 DMAC_CRC_SETUP Struct Reference

Data Fields

- bool **append_mode**
- bool **reverse_crc_input**
- uint8_t **polynomial_length**
- uint32_t **polynomial**
- uint32_t **non_direct_seed**
- bool **reverse_crc_output**
- uint32_t **final_xor_value**

The documentation for this struct was generated from the following file:

- src/config/bma400/peripheral/dmac/plib_dmac.h

3.22 EXT_INT_PIN_CALLBACK_OBJ Struct Reference

Data Fields

- EXTERNAL_INT_PIN_CALLBACK **callback**
- uintptr_t **context**

The documentation for this struct was generated from the following file:

- src/config/bma400/peripheral/evic/plib_evic.h

3.23 GPIO_PIN_CALLBACK_OBJ Struct Reference

Data Fields

- GPIO_PIN **pin**
- GPIO_PIN_CALLBACK **callback**
- uintptr_t **context**

The documentation for this struct was generated from the following file:

- src/config/bma400/peripheral/gpio/plib_gpio.h

3.24 H_type Struct Reference

Data Fields

- H_STATE **hid_state**
- volatile bool **dis_alt**
- volatile bool **show_la**
- volatile bool **la_mod**
- volatile bool **dis_unblank**
- volatile bool **dis_reset**
- volatile bool **silent**

The documentation for this struct was generated from the following file:

- src/hid.h

3.25 MCPWM_CH_OBJECT Struct Reference

Data Fields

- MCPWM_CH_CALLBACK **callback**
- uintptr_t **context**

The documentation for this struct was generated from the following file:

- src/config/bma400/peripheral/mcpwm/plib_mcpwm_common.h

3.26 MCPWM_OBJECT Struct Reference

Data Fields

- MCPWM_CALLBACK **callback**
- uintptr_t **context**

The documentation for this struct was generated from the following file:

- src/config/bma400/peripheral/mcpwm/plib_mcpwm_common.h

3.27 QEI_CH_OBJECT Struct Reference

Data Fields

- QEI_CALLBACK **callback**
- uintptr_t **context**

The documentation for this struct was generated from the following file:

- src/config/bma400/peripheral/qei/plib_qei_common.h

3.28 Quaternion Struct Reference

```
#include <quat.h>
```

Data Fields

- double **w**
- double **v** [3]

3.28.1 Detailed Description

Data structure to hold a quaternion.

3.28.2 Field Documentation

3.28.2.1 v

```
double v[3]
```

Vector part

3.28.2.2 w

```
double w
```

Scalar part

The documentation for this struct was generated from the following file:

- src/quat.h

3.29 sca3300_data Struct Reference

Data Fields

- `const uint16_t id`
- `struct {
 int16_t channels [9]
 uint16_t ret_status
 uint32_t ts
} scan`
- `uint8_t rs`
- `uint8_t ss`

The documentation for this struct was generated from the following file:

- `src/imu.h`

3.30 sFFTData_t Struct Reference

Data Fields

- `uint16_t id`
- `uint8_t buffer [64]`

The documentation for this struct was generated from the following file:

- `src/imu.h`

3.31 sh2_Accelerometer Struct Reference

Accelerometer.

```
#include <bno086_reg.h>
```

Data Fields

- `float x`
- `float y`
- `float z`

3.31.1 Detailed Description

Accelerometer.

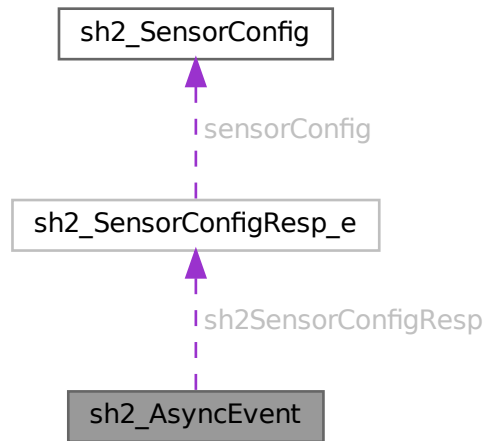
See the SH-2 Reference Manual for more detail.

The documentation for this struct was generated from the following file:

- `src/bno086_reg.h`

3.32 sh2_AsyncEvent Struct Reference

Collaboration diagram for sh2_AsyncEvent:



Data Fields

- uint32_t **eventId**
- union {
 - sh2_ShtpEvent_t **shtpEvent**
 - [sh2_SensorConfigResp_t](#) **sh2SensorConfigResp**

The documentation for this struct was generated from the following file:

- `src/bno086_reg.h`

3.33 sh2_Counts Struct Reference

SensorHub Counter Record.

```
#include <bno086_reg.h>
```

Data Fields

- uint32_t **offered**
[events]
- uint32_t **accepted**
[events]
- uint32_t **on**
[events]
- uint32_t **attempted**
[events]

3.33.1 Detailed Description

SensorHub Counter Record.

See the SH-2 Reference Manual for more detail.

The documentation for this struct was generated from the following file:

- src/bno086_reg.h

3.34 sh2_ErrorRecord Struct Reference

SensorHub Error Record.

```
#include <bno086_reg.h>
```

Data Fields

- **uint8_t severity**
Error severity, 0: most severe.
- **uint8_t sequence**
Sequence number (by severity)
- **uint8_t source**
1-MotionEngine, 2-MotionHub, 3-SensorHub, 4-Chip
- **uint8_t error**
See SH-2 Reference Manual.
- **uint8_t module**
See SH-2 Reference Manual.
- **uint8_t code**
See SH-2 Reference Manual.

3.34.1 Detailed Description

SensorHub Error Record.

See the SH-2 Reference Manual for more detail.

The documentation for this struct was generated from the following file:

- src/bno086_reg.h

3.35 sh2_Gyroscope Struct Reference

Gyroscope.

```
#include <bno086_reg.h>
```

Data Fields

- float **x**
- float **y**
- float **z**

3.35.1 Detailed Description

Gyroscope.

See the SH-2 Reference Manual for more detail.

The documentation for this struct was generated from the following file:

- src/bno086_reg.h

3.36 sh2_GyroscopeUncalibrated Struct Reference

Uncalibrated gyroscope.

```
#include <bno086_reg.h>
```

Data Fields

- float **x**
[rad/s]
- float **y**
[rad/s]
- float **z**
[rad/s]
- float **biasX**
[rad/s]
- float **biasY**
[rad/s]
- float **biasZ**
[rad/s]

3.36.1 Detailed Description

Uncalibrated gyroscope.

See the SH-2 Reference Manual for more detail.

The documentation for this struct was generated from the following file:

- src/bno086_reg.h

3.37 sh2_MagneticField Struct Reference

Magnetic field.

```
#include <bno086_reg.h>
```

Data Fields

- float **x**
[uTesla]
- float **y**
[uTesla]
- float **z**
[uTesla]

3.37.1 Detailed Description

Magnetic field.

See the SH-2 Reference Manual for more detail.

The documentation for this struct was generated from the following file:

- src/bno086_reg.h

3.38 sh2_MagneticFieldUncalibrated Struct Reference

Uncalibrated magnetic field.

```
#include <bno086_reg.h>
```

Data Fields

- float **x**
[uTesla]
- float **y**
[uTesla]
- float **z**
[uTesla]
- float **biasX**
[uTesla]
- float **biasY**
[uTesla]
- float **biasZ**
[uTesla]

3.38.1 Detailed Description

Uncalibrated magnetic field.

See the SH-2 Reference Manual for more detail.

The documentation for this struct was generated from the following file:

- src/bno086_reg.h

3.39 sh2_ProductId_s Struct Reference

Product Id value.

```
#include <bno086_reg.h>
```

Data Fields

- uint8_t **resetCause**
- uint8_t **swVersionMajor**
- uint8_t **swVersionMinor**
- uint32_t **swPartNumber**
- uint32_t **swBuildNumber**
- uint16_t **swVersionPatch**
- uint8_t **reserved0**
- uint8_t **reserved1**

3.39.1 Detailed Description

Product Id value.

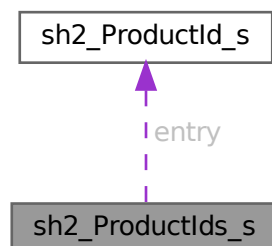
See the SH-2 Reference Manual for more detail.

The documentation for this struct was generated from the following file:

- src/bno086_reg.h

3.40 sh2_ProductIds_s Struct Reference

Collaboration diagram for sh2_ProductIds_s:



Data Fields

- [sh2_ProductId_t](#) **entry** [SH2_MAX_PROD_ID_ENTRIES]
- `uint8_t` **numEntries**

The documentation for this struct was generated from the following file:

- `src/bno086_reg.h`

3.41 sh2_Quaternion Struct Reference

[Quaternion](#) (double precision floating point representation.)

```
#include <bno086_reg.h>
```

Data Fields

- double **x**
- double **y**
- double **z**
- double **w**

3.41.1 Detailed Description

[Quaternion](#) (double precision floating point representation.)

See the SH-2 Reference Manual for more detail.

The documentation for this struct was generated from the following file:

- `src/bno086_reg.h`

3.42 sh2_RawAccelerometer Struct Reference

Raw Accelerometer.

```
#include <bno086_reg.h>
```

Data Fields

- `int16_t` **x**
[ADC counts]
- `int16_t` **y**
[ADC counts]
- `int16_t` **z**
[ADC counts]
- `uint32_t` **timestamp**
[μ S]

3.42.1 Detailed Description

Raw Accelerometer.

See the SH-2 Reference Manual for more detail.

The documentation for this struct was generated from the following file:

- src/bno086_reg.h

3.43 sh2_RawGyroscope Struct Reference

Raw gyroscope.

```
#include <bno086_reg.h>
```

Data Fields

- int16_t **x**
[ADC Counts]
- int16_t **y**
[ADC Counts]
- int16_t **z**
[ADC Counts]
- int16_t **temperature**
[ADC Counts]
- uint32_t **timestamp**
[uS]

3.43.1 Detailed Description

Raw gyroscope.

See the SH-2 Reference Manual for more detail.

The documentation for this struct was generated from the following file:

- src/bno086_reg.h

3.44 sh2_RawMagnetometer Struct Reference

Raw Magnetometer.

```
#include <bno086_reg.h>
```

Data Fields

- int16_t **x**
[ADC Counts]
- int16_t **y**
[ADC Counts]
- int16_t **z**
[ADC Counts]
- uint32_t **timestamp**
[uS]

3.44.1 Detailed Description

Raw Magnetometer.

See the SH-2 Reference Manual for more detail.

The documentation for this struct was generated from the following file:

- src/bno086_reg.h

3.45 sh2_RotationVector Struct Reference

Rotation Vector.

```
#include <bno086_reg.h>
```

Data Fields

- float **i**
Quaternion component i.
- float **j**
Quaternion component j.
- float **k**
Quaternion component k.
- float **real**
Quaternion component real.

3.45.1 Detailed Description

Rotation Vector.

See the SH-2 Reference Manual for more detail.

The documentation for this struct was generated from the following file:

- src/bno086_reg.h

3.46 sh2_RotationVectorWAcc Struct Reference

Rotation Vector with Accuracy.

```
#include <bno086_reg.h>
```

Data Fields

- float **i**
Quaternion component i.
- float **j**
Quaternion component j.
- float **k**
Quaternion component k.
- float **real**
Quaternion component, real.
- float **accuracy**
Accuracy estimate [radians].

3.46.1 Detailed Description

Rotation Vector with Accuracy.

See the SH-2 Reference Manual for more detail.

The documentation for this struct was generated from the following file:

- src/bno086_reg.h

3.47 sh2_SensorConfig Struct Reference

Sensor Configuration settings.

```
#include <bno086_reg.h>
```

Data Fields

- bool **changeSensitivityEnabled**
Enable reports on change.
- bool **changeSensitivityRelative**
Change reports relative (vs absolute)
- bool **wakeupEnabled**
Wake host on event.
- bool **alwaysOnEnabled**
Sensor remains on in sleep state.
- bool **sniffEnabled**
Any output from this sensor should be sent to host, but reportInterval_us and sensorSpecific do not influence sensor operation. Not supported by all sensors.
- uint16_t **changeSensitivity**
Report-on-change threshold.
- uint32_t **reportInterval_us**
[uS] Report interval
- uint32_t **batchInterval_us**
[uS] Batch interval
- uint32_t **sensorSpecific**
See SH-2 Reference Manual for details.

3.47.1 Detailed Description

Sensor Configuration settings.

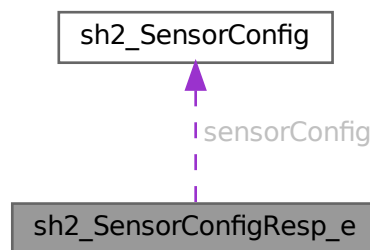
See the SH-2 Reference Manual for more detail.

The documentation for this struct was generated from the following file:

- src/bno086_reg.h

3.48 sh2_SensorConfigResp_e Struct Reference

Collaboration diagram for sh2_SensorConfigResp_e:



Data Fields

- sh2_SensorId_t **sensorId**
- [sh2_SensorConfig_t](#) **sensorConfig**

The documentation for this struct was generated from the following file:

- src/bno086_reg.h

3.49 sh2_SensorEvent Struct Reference

Data Fields

- uint64_t **timestamp_uS**
- int64_t **delay_uS**
- uint8_t **len**
- uint8_t **reportId**
- uint8_t **report** [SH2_MAX_SENSOR_EVENT_LEN]

The documentation for this struct was generated from the following file:

- src/bno086_reg.h

3.50 sh2_SensorMetadata Struct Reference

Sensor Metadata Record.

```
#include <bno086_reg.h>
```

Data Fields

- uint8_t **meVersion**
Motion Engine Version.
- uint8_t **mhVersion**
Motion Hub Version.
- uint8_t **shVersion**
SensorHub Version.
- uint32_t **range**
Same units as sensor reports.
- uint32_t **resolution**
Same units as sensor reports.
- uint16_t **revision**
Metadata record format revision.
- uint16_t **power_mA**
[mA] Fixed point 16Q10 format
- uint32_t **minPeriod_uS**
[uS]
- uint32_t **maxPeriod_uS**
[uS]
- uint32_t **fifoReserved**
(Unused)
- uint32_t **fifoMax**
(Unused)
- uint32_t **batchBufferBytes**
(Unused)
- uint16_t **qPoint1**
q point for sensor values
- uint16_t **qPoint2**
q point for accuracy or bias fields
- uint16_t **qPoint3**
q point for sensor data change sensitivity
- uint32_t **vendorIdLen**
[bytes]
- char **vendorId** [48]
Vendor name and part number.
- uint32_t **sensorSpecificLen**
[bytes]
- uint8_t **sensorSpecific** [48]
See SH-2 Reference Manual.

3.50.1 Detailed Description

Sensor Metadata Record.

See the SH-2 Reference Manual for more detail.

The documentation for this struct was generated from the following file:

- src/bno086_reg.h

3.51 SPI_OBJECT Struct Reference

Data Fields

- void * **txBuffer**
- void * **rxBuffer**
- size_t **txSize**
- size_t **rxSize**
- size_t **dummySize**
- size_t **rxCount**
- size_t **txCount**
- bool **transfersBusy**
- SPI_CALLBACK **callback**
- uintptr_t **context**

The documentation for this struct was generated from the following file:

- src/config/bma400/peripheral/spi/spi_master/plib_spi_master_common.h

3.52 SPI_TRANSFER_SETUP Struct Reference

Data Fields

- uint32_t **clockFrequency**
- SPI_CLOCK_PHASE **clockPhase**
- SPI_CLOCK_POLARITY **clockPolarity**
- SPI_DATA_BITS **dataBits**

The documentation for this struct was generated from the following file:

- src/config/bma400/peripheral/spi/spi_master/plib_spi_master_common.h

3.53 TMR_TIMER_OBJECT Struct Reference

Data Fields

- TMR_CALLBACK **callback_fn**
- uintptr_t **context**

The documentation for this struct was generated from the following file:

- src/config/bma400/peripheral/tmr/plib_tmr_common.h

3.54 UART_OBJECT Struct Reference

Data Fields

- uint8_t * **txBuffer**
- size_t **txSize**
- size_t **txProcessedSize**
- UART_CALLBACK **txCallback**
- uintptr_t **txContext**
- bool **txBusyStatus**
- uint8_t * **rxBuffer**
- size_t **rxSize**
- size_t **rxProcessedSize**
- UART_CALLBACK **rxCallback**
- uintptr_t **rxContext**
- bool **rxBusyStatus**
- volatile UART_ERROR **errors**

The documentation for this struct was generated from the following file:

- src/config/bma400/peripheral/uart/plib_uart_common.h

3.55 UART_RING_BUFFER_OBJECT Struct Reference

Data Fields

- UART_RING_BUFFER_CALLBACK **wrCallback**
- uintptr_t **wrContext**
- volatile uint32_t **wrInIndex**
- volatile uint32_t **wrOutIndex**
- uint32_t **wrThreshold**
- bool **isWrNotificationEnabled**
- bool **isWrNotifyPersistently**
- uint32_t **wrBufferSize**
- UART_RING_BUFFER_CALLBACK **rdCallback**
- uintptr_t **rdContext**
- volatile uint32_t **rdInIndex**
- volatile uint32_t **rdOutIndex**
- uint32_t **rdBufferSize**
- bool **isRdNotificationEnabled**
- uint32_t **rdThreshold**
- bool **isRdNotifyPersistently**
- volatile UART_ERROR **errors**

The documentation for this struct was generated from the following file:

- src/config/bma400/peripheral/uart/plib_uart_common.h

3.56 UART_SERIAL_SETUP Struct Reference

Data Fields

- uint32_t **baudRate**
- UART_PARITY **parity**
- UART_DATA **dataWidth**
- UART_STOP **stopBits**

The documentation for this struct was generated from the following file:

- src/config/bma400/peripheral/uart/plib_uart_common.h

Chapter 4

File Documentation

4.1 display_type.h

```
00001 /*
00002  * File:    display_type.h
00003  * Author:  root
00004  *
00005  * Created on August 18, 2020, 3:04 PM
00006  */
00007
00008 #ifndef DISPLAY_TYPE_H
00009 #define DISPLAY_TYPE_H
00010
00011 #ifdef __cplusplus
00012 extern "C" {
00013 #endif
00014
00015     /*
00016     * LCD hardware selection
00017     */
00018     //Select the display type: DOGS102: 102, DOGM128/DOGL128: 128, DOGM132: 132, DOGXL160: 160,
    DOGXL240: 240
00019 #define DISPLAY_TYPE 240
00020
00021
00022 #ifdef __cplusplus
00023 }
00024 #endif
00025
00026 #endif /* DISPLAY_TYPE_H */
00027
```

4.2 dogm-graphic.h

```
00001 #ifndef DOGMGRAPHIC_H_INCLUDED
00002 #define DOGMGRAPHIC_H_INCLUDED
00003
00004 #include "lcd_drv.h"
00005 #include "timers.h"
00006
00007 /*****
00008  * BEGIN CONFIG BLOCK
00009  *****/
00010
00011 //Display Orientation: Normal (0) or upside-down (1)?
00012 #define ORIENTATION_UPSIDEDOWN 0
00013
00014 //Should chip select (CS) be used?
00015 #define LCD_USE_CHIPSELECT 1
00016
00017 //Use Backlight? (0: no backlight, 1: backlight (on when pin is high), 2: backlight (on when pin is
    low)
00018 #define LCD_USE_BACKLIGHT 0
00019
00020 //Define this if LCD Output should continue in next line when reaching edge of display
00021 //Used for all outputs. To enable this feature for text only, use the appropriate flag in font.h
```

```

00022 #define LCD_WRAP_AROUND 0
00023
00024 //Include graphic functions, i.e. lcd_draw_image_P, lcd_draw_image_xy_P, lcd_clear_area ?
00025 #define LCD_INCLUDE_GRAPHIC_FUNCTIONS 0
00026
00027 /*****
00028  * END CONFIG BLOCK
00029  *****/
00030
00031 extern volatile uint8_t __attribute__((coherent)) rgbOledBmp_page[];
00032
00033 extern void ReSet_SetLow(void);
00034 extern void ReSet_SetHigh(void);
00035 extern void RS_SetLow(void);
00036 extern void RS_SetHigh(void);
00037 extern void CSB_SetLow(void);
00038 extern void CSB_SetHigh(void);
00039 extern void SPI1_Exchange8bit(uint8_t);
00040
00041
00042 /*****
00043  * Public Functions
00044  *****/
00045
00046 //initializes the display in standard settings
00047 //for DOGS, DOGM, DOGL and DOGXL
00048 void lcd_init(void);
00049
00050 //write data word or command to the LCD
00051 void lcd_data(uint8_t data);
00052 void lcd_command(uint8_t cmd);
00053 void lcd_command_s(uint8_t cmd);
00054 uint8_t lcd_inc_column(int16_t);
00055
00056 //Function to read the current position
00057 extern uint8_t lcd_current_page;
00058 extern uint8_t lcd_current_column;
00059
00060 static inline uint8_t lcd_get_position_page(void)
00061 {
00062     return lcd_current_page;
00063 }
00064
00065 static inline uint8_t lcd_get_position_column(void)
00066 {
00067     return lcd_current_column;
00068 }
00069
00070 //User functions to write raw data from flash
00071 #if LCD_INCLUDE_GRAPHIC_FUNCTIONS >= 1
00072 void lcd_draw_image_P(uint8_t * progmem_image, uint8_t pages, uint8_t columns, uint8_t style);
00073 void lcd_draw_image_xy_P(uint8_t * progmem_image, uint8_t x, uint8_t y, uint8_t pages, uint8_t
columns, uint8_t style);
00074 #endif
00075
00076 void lcd_clear_area(uint8_t pages, uint8_t columns, uint8_t style);
00077 void lcd_clear_area_xy(uint8_t pages, uint8_t columns, uint8_t style, uint8_t page, uint8_t col);
00078
00079
00080 //Move cursor
00081 void lcd_moveto_xy(uint8_t page, uint8_t column);
00082 void lcd_move_xy(int8_t pages, int16_t columns);
00083
00084 //Set display contrast
00085 #if DISPLAY_TYPE == 240
00086 void lcd_set_contrast(uint8_t value);
00087 #endif
00088
00089 //Text functions are included in font.c / font.h
00090
00091 /*****
00092  * LCD Size, based on type selection above
00093  *****/
00094 #if DISPLAY_TYPE == 240 // Autor: Robert Steigemann
00095 #define LCD_WIDTH 240 //width of the LCD
00096 #define LCD_HEIGHT 128 // height of the LCD
00097 #define LCD_RAM_PAGES 16 //size of LCD RAM
00098 #define LCD_PIXEL_PER_BYTE 8 //using single pixels
00099 #endif
00100
00101 #if DISPLAY_TYPE == 160
00102 #define LCD_WIDTH 160 //width of the LCD
00103 #define LCD_HEIGHT 104 //height of the LCD
00104 #define LCD_RAM_PAGES 26 //size of LCD RAM
00105 #define LCD_PIXEL_PER_BYTE 4 //using double pixels
00106 #define LCD_DOUBLE_PIXEL 1
00107 #define SHIFT_ADDR_NORMAL 0 //column offset for normal orientation

```



```

00108 #define SHIFT_ADDR_TOPVIEW 0 //column offset for bottom view orientation
00109 #endif
00110
00111 #if DISPLAY_TYPE == 132
00112 #define LCD_WIDTH 132 //width of the LCD
00113 #define LCD_HEIGHT 32 //height of the LCD
00114 #define LCD_RAM_PAGES 4 //size of LCD RAM
00115 #define LCD_PIXEL_PER_BYTE 8 //using single pixels
00116 #define SHIFT_ADDR_NORMAL 0 //column offset for normal orientation
00117 #define SHIFT_ADDR_TOPVIEW 0 //column offset for bottom view orientation
00118 #endif
00119
00120 #if DISPLAY_TYPE == 128
00121 #define LCD_WIDTH 128 //width of the LCD
00122 #define LCD_HEIGHT 64 //height of the LCD
00123 #define LCD_RAM_PAGES 8 //size of LCD RAM
00124 #define LCD_PIXEL_PER_BYTE 8 //using single pixels
00125 #define SHIFT_ADDR_NORMAL 0 //column offset for normal orientation
00126 #define SHIFT_ADDR_TOPVIEW 4 //column offset for bottom view orientation
00127 #endif
00128
00129 #if DISPLAY_TYPE == 102
00130 #define LCD_WIDTH 102 //width of the LCD
00131 #define LCD_HEIGHT 64 //height of the LCD
00132 #define LCD_RAM_PAGES 8 //size of LCD RAM
00133 #define LCD_PIXEL_PER_BYTE 8 //using single pixels
00134 #define SHIFT_ADDR_NORMAL 0 //column offset for normal orientation
00135 #define SHIFT_ADDR_TOPVIEW 30 //column offset for bottom view orientation
00136 #endif
00137
00138 #if ORIENTATION_UPSIDEDOWN == 0
00139 #define SHIFT_ADDR SHIFT_ADDR_NORMAL
00140 #endif
00141
00142 #if ORIENTATION_UPSIDEDOWN == 1
00143 #define SHIFT_ADDR SHIFT_ADDR_TOPVIEW
00144 #endif
00145
00146 /*****
00147  * Command Codes
00148  *****/
00149 #if DISPLAY_TYPE == 128 || DISPLAY_TYPE == 132 || DISPLAY_TYPE == 102
00150 #define LCD_DISPLAY_ENABLE 0xAE //1: Display on/off
00151 #define LCD_START_LINE 0x40 //2: display start line set
00152 #define LCD_PAGE_ADDRESS 0xB0 //3: Page address set (lower 4 bits select one of the pages)
00153 #define LCD_COL_ADDRESS 0x10 //4: column address
00154 #define LCD_BOTTOMVIEW 0xA0 //8: select orientation
00155 #define LCD_DISPLAY_INVERT 0xA6 //9: inverted display
00156 #define LCD_ALL_PIXEL 0xA4 //10: show memory content or switch all pixels on
00157 #define LCD_BIAS 0xA2 //11: lcd bias set
00158 #define LCD_RESET_CMD 0xE2 //14: Reset Controller
00159 #define LCD_SCAN_DIR 0xC0 //15: output mode select (turns display upside-down)
00160 #define LCD_POWER_CONTROL 0x28 //16: power control set
00161 #define LCD_VOLTAGE 0x20 //17: voltage regulator resistor ratio set
00162 #define LCD_VOLUME_MODE 0x81 //18: Volume mode set
00163 #define LCD_NO_OP 0xE3 //22: NOP command
00164 #endif
00165
00166 #if DISPLAY_TYPE == 128 || DISPLAY_TYPE == 132
00167 #define LCD_INDICATOR 0xAC //19: static indicator (2-byte command)
00168 #define LCD_BOOSTER_SET 0xF8 //20: booster ratio set
00169 #endif
00170
00171 #if DISPLAY_TYPE == 102
00172 #define LCD_ADV_PROG_CTRL 0xFA //25: advanced program control
00173 #define LCD_ADV_PROG_CTRL2 0x10 //25: advanced program control
00174 #endif
00175
00176 #if DISPLAY_TYPE == 160
00177 #define LCD_COL_ADDRESS 0x10 //4: column address
00178 #define LCD_TEMP_COMP 0x24 //5: Set Temperature Compensation
00179 #define LCD_PANEL_LOAD 0x28 //6: Set Panel loading
00180 #define LCD_PUMP_CTRL 0x2C //7: Set pump control
00181 #define LCD_ADV_PROG_CTRL 0x30 //8: advanced program control first word
00182 #define LCD_START_LINE 0x40 //9: display scroll line set LSB
00183 #define LCD_START_LINE2 0x50 //9: display scroll line set MSB
00184 #define LCD_PAGE_ADDRESS 0x60 //10: Page address set
00185 #define LCD_VOLTAGE_BIAS 0x81 //11: Bias set
00186 #define LCD_PARTIAL_CTRL 0x84 //12: Set partial display control
00187 #define LCD_RAM_ADDR_CTRL 0x88 //13: Set RAM address control
00188 #define LCD_FIXED_LINES 0x90 //14: Set display fixed lines
00189 #define LCD_LINE_RATE 0xA0 //15: Set line rate
00190 #define LCD_ALL_PIXEL 0xA4 //16: show all points
00191 #define LCD_INVERSE 0xA6 //17: Inverse display
00192 #define LCD_DISPLAY_ENABLE 0xAE //18: Display enable
00193 #define LCD_MAPPING_CTRL 0xC0 //19: LCD mapping control
00194 #define LCD_GRAY_SHADE 0xD0 //20: LCD gray shade

```

```

00195 #define LCD_RESET_CMD          0xE2 //21: System reset
00196 #define LCD_NO_OP                0xE3 //22: NOP
00197 #define LCD_BIAS_RATIO           0xE8 //24: Bias Ratio
00198 #define LCD_CURSOR_MODE_RESET   0xEE //25: Reset cursor update mode
00199 #define LCD_CURSOR_MODE_SET      0xEF //26: Set cursor update mode
00200 #define LCD_COM_END              0xF1 //27: Set COM End
00201 #define LCD_PARTIAL_START        0xF2 //28: Set partial display start
00202 #define LCD_PARTIAL_END          0xF3 //29: Set partial display end
00203 #define LCD_WINDOW_START_COL     0xF4 //30: Window program start column
00204 #define LCD_WINDOW_START_PAGE    0xF5 //31: Window program start column
00205 #define LCD_WINDOW_END_COL       0xF6 //32: Window program start column
00206 #define LCD_WINDOW_END_PAGE     0xF7 //33: Window program start column
00207 #define LCD_WINDOW_PROGRAM       0xF8 //34: Enable window programming
00208 #endif
00209
00210 #if DISPLAY_TYPE == 240
00211 #define LCD_COL_ADDRESS          0x10 //4: column address MSB
00212 #define LCD_TEMP_COMP            0x24 //5: Set Temperature Compensation
00213 #define LCD_PANEL_LOAD           0x28 //6: Set Panel loading
00214 #define LCD_PUMP_CTRL            0x2C //7: Set pump control
00215 #define LCD_START_LINE           0x40 //9: display scroll line set LSB
00216 #define LCD_START_LINE2         0x50 //9: display scroll line set MSB
00217 #define LCD_PAGE_ADDRESS_LSB     0x60 //10: Page address set
00218 #define LCD_PAGE_ADDRESS_MSB     0x70 //10: Page address set
00219 #define LCD_POTENTIOMETER        0x81 //11: Potentiometer set
00220 #define LCD_PARTIAL_CTRL         0x84 //14: Set partial display control
00221 #define LCD_RAM_ADDR_CTRL        0x88 //15: Set RAM address control
00222 #define LCD_FIXED_LINES          0x90 //16: Set fixed lines
00223 #define LCD_LINE_RATE            0xA0 //17: Set line rate
00224 #define LCD_ALL_PIXEL            0xA4 //18: Set all pixel on
00225 #define LCD_INVERSE              0xA6 //19: Inverse display
00226 #define LCD_DISPLAY_ENABLE       0xA8 //20: Display enable
00227 #define LCD_MAPPING_CTRL         0xC0 //21: LCD mapping control
00228 #define LCD_NLINE_INV            0xC8 //22: Set N-Line Inversion
00229 #define LCD_SET_PATTERN          0xD0 //23: Set Display Pattern
00230 #define LCD_RESET_CMD            0xE2 //24: System reset
00231 #define LCD_NO_OP                0xE3 //25: NOP
00232 #define LCD_BIAS_RATIO           0xE8 //27: Bias Ratio
00233 #define LCD_COM_END              0xF1 //28: Set COM End
00234 #define LCD_PARTIAL_START        0xF2 //29: Set display start
00235 #define LCD_PARTIAL_END          0xF3 //30: Set display end
00236 #define LCD_WINDOW_START_COL     0xF4 //31: Window program start column addr
00237 #define LCD_WINDOW_START_PAGE    0xF5 //32: Window program start row adr
00238 #define LCD_WINDOW_END_COL       0xF6 //33: Window program end column adr
00239 #define LCD_WINDOW_END_PAGE     0xF7 //34: Window program end row adr
00240 #define LCD_WINDOW_PROGRAM       0xF8 //35: Enable window programming
00241 #endif
00242
00243 /*****
00244  * Makros to execute commands
00245  *****/
00246 #if DISPLAY_TYPE == 240
00247 #define LCD_SET_TEMP_COMP(i)      lcd_command(LCD_TEMP_COMP | ((i) & 0x3))
00248 #define LCD_SET_PANEL_LOAD(i)     lcd_command(LCD_PANEL_LOAD | ((i) & 0x3))
00249 #define LCD_SET_PUMP_CTRL(i)      lcd_command(LCD_PUMP_CTRL | ((i) & 0x3))
00250 #define LCD_SET_START_LINE(i)     lcd_command(LCD_START_LINE | ((i)&0xF)); \
00251                                   lcd_command(LCD_START_LINE2 | (((i)>4)&0x7))
00252 #define LCD_SET_PARTIAL_DISPLAY_CTRL(i) \
00253                                   lcd_command(LCD_PARTIAL_CTRL | ((i) & 0x3))
00254 #define LCD_SET_RAM_ADDR_CTRL(i)  lcd_command(LCD_RAM_ADDR_CTRL | ((i) & 0x7))
00255 #define LCD_SET_DISPLAY_PATTERN(i) lcd_command(LCD_SET_PATTERN | ((i) & 0x7))
00256 #define LCD_SET_FIXED_LINES(i)    lcd_command(LCD_FIXED_LINES | ((i) & 0xF))
00257 #define LCD_SET_LINE_RATE(i)      lcd_command(LCD_LINE_RATE | ((i) & 0x3))
00258 #define LCD_SHOW_ALL_PIXELS_ON()  lcd_command(LCD_ALL_PIXEL | 1)
00259 #define LCD_SHOW_ALL_PIXELS_OFF() lcd_command(LCD_ALL_PIXEL | 0)
00260 #define LCD_INVERT_DISPLAY(i)     lcd_command(LCD_INVERSE | ((i)&1))
00261 #define LCD_SWITCH_ON()           lcd_command(LCD_DISPLAY_ENABLE | 1)
00262 #define LCD_SWITCH_OFF()          lcd_command(LCD_DISPLAY_ENABLE | 0)
00263 #define LCD_SET_MAPPING_CTRL(i)   lcd_command(LCD_MAPPING_CTRL); \
00264                                   lcd_command((i) & 0xF)
00265 #define LCD_SET_PAGE_ADDR(i)      lcd_command(LCD_PAGE_ADDRESS_LSB | ((i) & 0x0F)); \
00266                                   lcd_command(LCD_PAGE_ADDRESS_MSB | (((i)>4) & 0x0F))
00267 #define LCD_SET_COLUMN_ADDR(i)    lcd_command(LCD_COL_ADDRESS | (((i)>4) & 0x0F)); \
00268                                   lcd_command((i) & 0x0F)
00269 #define LCD_GOTO_ADDRESS(page,col) lcd_command(LCD_PAGE_ADDRESS_LSB | ((page) & 0x0F)); \
00270                                   lcd_command(LCD_PAGE_ADDRESS_MSB | (((page)>4) & 0x0F)); \
00271                                   lcd_command(LCD_COL_ADDRESS | (((col)>4) & 0x0F)); \
00272                                   lcd_command((col) & 0x0F)
00273 #define LCD_GOTO_ADDRESS_S(page,col) lcd_command_s(LCD_PAGE_ADDRESS_LSB | ((page) & 0x0F)); \
00274                                   lcd_command_s(LCD_PAGE_ADDRESS_MSB | (((page)>4) & 0x0F)); \
00275                                   lcd_command_s(LCD_COL_ADDRESS | (((col)>4) & 0x0F)); \
00276                                   lcd_command_s((col) & 0x0F)
00277
00278 #define LCD_NOP()                 lcd_command(LCD_NO_OP)
00279 #define LCD_SET_BIAS_RATIO(i)     lcd_command(LCD_BIAS_RATIO | ((i) & 0x3))
00280 #define LCD_SET_COM_END(i)        lcd_command(LCD_COM_END); \
00281                                   lcd_command(i)

```

```

00282 #define LCD_SET_PARTIAL_DISPLAY(start,end) \
00283         lcd_command(LCD_PARTIAL_START); \
00284         lcd_command(start); \
00285         lcd_command(LCD_PARTIAL_END); \
00286         lcd_command(end)
00287 #define LCD_SET_PROGRAM_WINDOW(startpage,startcol,endpage,endcol) \
00288         lcd_command(LCD_WINDOW_START_PAGE); \
00289         lcd_command(startpage); \
00290         lcd_command(LCD_WINDOW_START_COL); \
00291         lcd_command(startcol); \
00292         lcd_command(LCD_WINDOW_END_PAGE); \
00293         lcd_command(endpage); \
00294         lcd_command(LCD_WINDOW_END_COL); \
00295         lcd_command(endcol)
00296 #define LCD_ENABLE_WINDOW_PROGRAM        lcd_command(LCD_WINDOW_PROGRAM | 1)
00297 #define LCD_DISABLE_WINDOW_PROGRAM        lcd_command(LCD_WINDOW_PROGRAM | 0)
00298 #define LCD_SET_POTI(i)                   lcd_command(LCD_POTENTIOMETER); \
00299         lcd_command(i)
00300 #define LCD_SYSTEM_RESET                  lcd_command(LCD_RESET_CMD)
00301 #endif
00302
00303 #if DISPLAY_TYPE == 160
00304 #define LCD_SET_TEMP_COMP(i)              lcd_command(LCD_TEMP_COMP | ((i) & 0x3))
00305 #define LCD_SET_PANEL_LOAD(i)             lcd_command(LCD_PANEL_LOAD | ((i) & 0x3))
00306 #define LCD_SET_PUMP_CTRL(i)              lcd_command(LCD_PUMP_CTRL | ((i) & 0x3))
00307 #define LCD_SET_ADV_PROG_CTRL(i)          lcd_command(LCD_ADV_PROG_CTRL); \
00308         lcd_command((i) & 0x3F)
00309 #define LCD_SET_START_LINE(i)             lcd_command(LCD_START_LINE | ((i) & 0xF)); \
00310         lcd_command(LCD_START_LINE2 | (((i) > 4) & 0x7))
00311 #define LCD_SET_VOLTAGE_BIAS(i)           lcd_command(LCD_VOLTAGE_BIAS); \
00312         lcd_command((i) & 0xFF)
00313 #define LCD_SET_PARTIAL_DISPLAY_CTRL(i)   lcd_command(LCD_PARTIAL_CTRL | ((i) & 0x3))
00314 #define LCD_SET_RAM_ADDR_CTRL(i)          lcd_command(LCD_RAM_ADDR_CTRL | ((i) & 0x7))
00315 #define LCD_SET_FIXED_LINES(i)           lcd_command(LCD_FIXED_LINES | ((i) & 0xF))
00316 #define LCD_SET_LINE_RATE(i)             lcd_command(LCD_LINE_RATE | ((i) & 0x3))
00317 #define LCD_SHOW_ALL_PIXELS_ON()          lcd_command(LCD_ALL_PIXEL | 1)
00318 #define LCD_SHOW_ALL_PIXELS_OFF()         lcd_command(LCD_ALL_PIXEL | 0)
00319 #define LCD_INVERT_DISPLAY(i)            lcd_command(LCD_INVERSE | ((i) & 1))
00320 #define LCD_SWITCH_ON()                  lcd_command(LCD_DISPLAY_ENABLE | 1)
00321 #define LCD_SWITCH_OFF()                 lcd_command(LCD_DISPLAY_ENABLE | 0)
00322 #define LCD_SET_MAPPING_CTRL(i)          lcd_command(LCD_MAPPING_CTRL | ((i) & 0x7))
00323 #define LCD_RESET_CURSOR_UPDATE_MODE     lcd_command(LCD_MAPPING_CTRL | 0)
00324 #define LCD_SET_TOP_VIEW()               lcd_command(LCD_MAPPING_CTRL | 6)
00325 #define LCD_SET_GRAY_SHADE(i)            lcd_command(LCD_GRAY_SHADE | ((i) & 0x3)) SPI3_Exchange8bit
00326 #define LCD_SET_PAGE_ADDR(i)             lcd_command(LCD_PAGE_ADDRESS | ((i) & 0x1F))
00327 #define LCD_SET_COLUMN_ADDR(col)          lcd_command(LCD_COL_ADDRESS | (((col)+SHIFT_ADDR)>4) & 0x0F)); \
00328         lcd_command(((col)+SHIFT_ADDR) & 0x0F))
00329 #define LCD_GOTO_ADDRESS(page,col)        lcd_command(LCD_PAGE_ADDRESS | ((page) & 0x1F)); \
00330         lcd_command(LCD_COL_ADDRESS | (((col)+SHIFT_ADDR)>4) & \
00331         0x0F)); \
00332         lcd_command(((col)+SHIFT_ADDR) & 0x0F))
00333 #define LCD_NOP()                        lcd_command(LCD_NO_OP)
00334 #define LCD_SET_BIAS_RATIO(i)            lcd_command(LCD_BIAS_RATIO | ((i) & 0x3))
00335 #define LCD_SET_CURSOR_UPDATE_MODE       lcd_command(LCD_CURSOR_MODE_SET)
00336 #define LCD_RESET_CURSOR_UPDATE_MODE     lcd_command(LCD_CURSOR_MODE_RESET)
00337 #define LCD_SET_COM_END(i)               lcd_command(LCD_COM_END); \
00338         lcd_command(i)
00339 #define LCD_SET_PARTIAL_DISPLAY(start,end) \
00340         lcd_command(LCD_PARTIAL_START); \
00341         lcd_command((start) & 0x7F) \
00342         lcd_command(LCD_PARTIAL_END); \
00343         lcd_command((end) & 0x7F)
00344 #define LCD_SET_PROGRAM_WINDOW(startpage,startcol,endpage,endcol) \
00345         lcd_command(LCD_WINDOW_START_PAGE); \
00346         lcd_command(startpage); \
00347         lcd_command(LCD_WINDOW_START_COL); \
00348         lcd_command(startcol); \
00349         lcd_command(LCD_WINDOW_END_PAGE); \
00350         lcd_command(endpage); \
00351         lcd_command(LCD_WINDOW_END_COL); \
00352         lcd_command(endcol)
00353 #define LCD_ENABLE_WINDOW_PROGRAM        lcd_command(LCD_WINDOW_PROGRAM | 1)
00354 #define LCD_DISABLE_WINDOW_PROGRAM        lcd_command(LCD_WINDOW_PROGRAM | 0)
00355 #endif
00356
00357 #if DISPLAY_TYPE == 128 || DISPLAY_TYPE == 132 || DISPLAY_TYPE == 102
00358 #define LCD_SWITCH_ON()                  lcd_command(LCD_DISPLAY_ENABLE | 1)
00359 #define LCD_SWITCH_OFF()                 lcd_command(LCD_DISPLAY_ENABLE | 0)
00360 #define LCD_SET_FIRST_LINE(i)            lcd_command(LCD_START_LINE | ((i) & 0x3F))
00361 #define LCD_SET_BOTTOM_VIEW()            lcd_command(LCD_BOTTOMVIEW | 1)
00362 #define LCD_SET_TOP_VIEW()               lcd_command(LCD_BOTTOMVIEW | 0)
00363 #define LCD_SET_MODE_POSITIVE()          lcd_command(LCD_DISPLAY_INVERT | 0)
00364 #define LCD_SET_MODE_INVERTED()          lcd_command(LCD_DISPLAY_INVERT | 1)
00365 #define LCD_SHOW_ALL_PIXELS_ON()         lcd_command(LCD_ALL_PIXEL | 1)
00366 #define LCD_SHOW_ALL_PIXELS_OFF()        lcd_command(LCD_ALL_PIXEL | 0)

```

```

00367 #define LCD_SET_BIAS_RATIO_1_7()      lcd_command(LCD_BIAS | 1)
00368 #define LCD_SET_BIAS_RATIO_1_9()      lcd_command(LCD_BIAS | 0)
00369 #define LCD_SEND_RESET()              lcd_command(LCD_RESET_CMD)
00370 #define LCD_ORIENTATION_NORMAL()      lcd_command(LCD_SCAN_DIR | 0x0)
00371 #define LCD_ORIENTATION_UPSIDEDOWN()  lcd_command(LCD_SCAN_DIR | 0x8)
00372 #define LCD_SET_POWER_CONTROL(i)      lcd_command(LCD_POWER_CONTROL | ((i) & 0x07))
00373 #define LCD_SET_LOW_POWER()            lcd_command(LCD_POWER_CONTROL | 0x7)
00374 #define LCD_SET_WIDE_RANGE()           lcd_command(LCD_POWER_CONTROL | 0x7)
00375 #define LCD_SET_LOW_VOLTAGE()          lcd_command(LCD_POWER_CONTROL | 0x3)
00376 #define LCD_SET_BIAS_VOLTAGE(i)        lcd_command(LCD_VOLTAGE | ((i) & 0x07))

00377 #define LCD_SET_VOLUME_MODE(i)         lcd_command(LCD_VOLUME_MODE); \
00378                                     lcd_command(((i) & 0x3F))
00379 #define LCD_NOP()                      lcd_command(LCD_NO_OP)
00380 #endif
00381
00382 #if DISPLAY_TYPE == 128 || DISPLAY_TYPE == 132
00383 #define LCD_SET_PAGE_ADDR(i)           lcd_command(LCD_PAGE_ADDRESS | ((i) & 0x0F))
00384 #define LCD_SET_COLUMN_ADDR(col)       lcd_command(LCD_COL_ADDRESS | (((col)+SHIFT_ADDR)>>4) & 0x0F)); \
00385                                     lcd_command(((col)+SHIFT_ADDR) & 0x0F))
00386 #define LCD_GOTO_ADDRESS(page,col)     lcd_command(LCD_PAGE_ADDRESS | ((page) & 0x1F)); \
00387                                     lcd_command(LCD_COL_ADDRESS | (((col)+SHIFT_ADDR)>>4) & 0x0F)); \
00388                                     lcd_command(((col)+SHIFT_ADDR) & 0x0F))
00389 #define LCD_SET_INDICATOR_OFF()         lcd_command(LCD_INDICATOR | 0); \
00390                                     lcd_command(0x00)
00391 #define LCD_SET_INDICATOR_STATIC()      lcd_command(LCD_INDICATOR | 1); \
00392                                     lcd_command(0x11)
00393 #define LCD_SET_INDICATOR_1HZ()         lcd_command(LCD_INDICATOR | 1); \
00394                                     lcd_command(0x01)
00395 #define LCD_SET_INDICATOR_2HZ()         lcd_command(LCD_INDICATOR | 1); \
00396                                     lcd_command(0x10)
00397 #define LCD_SET_INDICATOR(i,j)          lcd_command(LCD_INDICATOR | ((i) & 1)); \
00398                                     lcd_command(((j) & 2))
00399 #define LCD_SET_BOOSTER_MODE(i)         lcd_command(LCD_BOOSTER_SET); \
00400                                     lcd_command(((i) & 0x03))
00401 #define LCD_SET_BOOSTER_MODE_234       lcd_command(LCD_BOOSTER_SET); \
00402                                     lcd_command(0x0)
00403 #define LCD_SET_BOOSTER_MODE_5         lcd_command(LCD_BOOSTER_SET); \
00404                                     lcd_command(0x1)
00405 #define LCD_SET_BOOSTER_MODE_6         lcd_command(LCD_BOOSTER_SET); \
00406                                     lcd_command(0x3)
00407 #define LCD_SLEEP_MODE                 lcd_command(LCD_INDICATOR_OFF); \
00408                                     lcd_command(LCD_DISPLAY_ENABLE | 0); \
00409                                     lcd_command(LCD_ALL_PIXEL | 1)
00410 #endif
00411
00412 #if DISPLAY_TYPE == 102
00413 #define LCD_SET_PAGE_ADDR(i)           lcd_command(LCD_PAGE_ADDRESS | ((i) & 0x0F))
00414 #define LCD_SET_COLUMN_ADDR(col)       lcd_command(((col)+SHIFT_ADDR) & 0x0F)); \
00415                                     lcd_command(LCD_COL_ADDRESS | (((col)+SHIFT_ADDR)>>4) & 0x0F))
00416 #define LCD_GOTO_ADDRESS(page,col)     lcd_command(LCD_PAGE_ADDRESS | ((page) & 0x1F)); \
00417                                     lcd_command(((col)+SHIFT_ADDR) & 0x0F)); \
00418                                     lcd_command(LCD_COL_ADDRESS | (((col)+SHIFT_ADDR)>>4) & 0x0F))
00419 #define LCD_TEMP_COMP_HIGH             0x80
00420 #define LCD_COL_WRAP                   0x02
00421 #define LCD_PAGE_WRAP                  0x01
00422 #define LCD_SET_ADV_PROG_CTRL(i)       lcd_command(LCD_ADV_PROG_CTRL); \
00423                                     lcd_command(LCD_ADV_PROG_CTRL2 | i)
00424 #endif
00425
00426
00427
00428
00429
00430 /*****
00431  * Output pin controlling macros
00432  *****/
00433
00434 //Control A0 input of LCD
00435 #define LCD_DRAM()                     RS_SetHigh();
00436 #define LCD_CMD()                      RS_SetLow();
00437
00438 //Control reset input of LCD
00439 #define LCD_RESET_OFF()                PORT_RST |= _BV(PIN_RST)
00440 #define LCD_RESET_ON()                 PORT_RST &= ~_BV(PIN_RST)
00441 #define LCD_SET_OUTPUT_RST()           DDR_RST |= _BV(PIN_RST)
00442 #define LCD_RESET()                    LCD_RESET_ON(); _delay_ms(1); LCD_RESET_OFF(); _delay_ms(10)
00443
00444 //Control pin for Backlight
00445 #if LCD_USE_BACKLIGHT == 1            //normal
00446 #define LCD_SET_OUTPUT_LED()           BACKLIGHT_OutputEnable();
00447 #define BACKLIGHT_ON()                 BACKLIGHT_Set();
00448 #define BACKLIGHT_OFF()                BACKLIGHT_Clear();
00449 #elif LCD_USE_BACKLIGHT == 2          //inverted
00450 #define LCD_SET_OUTPUT_LED()           DDR_LED |= _BV(PIN_LED)
00451 #define BACKLIGHT_ON()                 PORT_LED &= ~(1<<PIN_LED)

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00452 #define BACKLIGHT_OFF()      PORT_LED |= (1<<PIN_LED)
00453 #else
00454 #define LCD_SET_OUTPUT_LED()
00455 #define BACKLIGHT_ON()
00456 #define BACKLIGHT_OFF()
00457 #endif
00458
00459 //Control pin for chip select
00460 #if LCD_USE_CHIPSELECT == 1
00461 #define LCD_SELECT()          CSB_SetLow();
00462 #define LCD_UNSELECT()        CSB_SetHigh();
00463 #else
00464 #define LCD_SET_OUTPUT_CS()
00465 #define LCD_SELECT()          spi_wait_for_idle();
00466 #define LCD_UNSELECT()
00467 #endif
00468
00469 //Bit positions for style settings
00470 #define NORMAL      0
00471 #define INVERT      4
00472 #define INVERT_BIT  4
00473
00474
00475 #endif
00476

```

4.3 eadog.h

```

00001 /*
00002  * File:   eadog.h
00003  * Author: root
00004  *
00005  * Created on July 27, 2015, 2:05 PM
00006  */
00007
00008 /* Parts taken from:
00009  * This version for XC32 PIC32MK MCM
00010  * modified for DMA and 32-bit hardware
00011  *
00012  *      file: EA-DOGM_MIO.c
00013  *      version: 2.03
00014  *      description: Multi I/O driver for EA DOGM displays
00015  *                  : Uses 8Bit, SPI HW or SPI SW (bitbang)
00016  *      written by : Michael Bradley (mbradley@mcmlabs.com)
00017  *      contributions: Imaginos (CCS forum), Emil Nad (8Bit testing)
00018  *                  jgschmidt (CCS forum)
00019  */
00020
00021 /*
00022  * machine dependant hardware routines should be here
00023  */
00024
00025 #ifndef EADOG_H
00026 #define EADOG_H
00027
00028 #ifdef __cplusplus
00029 extern "C" {
00030 #endif
00031
00032 #include "lcd_drv.h"
00033
00034 #define LCD_CMD_MASK      0x01
00035 #define LCD_CMD_SET      0x100
00036 #define LCD_CLEAR_HOME    0x04
00037
00038 /*
00039  * for 2MHz SPI clock
00040  */
00041 #define IS_DELAYSHORT      4
00042 #define IS_DELAYMED      5
00043 #define IS_DELAYLONG      800
00044 #define IS_DELAYPOWERUP    400000
00045
00046 #define EADOGM_CMD_CLR      1
00047 #define EADOGM_CMD_CURSOR_ON      0b00001111
00048 #define EADOGM_CMD_CURSOR_OFF     0b00001100
00049 #define EADOGM_CMD_DISPLAY_ON     0b00001100
00050 #define EADOGM_CMD_DISPLAY_OFF    0b00001000
00051 #define EADOGM_CMD_DDRAM_ADDR     0b10000000
00052 #define EADOGM_CMD_CGRAM_ADDR     0b01000000
00053 #define EADOGM_CMD_SELECT_R0      0b00011000
00054 #define EADOGM_CMD_SELECT_R1      0b00010000
00055 #define EADOGM_CMD_SET_TABLE2     0b00101010
00056 #define EADOGM_COLSPAN      16

```

```

00056
00057     void wdtDelay(uint32_t);
00058     void init_display(void);
00059     void eaDogM_WriteChr(int8_t);
00060     void eaDogM_WriteCommand(uint8_t);
00061     void eaDogM_SetPos(uint8_t, uint8_t);
00062     void eaDogM_ClearRow(uint8_t);
00063     void eaDogM_WriteString(char *);
00064     void eaDogM_WriteStringAtPos(uint8_t, uint8_t, char *);
00065     void eaDogM_WriteIntAtPos(uint8_t, uint8_t, uint8_t);
00066     void eaDogM_WriteByteToCGRAM(uint8_t, uint8_t);
00067     void SPI1_Initialize_edogm(void);
00068     void SPI1_Initialize_edogs(void);
00069     void SPI1_ExchangeBuffer(uint8_t *, uint16_t);
00070
00071 #define eaDogM_Cls()          eaDogM_WriteCommand(EADOGM_CMD_CLR)
00072 #define eaDogM_CursorOn()    eaDogM_WriteCommand(EADOGM_CMD_CURSOR_ON)
00073 #define eaDogM_CursorOff()   eaDogM_WriteCommand(EADOGM_CMD_CURSOR_OFF)
00074 #define eaDogM_DisplayOn()   eaDogM_WriteCommand(EADOGM_CMD_DISPLAY_ON)
00075 #define eaDogM_DisplayOff()  eaDogM_WriteCommand(EADOGM_CMD_DISPLAY_OFF)
00076
00077 #ifdef __cplusplus
00078 }
00079 #endif
00080
00081 #endif /* EADOG_H */
00082

```

4.4 lcd_drv.h

```

00001 /*
00002  * File:   lcd_drv.h
00003  * Author: root
00004  *
00005  * Created on October 12, 2021, 7:01 PM
00006  */
00007
00008 #ifndef LCD_DRV_H
00009 #define LCD_DRV_H
00010
00011 #ifdef __cplusplus
00012 extern "C" {
00013 #endif
00014
00015 #define DMA_MAGIC    1957
00016 #define USE_DMA // use DMA spi driver
00017
00018 #ifdef USE_DMA
00019 #define DMA_GAP      1 // set to 0 for SPI byte gaps in DMA transmissions
00020 #define DMA_STATE_M
00021 #endif
00022
00023 #define EDOGS
00024
00025 /*
00026  * DMA driver for DOGXL240 @15MHz SPI clock
00027  * 2021 XC32 and H3
00028  */
00029 #include <stdint.h>
00030 #include <stdbool.h>
00031 #include <stdio.h>
00032 #include "definitions.h" // SYS function prototypes
00033 #ifdef __32MK0512MCJ048__X
00034 #include "../src/config/mcj/peripheral/spi/spi_master/plib_spi_master_common.h"
00035 #include "../src/config/mcj/peripheral/gpio/plib_gpio.h"
00036 #include "../src/config/mcj/peripheral/dmac/plib_dmac.h"
00037 #include "../src/config/mcj/peripheral/coretimer/plib_coretimer.h"
00038 #endif
00039 #ifdef __32MZ1025W104132__X
00040 #include "../src/config/default/peripheral/spi/spi_master/plib_spi_master_common.h"
00041 #include "../src/config/default/peripheral/gpio/plib_gpio.h"
00042 #include "../src/config/default/peripheral/dmac/plib_dmac.h"
00043 #include "../src/config/default/peripheral/coretimer/plib_coretimer.h"
00044 #endif
00045 #include "display_type.h"
00046 #include "eadog.h"
00047 #include "dogm-graphic.h"
00048 #include "OledDriver.h"
00049 #include "OledChar.h"
00050 #include "OledGrph.h"
00051
00052 #include "device.h"
00053

```

```

00054 #define BMP_DELAY    1000000 // image display delay counts
00055 #define bmp_x          60      // screen positions of image
00056 #define bmp_y          24
00057 #define bmp_size       100     // x,y size of image
00058 #define BANK1          0xA000A000 // bank 1 frame buffer memory address
00059 #define BANK2          0x80030000 // bank 2 frame buffer memory address
00060
00061 #define LCD_DRIVER      "V1.003"
00062 #define LCD_ALIAS      "EADOG LCD"
00063
00064     typedef enum {
00065         D_INIT,
00066         D_MISC,
00067         D_BLANK,
00068     } LCD_DVR_STATE;
00069
00070 #define lcd_frame      0      // frame buffer 1 or 0
00071
00072     void init_lcd_drv(LCD_DVR_STATE init_type);
00073     void lcd_version(void);
00074
00075     extern const uint8_t foo_map[]; // image in flash array
00076
00077 #ifdef __cplusplus
00078 }
00079 #endif
00080
00081 #endif /* LCD_DRV_H */
00082

```

4.5 OledChar.h

```

00001 /*****
00002  */
00003  /* OledChar.h -- Interface Declarations for OledChar.c
00004  */
00005  *****/
00006  /* Author: Gene Apperson
00007  /* Copyright 2011, Digilent Inc.
00008  *****/
00009  /*
00010     This library is free software; you can redistribute it and/or
00011     modify it under the terms of the GNU Lesser General Public
00012     License as published by the Free Software Foundation; either
00013     version 2.1 of the License, or (at your option) any later version.
00014
00015     This library is distributed in the hope that it will be useful,
00016     but WITHOUT ANY WARRANTY; without even the implied warranty of
00017     MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the GNU
00018     Lesser General Public License for more details.
00019
00020     You should have received a copy of the GNU Lesser General Public
00021     License along with this library; if not, write to the Free Software
00022     Foundation, Inc., 51 Franklin St, Fifth Floor, Boston, MA 02110-1301 USA
00023  */
00024  *****/
00025  /* File Description:
00026  */
00027  /* Interface to OledChar.c
00028  */
00029  *****/
00030  /* Revision History:
00031  */
00032  /* 06/01/2011(GeneA): created
00033  */
00034  *****/
00035
00036 #if !defined(OLEDCHAR_H)
00037 #define OLEDCHAR_H
00038
00039 #include "lcd_drv.h"
00040
00041 extern int32_t xcoOledCur;
00042 extern int32_t ycoOledCur;
00043
00044 extern uint8_t * pbOledCur;
00045 extern uint8_t mskOledCur;
00046 extern int32_t bnOledCur;
00047 extern int32_t fOledCharUpdate;
00048
00049 extern uint8_t rgbOledBmp[cbOledDispMax];
00050
00051 extern int32_t dxcoOledFontCur;

```



```

00052 extern int32_t dycoOledFontCur;
00053
00054 extern uint8_t * pbOledFontCur;
00055 extern uint8_t * pbOledFontUser;
00056
00057 void OledSetCursor(int32_t xch, int32_t ych);
00058 void OledGetCursor(int32_t * pxcy, int32_t * pych);
00059 int32_t OledDefUserChar(char ch, uint8_t * pbDef);
00060 void OledSetCharUpdate(int32_t f);
00061 int32_t OledGetCharUpdate();
00062 void OledPutChar(char ch);
00063 void OledPutString(char * sz);
00064
00065 /* ----- */
00066
00067 #endif
00068
00069 /*****/

```

4.6 OledDriver.h

```

00001 /*****/
00002 /* */
00003 /* OledDriver.h -- Interface Declarations for OLED Display Driver */
00004 /* */
00005 /*****/
00006 /* Author: Gene Apperson */
00007 /* Copyright 2011, Digilent Inc. */
00008 /*****/
00009 /*
00010 This library is free software; you can redistribute it and/or
00011 modify it under the terms of the GNU Lesser General Public
00012 License as published by the Free Software Foundation; either
00013 version 2.1 of the License, or (at your option) any later version.
00014
00015 This library is distributed in the hope that it will be useful,
00016 but WITHOUT ANY WARRANTY; without even the implied warranty of
00017 MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the GNU
00018 Lesser General Public License for more details.
00019
00020 You should have received a copy of the GNU Lesser General Public
00021 License along with this library; if not, write to the Free Software
00022 Foundation, Inc., 51 Franklin St, Fifth Floor, Boston, MA 02110-1301 USA
00023 */
00024 /*****/
00025 /* File Description: */
00026 /* */
00027 /* Interface to OledDriver.c */
00028 /* */
00029 /*****/
00030 /* Revision History: */
00031 /* */
00032 /* 04/29/2011(GeneA): created */
00033 /* */
00034 /*****/
00035
00036 /*
00037 * machine dependant hardware routines should be here
00038 */
00039
00040
00041 #if !defined(OLEDDRIVER_INC)
00042 #define OLEDDRIVER_INC
00043
00044 #include "lcd_drv.h"
00045
00046 extern int32_t xchOledMax;
00047 extern int32_t ychOledMax;
00048
00049 /* ----- */
00050 /* Miscellaneous Declarations */
00051 /* ----- */
00052 #if DISPLAY_TYPE == 240 // Autor: Robert Steigemann
00053 #define cbOledDispMax 3840 //max number of bytes in display buffer
00054 #define cbOledDispMax32 cbOledDispMax/4 //max number of 32-bit words in display buffer
00055 #define ccolOledMax 240 //number of display columns
00056 #define crowOledMax 128 //number of display rows
00057 #define cpagOledMax 16 //number of display memory pages
00058 #define STR_BUF_SIZE 160 //number of chars for display strings
00059 #endif
00060
00061 #if DISPLAY_TYPE == 102
00062 #define cbOledDispMax 816 //max number of bytes in display buffer

```



```

00063 #define ccolOledMax      102      //number of display columns
00064 #define crowOledMax      64      //number of display rows
00065 #define cpagOledMax      8        //number of display memory pages
00066 #define STR_BUF_SIZE     20        //number of chars for display strings
00067 #endif
00068
00069 #define max_strlen  STR_BUF_SIZE-1
00070
00071 #define cbOledChar      8          //font glyph definitions is 8 bytes long
00072 #define chOledUserMax   0x20      //number of character defs in user font table
00073 #define cbOledFontUser  (chOledUserMax*cbOledChar)
00074
00075 /* Graphics drawing modes.
00076 */
00077 #define modOledSet      0
00078 #define modOledOr       1
00079 #define modOledAnd      2
00080 #define modOledXor      3
00081
00082 /* ----- */
00083 /* General Type Declarations */
00084 /* ----- */
00085
00086 /* Pin definitions for access to OLED control signals on ChipKitUno
00087 */
00088
00089
00090 /* ----- */
00091 /* Object Class Declarations */
00092 /* ----- */
00093
00094
00095
00096 /* ----- */
00097 /* Variable Declarations */
00098 /* ----- */
00099
00100
00101
00102 /* ----- */
00103 /* Procedure Declarations */
00104 /* ----- */
00105
00106
00107 typedef enum {
00108     D_init,
00109     D_page,
00110     D_buffer,
00111     D_idle,
00112 } DMA_RUN_STATE;
00113
00114 void OledInit(void);
00115 void OledTerm(void);
00116 void OledDisplayOn(void);
00117 void OledDisplayOff(void);
00118 void OledClear(void);
00119 void OledClearBuffer(void);
00120 void OledUpdate(void);
00121
00122 void wait_lcd_done(void);
00123
00124 /* ----- */
00125
00126 #endif
00127
00128 /*****

```

4.7 OledGrph.h

```

00001 /*****
00002 */
00003 /* OledGrph.h -- Declarations for OLED Graphics Routines */
00004 /* */
00005 /*****
00006 /* Author:      Gene Apperson */
00007 /* Copyright 2011, Digilent Inc. */
00008 /*****
00009 */
00010 This library is free software; you can redistribute it and/or
00011 modify it under the terms of the GNU Lesser General Public
00012 License as published by the Free Software Foundation; either
00013 version 2.1 of the License, or (at your option) any later version.
00014

```

```

00015  This library is distributed in the hope that it will be useful,
00016  but WITHOUT ANY WARRANTY; without even the implied warranty of
00017  MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the GNU
00018  Lesser General Public License for more details.
00019
00020  You should have received a copy of the GNU Lesser General Public
00021  License along with this library; if not, write to the Free Software
00022  Foundation, Inc., 51 Franklin St, Fifth Floor, Boston, MA 02110-1301 USA
00023  */
00024  /*****
00025  /* File Description:
00026  /*
00027  /* Interface declarations for OledGrph.c
00028  /*
00029  /*****
00030  /* Revision History:
00031  /*
00032  /* 06/03/2011(GeneA): created
00033  /*
00034  /*****
00035
00036  #if !defined(OLEDGRPH_H)
00037  #define OLEDGRPH_H
00038
00039  #include "lcd_drv.h"
00040
00041  extern int32_t xcoOledCur;
00042  extern int32_t ycoOledCur;
00043  extern uint8_t * pbOledCur;
00044  extern uint8_t rgbOledBmp0[];
00045  extern uint8_t rgbOledBmp1[];
00046  extern const uint8_t rgbFillPat[];
00047  extern int32_t bnOledCur;
00048  extern uint8_t clrOledCur;
00049  extern uint8_t * pbOledPatCur;
00050  extern uint8_t * pbOledFontUser;
00051  extern uint8_t * pbOledFontCur;
00052  extern int32_t dxcoOledFontCur;
00053  extern int32_t dycoOledFontCur;
00054
00055  extern volatile uint8_t disp_frame;
00056
00057  void OledSetDrawColor(uint8_t clr);
00058  void OledSetDrawMode(int32_t mod);
00059  int32_t OledGetDrawMode();
00060  uint8_t * OledGetStdPattern(int32_t ipat);
00061  void OledSetFillPattern(uint8_t * pbPat);
00062
00063  void OledMoveTo(int32_t xco, int32_t yco);
00064  void OledGetPos(int32_t * pxco, int32_t * pyco);
00065  void OledDrawPixel();
00066  uint8_t OledGetPixel();
00067  void OledLineTo(int32_t xco, int32_t yco);
00068  void OledDrawRect(int32_t xco, int32_t yco);
00069  void OledFillRect(int32_t xco, int32_t yco);
00070  void OledGetBmp(int32_t dxco, int32_t dyco, uint8_t * pbBmp);
00071  void OledPutBmp(int32_t dxco, int32_t dyco, uint8_t * pbBmp);
00072  void OledDrawChar(char ch);
00073  void OledDrawString(char * sz);
00074
00075  /* ----- */
00076
00077  #endif
00078
00079  /*****

```

4.8 bma490l.h

```

00001  /*
00002  * File: bma490.h
00003  * Author: nsasp
00004  *
00005  * Created on July 29, 2022, 12:37 PM
00006  * SPI mode 0, 8MHz, 8-bit transfers
00007  * 100Hz sensor sample rate
00008  */
00009
00010  #ifndef BMA490L_H
00011  #define BMA490L_H
00012
00013  #ifdef __cplusplus
00014  extern "C" {
00015  #endif

```

```

00016
00017 #include <stddef.h>                // Defines NULL
00018 #include <stdbool.h>                // Defines true
00019 #include <stdlib.h>                 // Defines EXIT_FAILURE
00020 #include <math.h>
00021 #include "definitions.h"           // SYS function prototypes
00022 #include "imupic32mcj.h"
00023 #include "bma490l_reg.h"
00024 #include "imu.h"
00025
00026 #define BMA490_DRIVER "V1.100"
00027 #define BMA490_ALIAS "BMA490L"
00028 #define BMA400_DRIVER "V1.604"
00029 #define BMA400_ALIAS "BMA400"
00030
00031 #define RBIT 0x80
00032 #define WBIT 0x7f
00033 #define CHIP_ID 0
00034 #define CHIP_STATUS 3
00035 #define CHIP_IS 0x2A
00036 #define CHIP_ID_INDEX 1
00037 #define CHIP_ID_DATA 2
00038 #define BMA490L_ID 0x1A
00039 #define BMA400_ID 0x90
00040 #define CHIP_ID_DELAY 100000
00041
00042 #define BMA490_ID_LEN 2
00043 #define BMA490_REG_LEN 2
00044 #define BMA490_DATA_LEN 12
00045 #define BMA400_DATA_LEN 12
00046 #define BMA490_DATA_BUFFER_LEN 64
00047 #define BMA400_DATA_BUFFER_LEN 64
00048 #define BMA490_DATA_INDEX 0x12
00049 #define BMA400_DATA_INDEX 0x04
00050
00051 #define BMA490_RD_WR_MAX_LEN 0x0514 // IMU ASIC firmware file size
00052
00053 #define ACCEL_CONFIG 0xa9 // 10101001 200Hz ODR 80Hz filter
00054 // #define ACCEL_CONFIG 0xa5 // 10100101 12.5Hz ODR 5.06Hz filter
00055 // #define ACCEL_CONFIG 0x55 // 01010101 12.5Hz ODR avg 32
00056 #define INT_MAP_DATA 0x04
00057 #define INT1_IO_CTRL 0x08
00058 #define REG_POWER_CTRL 0x04
00059
00061 #define BMA490L_RESERVED_REG_5B_ADDR UINT8_C(0x5B)
00062 #define BMA490L_RESERVED_REG_5C_ADDR UINT8_C(0x5C)
00063 #define BMA490L_FEATURE_CONFIG_ADDR UINT8_C(0x5E)
00064 #define BMA490L_INTERNAL_ERROR UINT8_C(0x5F)
00065
00066 #define imu_timeout 2000 // timeout for IMU ID data from query
00067
00068 /*
00069  * function pointer templates
00070  */
00071 void bma490l_set_spimode(void *);
00072 bool bma490l_getid(void *);
00073 bool bma490l_getdata(void *);
00074 void bma490_version(void);
00075
00076 bool imu_getis(imu_cmd_t *);
00077
00078 /*
00079  * BMA490L chip instance
00080  */
00081 extern imu_cmd_t imu0;
00082
00083 #ifdef __cplusplus
00084 }
00085 #endif
00086
00087 #endif /* BMA490L_H */
00088

```

4.9 bma490l_reg.h

```

00001 /*
00002  * File: bma490l_reg.h
00003  * Author: nsasp
00004  *
00005  * Created on August 4, 2022, 11:18 AM
00006  */
00007
00008 #ifndef BMA490L_REG_H

```

```

00009 #define BMA490L_REG_H
00010
00011 #ifdef __cplusplus
00012 extern "C" {
00013 #endif
00014
00015 #define BMA490L_REG_CHIP_ID 0x00
00016 #define BMA490L_REG_ERROR 0x02
00017 #define BMA490L_REG_STATUS 0x03
00018 #define BMA490L_REG_DATA_0 0x0A
00019 #define BMA490L_REG_DATA_8 0x12
00020 #define BMA490L_REG_SENSORTIME_0 0x18
00021 #define BMA490L_REG_EVENT 0x1B
00022 #define BMA490L_REG_INT_STAT_0 0x1C
00023 #define BMA490L_REG_INT_STAT_1 0x1D
00024 #define BMA490L_REG_TEMPERATURE 0x22
00025 #define BMA490L_REG_FIFO_LENGTH_0 0x24
00026 #define BMA490L_REG_FIFO_DATA 0x26
00027 #define BMA490L_REG_INTERNAL_STAT 0x2A
00028 #define BMA490L_REG_ACCEL_CONFIG 0x40
00029 #define BMA490L_REG_ACCEL_RANGE 0x41
00030 #define BMA490L_REG_AUX_CONF 0x44
00031 #define BMA490L_REG_FIFO_DOWN 0x45
00032 #define BMA490L_REG_FIFO_WTM_0 0x46
00033 #define BMA490L_REG_FIFO_CONFIG_0 0x48
00034 #define BMA490L_REG_FIFO_CONFIG_1 0x49
00035 #define BMA490L_REG_AUX_DEV_ID 0x4B
00036 #define BMA490L_REG_AUX_IF_CONF 0x4C
00037 #define BMA490L_REG_AUX_RD 0x4D
00038 #define BMA490L_REG_INT1_IO_CTRL 0x53
00039 #define BMA490L_REG_INT2_IO_CTRL 0x54
00040 #define BMA490L_REG_INT_LATCH 0x55
00041 #define BMA490L_REG_INT_MAP_1 0x56
00042 #define BMA490L_REG_INT_MAP_2 0x57
00043 #define BMA490L_REG_INT_MAP_DATA 0x58
00044 #define BMA490L_REG_INIT_CTRL 0x59
00045 #define BMA490L_REG_RESERVED_REG_5B 0x5B
00046 #define BMA490L_REG_RESERVED_REG_5C 0x5C
00047 #define BMA490L_REG_FEATURE_CONFIG 0x5E
00048 #define BMA490L_REG_INTERNAL_ERROR 0x5F
00049 #define BMA490L_REG_ACC_SELF_TEST 0x6D
00050 #define BMA490L_REG_ACC_SELF_TEST 0x6D
00051 #define BMA490L_REG_NV_CONFIG 0x70
00052 #define BMA490L_REG_OFFSET_0 0x71
00053 #define BMA490L_REG_OFFSET_1 0x72
00054 #define BMA490L_REG_OFFSET_2 0x73
00055 #define BMA490L_REG_POWER_CONF 0x7C
00056 #define BMA490L_REG_POWER_CTRL 0x7D
00057 #define BMA490L_REG_CMD 0x7E
00058
00059 #define BMA490L_INIT_START 0x00
00060 #define BMA490L_INIT_STOP 0x01
00061
00062 #define BMA490L_SOFT_RESET 0xB6
00063
00064 #define BMA490L_APS_OFF 0x00
00065 #define BMA490L_APS_ON 0x01
00066 #define BMA490L_FSW_OFF 0x00
00067 #define BMA490L_FSW_ON 0x02
00068
00069 #define BMA490L_NV_ENABLE_I2C 0x00
00070 #define BMA490L_NV_DISABLE_I2C 0x01
00071 #define BMA490L_NV_ACCEL_OFFSET_POS 3
00072 #define BMA490L_NV_ACCEL_OFFSET_MSK 0x08
00073
00074 /* BMA400 Register Address */
00075 #define BMA400_REG_CHIP_ID UINT8_C(0x00)
00076 #define BMA400_REG_STATUS UINT8_C(0x03)
00077 #define BMA400_REG_ACCEL_DATA UINT8_C(0x04)
00078 #define BMA400_REG_INT_STAT0 UINT8_C(0x0E)
00079 #define BMA400_REG_TEMP_DATA UINT8_C(0x11)
00080 #define BMA400_REG_FIFO_LENGTH UINT8_C(0x12)
00081 #define BMA400_REG_FIFO_DATA UINT8_C(0x14)
00082 #define BMA400_REG_STEP_CNT_0 UINT8_C(0x15)
00083 #define BMA400_REG_ACCEL_CONFIG_0 UINT8_C(0x19)
00084 #define BMA400_REG_ACCEL_CONFIG_1 UINT8_C(0x1A)
00085 #define BMA400_REG_ACCEL_CONFIG_2 UINT8_C(0x1B)
00086 #define BMA400_REG_INT_CONF_0 UINT8_C(0x1F)
00087 #define BMA400_REG_INT_12_IO_CTRL UINT8_C(0x24)
00088 #define BMA400_REG_INT_MAP UINT8_C(0x21)
00089 #define BMA400_REG_FIFO_CONFIG_0 UINT8_C(0x26)
00090 #define BMA400_REG_FIFO_READ_EN UINT8_C(0x29)
00091 #define BMA400_REG_AUTO_LOW_POW_0 UINT8_C(0x2A)
00092 #define BMA400_REG_AUTO_LOW_POW_1 UINT8_C(0x2B)
00093 #define BMA400_REG_AUTOWAKEUP_0 UINT8_C(0x2C)
00094 #define BMA400_REG_AUTOWAKEUP_1 UINT8_C(0x2D)
00095 #define BMA400_REG_WAKEUP_INT_CONF_0 UINT8_C(0x2F)

```

```

00096 #define BMA400_REG_ORIENTCH_INT_CONFIG      UINT8_C(0x35)
00097 #define BMA400_REG_GEN1_INT_CONFIG            UINT8_C(0x3F)
00098 #define BMA400_REG_GEN2_INT_CONFIG            UINT8_C(0x4A)
00099 #define BMA400_REG_ACT_CH_CONFIG_0            UINT8_C(0x55)
00100 #define BMA400_REG_TAP_CONFIG                 UINT8_C(0x57)
00101 #define BMA400_REG_SELF_TEST                  UINT8_C(0x7D)
00102 #define BMA400_REG_COMMAND                    UINT8_C(0x7E)
00103
00104 #define BMA400_NORM_MODE                       0x02
00105 #define BMA400_RANGE_MODE                     0x38
00106 #define BMA400_FILTER_MODE                    0x04
00107 #define BMA400_INT_CONF                       0x80
00108 #define BMA400_INT_MAP                        0x80
00109 #define BMA400_CMD_DUMMY                      0x7F
00110
00111 #ifdef __cplusplus
00112 }
00113 #endif
00114
00115 #endif /* BMA490L_REG_H */
00116

```

4.10 bno086.h

```

00001 /*
00002  * File:   bno086.h
00003  * Author: root
00004  *
00005  * Created on April 20, 2023, 11:51 AM
00006  * code model used and copied https://os.mbed.com/users/MultipleMonomials/code/BN0080/
00007  * https://github.com/ceva-dsp/sh2
00008  *
00009  */
00010
00011 #ifndef BNO086_H
00012 #define BNO086_H
00013
00014 #ifdef __cplusplus
00015 extern "C" {
00016 #endif
00017
00018 #include <stddef.h>           // Defines NULL
00019 #include <stdbool.h>          // Defines true
00020 #include <stdlib.h>           // Defines EXIT_FAILURE
00021 #include <ctype.h>
00022 #include <stdio.h>
00023 #include <math.h>
00024 #include <stdarg.h>
00025 #include "definitions.h"     // SYS function prototypes
00026 #include "imupic32mcj.h"
00027 #include "bno086_reg.h"
00028 #include "imu.h"
00029
00030 #define BNO086_DRIVER        "V1.015"
00031 #define BNO086_ALIAS         "BNO086"
00032
00033 #define UPDATE_MS_T 20
00034 #define UPDATE_MS_R 20
00035 #define UPDATE_MS_L 20
00036 #define UPDATE_MS_ENV 1000
00037 #define UPDATE_MS_CAL 500
00038 #define UPDATE_MS_MISC 100
00039 #define UPDATE_MS_PROX 333
00040
00041 /*
00042  * fake IMU for code testing
00043  */
00044 //#define FAKE_IMU
00045
00046 /*
00047  * Use terminal program TIO for 460800 speed serial debugging
00048  * https://github.com/tio/tio
00049  */
00050 //#define PDEBUG
00051 int printf_stub(const char*, ...);
00052
00053 #ifdef PDEBUG
00054 #define dprintf printf
00055 #else
00056 #define dprintf printf_stub
00057 #endif
00058
00059 #define SQRT_2 1.414213562f

```

```

00060 #define SQT_2_DIV_2    SQT_2/1.0f
00061
00062 #define BNO_INIT_DELAY_INT  400
00063
00064 #define SHTP_HEADER_SIZE 4
00065
00066     // Size of the largest individual packet we can receive.
00067     // Min value is set by the advertisement packet (272 bytes)
00068     // If you enable lots of sensor reports and get an error, you might need to increase this.
00069 #define SHTP_RX_PACKET_SIZE 274
00070
00071     // Size of largest packet that we need to transmit (not including header)
00072 #define SHTP_MAX_TX_PACKET_SIZE 17
00073
00074
00075     // Channels
00076 #define CHANNEL_COMMAND 0
00077 #define CHANNEL_EXECUTABLE 1
00078 #define CHANNEL_CONTROL 2
00079 #define CHANNEL_REPORTS 3
00080 #define CHANNEL_WAKE_REPORTS 4
00081 #define CHANNEL_GYRO 5
00082
00083     // Report IDs on the command channel.
00084     // Unlike the other constants, these come from the Sensor Hub Transport Protocol datasheet,
00085     section 5.1
00086 #define COMMAND_REPORTID_ADVERTISEMENT 0x0
00087 #define COMMAND_REPORTID_ERRORLIST 0x1
00088
00089     //All the ways we can configure or talk to the BNO080, figure 34, page 36 reference manual
00090     //These are used for low level communication with the sensor, on channel 2
00091 #define SHTP_REPORT_COMMAND_RESPONSE 0xF1
00092 #define SHTP_REPORT_COMMAND_REQUEST 0xF2
00093 #define SHTP_REPORT_FRS_READ_RESPONSE 0xF3
00094 #define SHTP_REPORT_FRS_READ_REQUEST 0xF4
00095 #define SHTP_REPORT_FRS_WRITE_RESPONSE 0xF5
00096 #define SHTP_REPORT_FRS_WRITE_DATA 0xF6
00097 #define SHTP_REPORT_FRS_WRITE_REQUEST 0xF7
00098 #define SHTP_REPORT_PRODUCT_ID_RESPONSE 0xF8
00099 #define SHTP_REPORT_PRODUCT_ID_REQUEST 0xF9
00100 #define SHTP_REPORT_BASE_TIMESTAMP 0xFB
00101 #define SHTP_REPORT_SET_FEATURE_COMMAND 0xFD
00102 #define SHTP_REPORT_GET_FEATURE_RESPONSE 0xFC
00103
00104     //All the different sensors and features we can get reports from
00105     //These are used when enabling a given sensor
00106 #define SENSOR_REPORTID_TIMESTAMP_REBASE 0xFA
00107 #define SENSOR_REPORTID_ACCELEROMETER 0x01
00108 #define SENSOR_REPORTID_GYROSCOPE_CALIBRATED 0x02
00109 #define SENSOR_REPORTID_MAGNETIC_FIELD_CALIBRATED 0x03
00110 #define SENSOR_REPORTID_LINEAR_ACCELERATION 0x04
00111 #define SENSOR_REPORTID_ROTATION_VECTOR 0x05
00112 #define SENSOR_REPORTID_GRAVITY 0x06
00113 #define SENSOR_REPORTID_GAME_ROTATION_VECTOR 0x08
00114 #define SENSOR_REPORTID_GEOMAGNETIC_ROTATION_VECTOR 0x09
00115 #define SENSOR_REPORTID_PRESSURE_DETECTOR 0x0A
00116 #define SENSOR_REPORTID_AMBIENT_DETECTOR 0x0B
00117 #define SENSOR_REPORTID_HUMIDITY_DETECTOR 0x0C
00118 #define SENSOR_REPORTID_PROXIMITY_DETECTOR 0x0D
00119 #define SENSOR_REPORTID_TEMPERATURE_DETECTOR 0x0E
00120 #define SENSOR_REPORTID_MAGNETIC_FIELD_UNCALIBRATED 0x0F
00121 #define SENSOR_REPORTID_TAP_DETECTOR 0x10
00122 #define SENSOR_REPORTID_STEP_COUNTER 0x11
00123 #define SENSOR_REPORTID_SIGNIFICANT_MOTION 0x12
00124 #define SENSOR_REPORTID_STABILITY_CLASSIFIER 0x13
00125 #define SENSOR_REPORTID_STEP_DETECTOR 0x18
00126 #define SENSOR_REPORTID_SHAKE_DETECTOR 0x19
00127 #define SENSOR_REPORTID_X20 0x20
00128 #define SENSOR_REPORTID_X21 0x21
00129 #define SENSOR_REPORTID_CIRCLE_DETECTOR 0x22
00130
00131     // sensor report ID with the largest numeric value
00132 #define MAX_SENSOR_REPORTID SENSOR_REPORTID_CIRCLE_DETECTOR
00133
00134     // Q points for various sensor data elements
00135 #define ACCELEROMETER_Q_POINT 8 // for accelerometer based data
00136 #define GYRO_Q_POINT 9 // for gyroscope data
00137 #define MAGNETOMETER_Q_POINT 4 // for magnetometer data
00138 #define ROTATION_Q_POINT 14 // for rotation data
00139 #define ROTATION_ACCURACY_Q_POINT 12 // for rotation accuracy data
00140 #define POWER_Q_POINT 10 // for power information in the metadata
00141 #define ORIENTATION_QUAT_Q_POINT 14 // for the set orientation command
00142 #define FRS_ORIENTATION_Q_POINT 30 // for the sensor orientation FRS record
00143
00144     // Report IDs on the Executable channel
00145     // See Figure 1-27 in the BNO080 datasheet
00146 #define EXECUTABLE_REPORTID_RESET 0x1

```

```

00146
00147 //Record IDs from SH-2 figure 28
00148 //These are used to read and set various configuration options
00149 #define FRS_RECORDID_SERIAL_NUMBER 0x4B4B
00150 #define FRS_RECORDID_SYSTEM_ORIENTATION 0x2D3E
00151
00152 //Command IDs from section 6.4, page 42
00153 //These are used to calibrate, initialize, set orientation, tare etc the sensor
00154 #define COMMAND_ERRORS 1
00155 #define COMMAND_COUNTER 2
00156 #define COMMAND_TARE 3
00157 #define COMMAND_INITIALIZE 4
00158 #define COMMAND_SAVE_DCD 6
00159 #define COMMAND_ME_CALIBRATE 7
00160 #define COMMAND_DCD_PERIOD_SAVE 9
00161 #define COMMAND_OSCILLATOR 10
00162 #define COMMAND_CLEAR_DCD 11
00163 #define COMMAND_UNSOLICITED_INITIALIZE 0x84
00164
00165 #define CALIBRATE_ACCEL 0
00166 #define CALIBRATE_GYRO 1
00167 #define CALIBRATE_MAG 2
00168 #define CALIBRATE_PLANAR_ACCEL 3
00169 #define CALIBRATE_ACCEL_GYRO_MAG 4
00170 #define CALIBRATE_STOP 5
00171
00172 #define TARE_NOW 0
00173 #define TARE_PERSIST 1
00174 #define TARE_SET_REORIENTATION 2
00175
00176 #define TARE_AXIS_ALL 0x07
00177 #define TARE_AXIS_Z 0x04
00178
00179 #define QEI_TARE_ALL 20
00180 #define QEI_TARE_Z 10
00181
00182 #define BNO080_RESET_TIMEOUT 180ms
00183
00184
00185
00186
00187
00188 enum Report {
00189     TOTAL_ACCELERATION = SENSOR_REPORTID_ACCELEROMETER,
00190     LINEAR_ACCELERATION = SENSOR_REPORTID_LINEAR_ACCELERATION,
00191     GRAVITY_ACCELERATION = SENSOR_REPORTID_GRAVITY,
00192     GYROSCOPE = SENSOR_REPORTID_GYROSCOPE_CALIBRATED,
00193     MAG_FIELD = SENSOR_REPORTID_MAGNETIC_FIELD_CALIBRATED,
00194     MAG_FIELD_UNCALIBRATED = SENSOR_REPORTID_MAGNETIC_FIELD_UNCALIBRATED,
00195     ROTATION = SENSOR_REPORTID_ROTATION_VECTOR,
00196     GEOMAGNETIC_ROTATION = SENSOR_REPORTID_GEOMAGNETIC_ROTATION_VECTOR,
00197     GAME_ROTATION = SENSOR_REPORTID_GAME_ROTATION_VECTOR,
00198     TAP_DETECTOR = SENSOR_REPORTID_TAP_DETECTOR,
00199     STABILITY_CLASSIFIER = SENSOR_REPORTID_STABILITY_CLASSIFIER,
00200     STEP_DETECTOR = SENSOR_REPORTID_STEP_DETECTOR,
00201     STEP_COUNTER = SENSOR_REPORTID_STEP_COUNTER,
00202     SIGNIFICANT_MOTION = SENSOR_REPORTID_SIGNIFICANT_MOTION,
00203     SHAKE_DETECTOR = SENSOR_REPORTID_SHAKE_DETECTOR,
00204     X20 = SENSOR_REPORTID_X20,
00205     X21 = SENSOR_REPORTID_X21,
00206     CIRCLE_DETECTOR = SENSOR_REPORTID_CIRCLE_DETECTOR,
00207     AMBIENT_DETECTOR = SENSOR_REPORTID_AMBIENT_DETECTOR,
00208     PRESSURE_DETECTOR = SENSOR_REPORTID_PRESSURE_DETECTOR,
00209     HUMIDITY_DETECTOR = SENSOR_REPORTID_HUMIDITY_DETECTOR,
00210     PROXIMITY_DETECTOR = SENSOR_REPORTID_PROXIMITY_DETECTOR,
00211     TEMPERATURE_DETECTOR = SENSOR_REPORTID_TEMPERATURE_DETECTOR,
00212 };
00213
00214 enum Stability {
00215     UNKNOWN = 0,
00216
00217     ON_TABLE = 1,
00218
00219     STATIONARY = 2,
00220
00221     STABLE = 3,
00222
00223     MOTION = 4
00224 };
00225
00226 typedef struct _BnoData_t {
00227     sh2_Quaternion_t fusion_q;
00228     Quaternion rotationVector, totalAcceleration, linearAcceleration, gravityAcceleration,
00229     gyroRotation, magField,
00230     magFieldUncalibrated, hardIronOffset, gameRotationVector, geomagneticRotationVector;
00231
00232     bool tapDetected;
00233     bool doubleTap;
00234

```

```

00239
00240     enum Stability stability;
00241     bool stepDetected;
00242     uint16_t stepCount;
00243     bool significantMotionDetected;
00244     bool shakeDetected;
00245     bool circleDetected;
00246     bool xAxisShake;
00247     bool yAxisShake;
00248     bool zAxisShake;
00249
00250     uint8_t majorSoftwareVersion;
00251     uint8_t minorSoftwareVersion;
00252     uint16_t patchSoftwareVersion;
00253     uint32_t partNumber;
00254     uint32_t buildNumber;
00255     float rotationAccuracy, geomagneticRotationAccuracy;
00256     double ambient, pressure, humidity, proximity, temperature;
00257     uint8_t statusA, statusG, statusM;
00258 } _BnoData_t;
00259
00260 /*
00261  * function pointer templates
00262  */
00263 void bno086_set_spimode(void *);
00264 bool bno086_getid(void *);
00265 bool bno086_getdata(void *);
00266 void bno086_version(void);
00267 bool bno086_get_cpacket(size_t, void *);
00268 void bno086_get_header(void *);
00269 bool bno086_receive_packet(void *);
00270 bool bno086_hasNewData(const uint8_t);
00271 void bno086_flush(void);
00272 void processPacket(void);
00273 void parseSensorDataPacket(void);
00274 bool sendPacket(uint8_t, uint8_t, void *);
00275 bool waitForPacket(int, uint8_t, void *);
00276 void clearSendBuffer(void *);
00277 void sendCommand(uint8_t);
00278 void sendTareCommand(uint8_t, uint8_t, uint8_t);
00279
00280 double qToFloat(int16_t, uint8_t);
00281 double qToFloat_dword(uint32_t, uint16_t);
00282 uint16_t floatToQ(float, uint8_t);
00283 uint32_t floatToQ_dword(float t, uint16_t);
00284
00285 void setFeatureCommand(uint8_t, uint16_t, uint32_t);
00286 void enableReport(enum Report, uint16_t);
00287 bool bno086_updateData(void);
00288 void productIdReg(void);
00289 void softreset(void);
00290 void get_id_dummy(void);
00291
00292 bool enableCalibration(bool, bool, bool);
00293 bool saveCalibration(void);
00294 /*
00295  * BNO086 chip instance
00296  */
00297 extern imu_cmd_t imu0;
00298 extern char response_buffer[RBUFFER_SIZE];
00299 extern char cmd_buffer[RBUFFER_SIZE];
00300 extern _BnoData_t bno;
00301 extern const char *bno_Status;
00302
00303 #ifdef __cplusplus
00304 }
00305 #endif
00306
00307 #endif /* BNO086_H */
00308

```

4.11 bno086_reg.h

```

00001 /*
00002  * File:    bno086_reg.h
00003  * Author:  root
00004  *
00005  * Created on April 20, 2023, 11:55 AM
00006  */
00007
00008 #ifndef BNO086_REG_H
00009 #define BNO086_REG_H
00010

```



```

00011 #ifdef __cplusplus
00012 extern "C" {
00013 #endif
00014
00015 #define BNO08x_SHTP_HDR_SIZE      0x4
00016 #define BNO08x_SHTP_HDR_LEN_LSB  0x0
00017 #define BNO08x_SHTP_HDR_LEN_MSB  0x1
00018 #define BNO08x_SHTP_HDR_CHAN     0x2
00019 #define BNO08x_SHTP_HDR_SEQ      0x3
00020
00021 #define BNO08x_SHTP_MAX_SIZE      128
00022 #define BNO08x_SHTP_ADV_MAX_SIZE  512
00023
00024 #define BNO08x_SHTP_SOFT_RESET    0x01
00025 #define BNO08x_SHTP_PROD_ID      0xf8
00026 #define BNO08x_SHTP_PROD_ID_REQ  0xf9
00027
00028 #define BNO08x_SHTP_GET_FEATURE_RESP 0xfc
00029 #define BNO08x_SHTP_SET_FEATURE_CMD 0xfd
00030 #define BNO08x_SHTP_GET_FEATURE_REQ 0xfe
00031
00032 #define BNO08x_ROT_SCAN_INDEX      0x01
00033 #define BNO08x_REPORTID_ROTATION_VEC 0x05
00034 #define BNO08x_REPORTID_TIMESTAMP_BASE 0xfb
00035
00036 enum bno08x_shtp_channels {
00037     BNO08x_SHTP_COMMAND_CHAN,
00038     BNO08x_SHTP_EXECUTEABLE_CHAN,
00039     BNO08x_SHTP_CONTROL_CHAN,
00040     BNO08x_SHTP_REPORTS_CHAN,
00041     BNO08x_SHTP_WAKE_REPORTS_CHAN,
00042     BNO08x_SHTP_GYRO_CHAN,
00043     BNO08x_NUM_CHANNELS,
00044 };
00045
00046 #define SH2_MAX_SENSOR_EVENT_LEN (60)
00047 typedef struct sh2_SensorEvent {
00048     uint64_t timestamp_uS;
00049     int64_t delay_uS;
00050     uint8_t len;
00051     uint8_t reportId;
00052     uint8_t report[SH2_MAX_SENSOR_EVENT_LEN];
00053 } sh2_SensorEvent_t;
00054
00055 typedef void (sh2_SensorCallback_t)(void * cookie, sh2_SensorEvent_t *pEvent);
00056
00062 typedef struct sh2_ProductId_s {
00063     uint8_t resetCause;
00064     uint8_t swVersionMajor;
00065     uint8_t swVersionMinor;
00066     uint32_t swPartNumber;
00067     uint32_t swBuildNumber;
00068     uint16_t swVersionPatch;
00069     uint8_t reserved0;
00070     uint8_t reserved1;
00071 } sh2_ProductId_t;
00072
00073 #define SH2_MAX_PROD_ID_ENTRIES (5)
00074 typedef struct sh2_ProductIds_s {
00075     sh2_ProductId_t entry[SH2_MAX_PROD_ID_ENTRIES];
00076     uint8_t numEntries;
00077 } sh2_ProductIds_t;
00078
00084 enum sh2_SensorId_e {
00085     SH2_RAW_ACCELEROMETER = 0x14,
00086     SH2_ACCELEROMETER = 0x01,
00087     SH2_LINEAR_ACCELERATION = 0x04,
00088     SH2_GRAVITY = 0x06,
00089     SH2_RAW_GYROSCOPE = 0x15,
00090     SH2_GYROSCOPE_CALIBRATED = 0x02,
00091     SH2_GYROSCOPE_UNCALIBRATED = 0x07,
00092     SH2_RAW_MAGNETOMETER = 0x16,
00093     SH2_MAGNETIC_FIELD_CALIBRATED = 0x03,
00094     SH2_MAGNETIC_FIELD_UNCALIBRATED = 0x0f,
00095     SH2_ROTATION_VECTOR = 0x05,
00096     SH2_GAME_ROTATION_VECTOR = 0x08,
00097     SH2_GEOMAGNETIC_ROTATION_VECTOR = 0x09,
00098     SH2_PRESSURE = 0x0a,
00099     SH2_AMBIENT_LIGHT = 0x0b,
00100     SH2_HUMIDITY = 0x0c,
00101     SH2_PROXIMITY = 0x0d,
00102     SH2_TEMPERATURE = 0x0e,
00103     SH2_RESERVED = 0x17,
00104     SH2_TAP_DETECTOR = 0x10,
00105     SH2_STEP_DETECTOR = 0x18,
00106     SH2_STEP_COUNTER = 0x11,
00107     SH2_SIGNIFICANT_MOTION = 0x12,

```

```

00108     SH2_STABILITY_CLASSIFIER = 0x13,
00109     SH2_SHAKE_DETECTOR = 0x19,
00110     SH2_FLIP_DETECTOR = 0x1a,
00111     SH2_PICKUP_DETECTOR = 0x1b,
00112     SH2_STABILITY_DETECTOR = 0x1c,
00113     SH2_PERSONAL_ACTIVITY_CLASSIFIER = 0x1e,
00114     SH2_SLEEP_DETECTOR = 0x1f,
00115     SH2_TILT_DETECTOR = 0x20,
00116     SH2_POCKET_DETECTOR = 0x21,
00117     SH2_CIRCLE_DETECTOR = 0x22,
00118     SH2_HEART_RATE_MONITOR = 0x23,
00119     SH2_ARVR_STABILIZED_RV = 0x28,
00120     SH2_ARVR_STABILIZED_GRV = 0x29,
00121     SH2_GYRO_INTEGRATED_RV = 0x2A,
00122     SH2_I2RO_MOTION_REQUEST = 0x2B,
00123     SH2_RAW_OPTICAL_FLOW = 0x2C,
00124     SH2_DEAD_RECKONING_POSE = 0x2D,
00125     SH2_WHEEL_ENCODER = 0x2E,
00126
00127     // UPDATE to reflect greatest sensor id
00128     SH2_MAX_SENSOR_ID = 0x2E,
00129 };
00130 typedef uint8_t sh2_SensorId_t;
00131
00132 typedef struct sh2_SensorConfig {
00133     /* Change sensitivity enabled */
00134     bool changeSensitivityEnabled;
00135
00136     /* Change sensitivity - true if relative; false if absolute */
00137     bool changeSensitivityRelative;
00138
00139     /* Wake-up enabled */
00140     bool wakeupEnabled;
00141
00142     /* Always on enabled */
00143     bool alwaysOnEnabled;
00144
00145     bool sniffEnabled;
00146
00147     /* 16-bit signed fixed point integer representing the value a
00148      * sensor output must exceed in order to trigger another input
00149      * report. A setting of 0 causes all reports to be sent.
00150      */
00151     uint16_t changeSensitivity;
00152
00153     /* Interval in microseconds between asynchronous input reports. */
00154     uint32_t reportInterval_us;
00155
00156     /* Reserved field, not used. */
00157     uint32_t batchInterval_us;
00158
00159     /* Meaning is sensor specific */
00160     uint32_t sensorSpecific;
00161 } sh2_SensorConfig_t;
00162
00163 typedef struct sh2_SensorMetadata {
00164     uint8_t meVersion;
00165     uint8_t mhVersion;
00166     uint8_t shVersion;
00167     uint32_t range;
00168     uint32_t resolution;
00169     uint16_t revision;
00170     uint16_t power_mA;
00171     uint32_t minPeriod_us;
00172     uint32_t maxPeriod_us;
00173     uint32_t fifoReserved;
00174     uint32_t fifoMax;
00175     uint32_t batchSizeBytes;
00176     uint16_t qPoint1;
00177     uint16_t qPoint2;
00178     uint16_t qPoint3;
00179     uint32_t vendorIdLen;
00180     char vendorId[48];
00181     uint32_t sensorSpecificLen;
00182     uint8_t sensorSpecific[48];
00183 } sh2_SensorMetadata_t;
00184
00185 typedef struct sh2_ErrorRecord {
00186     uint8_t severity;
00187     uint8_t sequence;
00188     uint8_t source;
00189     uint8_t error;
00190     uint8_t module;
00191     uint8_t code;
00192 } sh2_ErrorRecord_t;
00193
00194 typedef struct sh2_Counts {

```

```

00218     uint32_t offered;
00219     uint32_t accepted;
00220     uint32_t on;
00221     uint32_t attempted;
00222 } sh2_Counts_t;
00223
00229 typedef enum sh2_TareBasis {
00230     SH2_TARE_BASIS_ROTATION_VECTOR = 0,
00231     SH2_TARE_BASIS_GAMING_ROTATION_VECTOR = 1,
00232     SH2_TARE_BASIS_GEOMAGNETIC_ROTATION_VECTOR = 2,
00233 } sh2_TareBasis_t;
00234
00240 typedef enum sh2_TareAxis {
00241     SH2_TARE_X = 1,
00242     SH2_TARE_Y = 2,
00243     SH2_TARE_Z = 4,
00244 } sh2_TareAxis_t;
00245
00251 typedef struct sh2_Quaternion {
00252     double x;
00253     double y;
00254     double z;
00255     double w;
00256 } sh2_Quaternion_t;
00257
00263 typedef enum {
00264     SH2_OSC_INTERNAL = 0,
00265     SH2_OSC_EXT_CRYSTAL = 1,
00266     SH2_OSC_EXT_CLOCK = 2,
00267 } sh2_OscType_t;
00268
00274 typedef enum {
00275     SH2_CAL_SUCCESS = 0,
00276     SH2_CAL_NO_ZRO,
00277     SH2_CAL_NO_STATIONARY_DETECTION,
00278     SH2_CAL_ROTATION_OUTSIDE_SPEC,
00279     SH2_CAL_ZRO_OUTSIDE_SPEC,
00280     SH2_CAL_ZGO_OUTSIDE_SPEC,
00281     SH2_CAL_GYRO_GAIN_OUTSIDE_SPEC,
00282     SH2_CAL_GYRO_PERIOD_OUTSIDE_SPEC,
00283     SH2_CAL_GYRO_DROPS_OUTSIDE_SPEC,
00284 } sh2_CalStatus_t;
00285
00286 // FRS Record Ids
00287 #define STATIC_CALIBRATION_AGM (0x7979)
00288 #define NOMINAL_CALIBRATION (0x4D4D)
00289 #define STATIC_CALIBRATION_SRA (0x8A8A)
00290 #define NOMINAL_CALIBRATION_SRA (0x4E4E)
00291 #define DYNAMIC_CALIBRATION (0x1F1F)
00292 #define ME_POWER_MGMT (0xD3E2)
00293 #define SYSTEM_ORIENTATION (0x2D3E)
00294 #define ACCEL_ORIENTATION (0x2D41)
00295 #define SCREEN_ACCEL_ORIENTATION (0x2D43)
00296 #define GYROSCOPE_ORIENTATION (0x2D46)
00297 #define MAGNETOMETER_ORIENTATION (0x2D4C)
00298 #define ARVR_STABILIZATION_RV (0x3E2D)
00299 #define ARVR_STABILIZATION_GRV (0x3E2E)
00300 #define TAP_DETECT_CONFIG (0xC269)
00301 #define SIG_MOTION_DETECT_CONFIG (0xC274)
00302 #define SHAKE_DETECT_CONFIG (0x7D7D)
00303 #define MAX_FUSION_PERIOD (0xD7D7)
00304 #define SERIAL_NUMBER (0x4B4B)
00305 #define ES_PRESSURE_CAL (0x39AF)
00306 #define ES_TEMPERATURE_CAL (0x4D20)
00307 #define ES_HUMIDITY_CAL (0x1AC9)
00308 #define ES_AMBIENT_LIGHT_CAL (0x39B1)
00309 #define ES_PROXIMITY_CAL (0x4DA2)
00310 #define ALS_CAL (0xD401)
00311 #define PROXIMITY_SENSOR_CAL (0xD402)
00312 #define PICKUP_DETECTOR_CONFIG (0x1B2A)
00313 #define FLIP_DETECTOR_CONFIG (0xFC94)
00314 #define STABILITY_DETECTOR_CONFIG (0xED85)
00315 #define ACTIVITY_TRACKER_CONFIG (0xED88)
00316 #define SLEEP_DETECTOR_CONFIG (0xED87)
00317 #define TILT_DETECTOR_CONFIG (0xED89)
00318 #define POCKET_DETECTOR_CONFIG (0xEF27)
00319 #define CIRCLE_DETECTOR_CONFIG (0xEE51)
00320 #define USER_RECORD (0x74B4)
00321 #define ME_TIME_SOURCE_SELECT (0xD403)
00322 #define UART_FORMAT (0xA1A1)
00323 #define GYRO_INTEGRATED_RV_CONFIG (0xA1A2)
00324 #define DR_IMU_CONFIG (0xDEDE2)
00325 #define DR_VEL_EST_CONFIG (0xDEDE3)
00326 #define DR_SYNC_CONFIG (0xDEDE4)
00327 #define DR_QUAL_CONFIG (0xDEDE5)
00328 #define DR_CAL_CONFIG (0xDEDE6)
00329 #define DR_LIGHT_REC_CONFIG (0xDEDE8)

```

```

00330 #define DR_FUSION_CONFIG (0xDEDE9)
00331 #define DR_OF_CONFIG (0xDEDA)
00332 #define DR_WHEEL_CONFIG (0xDEDB)
00333 #define DR_CAL (0xDEDC)
00334 #define DR_WHEEL_SELECT (0xDEDF)
00335 #define FRS_ID_META_RAW_ACCELEROMETER (0xE301)
00336 #define FRS_ID_META_ACCELEROMETER (0xE302)
00337 #define FRS_ID_META_LINEAR_ACCELERATION (0xE303)
00338 #define FRS_ID_META_GRAVITY (0xE304)
00339 #define FRS_ID_META_RAW_GYROSCOPE (0xE305)
00340 #define FRS_ID_META_GYROSCOPE_CALIBRATED (0xE306)
00341 #define FRS_ID_META_GYROSCOPE_UNCALIBRATED (0xE307)
00342 #define FRS_ID_META_RAW_MAGNETOMETER (0xE308)
00343 #define FRS_ID_META_MAGNETIC_FIELD_CALIBRATED (0xE309)
00344 #define FRS_ID_META_MAGNETIC_FIELD_UNCALIBRATED (0xE30A)
00345 #define FRS_ID_META_ROTATION_VECTOR (0xE30B)
00346 #define FRS_ID_META_GAME_ROTATION_VECTOR (0xE30C)
00347 #define FRS_ID_META_GEOMAGNETIC_ROTATION_VECTOR (0xE30D)
00348 #define FRS_ID_META_PRESSURE (0xE30E)
00349 #define FRS_ID_META_AMBIENT_LIGHT (0xE30F)
00350 #define FRS_ID_META_HUMIDITY (0xE310)
00351 #define FRS_ID_META_PROXIMITY (0xE311)
00352 #define FRS_ID_META_TEMPERATURE (0xE312)
00353 #define FRS_ID_META_TAP_DETECTOR (0xE313)
00354 #define FRS_ID_META_STEP_DETECTOR (0xE314)
00355 #define FRS_ID_META_STEP_COUNTER (0xE315)
00356 #define FRS_ID_META_SIGNIFICANT_MOTION (0xE316)
00357 #define FRS_ID_META_STABILITY_CLASSIFIER (0xE317)
00358 #define FRS_ID_META_SHAKE_DETECTOR (0xE318)
00359 #define FRS_ID_META_FLIP_DETECTOR (0xE319)
00360 #define FRS_ID_META_PICKUP_DETECTOR (0xE31A)
00361 #define FRS_ID_META_STABILITY_DETECTOR (0xE31B)
00362 #define FRS_ID_META_PERSONAL_ACTIVITY_CLASSIFIER (0xE31C)
00363 #define FRS_ID_META_SLEEP_DETECTOR (0xE31D)
00364 #define FRS_ID_META_TILT_DETECTOR (0xE31E)
00365 #define FRS_ID_META_POCKET_DETECTOR (0xE31F)
00366 #define FRS_ID_META_CIRCLE_DETECTOR (0xE320)
00367 #define FRS_ID_META_HEART_RATE_MONITOR (0xE321)
00368 #define FRS_ID_META_ARVR_STABILIZED_RV (0xE322)
00369 #define FRS_ID_META_ARVR_STABILIZED_GRV (0xE323)
00370 #define FRS_ID_META_GYRO_INTEGRATED_RV (0xE324)
00371 #define FRS_ID_META_RAW_OPTICAL_FLOW (0xE326)
00372
00378 typedef enum {
00379     SH2_IZRO_MI_UNKNOWN = 0,
00380     SH2_IZRO_MI_STATIONARY_NO_VIBRATION,
00381     SH2_IZRO_MI_STATIONARY_WITH_VIBRATION,
00382     SH2_IZRO_MI_IN_MOTION,
00383     SH2_IZRO_MI_ACCELERATING,
00384 } sh2_IzroMotionIntent_t;
00385
00391 typedef enum {
00392     SH2_IZRO_MR_NO_REQUEST = 0,
00393     SH2_IZRO_MR_STAY_STATIONARY,
00394     SH2_IZRO_MR_STATIONARY_NON_URGENT,
00395     SH2_IZRO_MR_STATIONARY_URGENT,
00396 } sh2_IzroMotionRequest_t;
00397
00398
00404
00405 enum sh2_AsyncEventId_e {
00406     SH2_RESET,
00407     SH2_SHTP_EVENT,
00408     SH2_GET_FEATURE_RESP,
00409 };
00410 typedef enum sh2_AsyncEventId_e sh2_AsyncEventId_t;
00411
00412 enum sh2_ShtpEvent_e {
00413     SH2_SHTP_TX_DISCARD = 0,
00414     SH2_SHTP_SHORT_FRAGMENT = 1,
00415     SH2_SHTP_TOO_LARGE_PAYLOADS = 2,
00416     SH2_SHTP_BAD_RX_CHAN = 3,
00417     SH2_SHTP_BAD_TX_CHAN = 4,
00418     SH2_SHTP_BAD_FRAGMENT = 5,
00419     SH2_SHTP_BAD_SN = 6,
00420     SH2_SHTP_INTERRUPTED_PAYLOAD = 7,
00421 };
00422 typedef uint8_t sh2_ShtpEvent_t;
00423
00424 typedef struct sh2_SensorConfigResp_e {
00425     sh2_SensorId_t sensorId;
00426     sh2_SensorConfig_t sensorConfig;
00427 } sh2_SensorConfigResp_t;
00428
00429 typedef struct sh2_AsyncEvent {
00430     uint32_t eventId;
00431     union {

```

```

00432         sh2_ShtpEvent_t shtpEvent;
00433         sh2_SensorConfigResp_t sh2SensorConfigResp;
00434     };
00435 } sh2_AsyncEvent_t;
00436
00437 typedef enum shtp_Event_e {
00438     SHTP_SHORT_FRAGMENT = 1,
00439     SHTP_TOO_LARGE_PAYLOADS = 2,
00440     SHTP_BAD_RX_CHAN = 3,
00441     SHTP_BAD_TX_CHAN = 4,
00442     SHTP_BAD_FRAGMENT = 5,
00443     SHTP_BAD_SN = 6,
00444     SHTP_INTERRUPTED_PAYLOAD = 7,
00445 } shtp_Event_t;
00446
00447 /* Note on quaternion naming conventions:
00448  * Quaternions are values with four real components that are usually
00449  * interpreted as coefficients in the complex quantity, Q.
00450  *
00451  * As in,  $Q = W + Xi + Yj + Zk$ 
00452  *
00453  * Where i, j and k represent the three imaginary dimensions.
00454  *
00455  * So W represents the Real components and X, Y and Z the Imaginary ones.
00456  *
00457  * In the Hillcrest datasheets and in this code, however, the four components
00458  * are named real, i, j and k, to make it explicit which is which. If you
00459  * need to translate these names into the "wxyz" or "xyzw" convention, then, the
00460  * appropriate mapping is this:
00461  *     w = real
00462  *     x = i
00463  *     y = j
00464  *     z = k
00465  */
00466
00472 typedef struct sh2_RawAccelerometer {
00473     /* Units are ADC counts */
00474     int16_t x;
00475     int16_t y;
00476     int16_t z;
00477
00478     /* Microseconds */
00479     uint32_t timestamp;
00480 } sh2_RawAccelerometer_t;
00481
00487 typedef struct sh2_Accelerometer {
00488     float x;
00489     float y;
00490     float z;
00491 } sh2_Accelerometer_t;
00492
00498 typedef struct sh2_RawGyroscope {
00499     /* Units are ADC counts */
00500     int16_t x;
00501     int16_t y;
00502     int16_t z;
00503     int16_t temperature;
00504
00505     /* Microseconds */
00506     uint32_t timestamp;
00507 } sh2_RawGyroscope_t;
00508
00514 typedef struct sh2_Gyroscope {
00515     /* Units are rad/s */
00516     float x;
00517     float y;
00518     float z;
00519 } sh2_Gyroscope_t;
00520
00526 typedef struct sh2_GyroscopeUncalibrated {
00527     /* Units are rad/s */
00528     float x;
00529     float y;
00530     float z;
00531     float biasX;
00532     float biasY;
00533     float biasZ;
00534 } sh2_GyroscopeUncalibrated_t;
00535
00541 typedef struct sh2_RawMagnetometer {
00542     /* Units are ADC counts */
00543     int16_t x;
00544     int16_t y;
00545     int16_t z;
00546
00547     /* Microseconds */
00548     uint32_t timestamp;

```

```

00549 } sh2_RawMagnetometer_t;
00550
00556 typedef struct sh2_MagneticField {
00557     /* Units are uTesla */
00558     float x;
00559     float y;
00560     float z;
00561 } sh2_MagneticField_t;
00562
00568 typedef struct sh2_MagneticFieldUncalibrated {
00569     /* Units are uTesla */
00570     float x;
00571     float y;
00572     float z;
00573     float biasX;
00574     float biasY;
00575     float biasZ;
00576 } sh2_MagneticFieldUncalibrated_t;
00577
00583 typedef struct sh2_RotationVectorWAcc {
00584     float i;
00585     float j;
00586     float k;
00587     float real;
00588     float accuracy;
00589 } sh2_RotationVectorWAcc_t;
00590
00596 typedef struct sh2_RotationVector {
00597     float i;
00598     float j;
00599     float k;
00600     float real;
00601 } sh2_RotationVector_t;
00602
00603
00604 #ifdef __cplusplus
00605 }
00606 #endif
00607
00608 #endif /* BNO086_REG_H */
00609

```

4.12 buzzer.h

```

00001 /*
00002  * TMR2 buzzer PWM update timer
00003  */
00004
00005 #ifndef BUZZER_H
00006 #define BUZZER_H
00007
00008 #ifdef __cplusplus
00009 extern "C" {
00010 #endif
00011 #include "imupic32mcj.h"
00012
00013 #define BUZZER_DRIVER "V1.002"
00014 #define BUZZER_ALIAS "BUZZER"
00015
00016 #define BZ0 0
00017 #define BZ1 1
00018 #define BZ2 11
00019 #define BZ3 21
00020
00021 #define BZ_OFF 255
00022
00023 #define BZ_ON_LOUD 100
00024 #define BZ_ON_LOW 101
00025
00026     typedef enum {
00027         B_init,
00028         B_abort,
00029     } B_STATE;
00030
00031     typedef struct B_type {
00032         volatile uint32_t arate, alimit;
00033         volatile uint32_t srate, slimit;
00034         volatile uint32_t drate, dlimit;
00035     } B_type;
00036
00037     void buzzer_init(B_STATE);
00038     void buzzer_trigger(uint32_t);
00039     uint32_t get_buzzer_trigger(void);

```

```

00040
00041     extern B_type tone1, tone2, tone3;
00042
00043 #ifdef __cplusplus
00044 }
00045 #endif
00046
00047 #endif /* BUZZER_H */
00048

```

4.13 canfd.h

```

00001 /*
00002  * File:   canfd.h
00003  * Author: root
00004  *
00005  * Created on September 22, 2022, 6:48 AM
00006  */
00007
00008 #ifndef CANFD_H
00009 #define CANFD_H
00010
00011 #ifdef __cplusplus
00012 extern "C" {
00013 #endif
00014
00015 #include <stddef.h>           // Defines NULL
00016 #include <stdbool.h>         // Defines true
00017 #include <stdlib.h>          // Defines EXIT_FAILURE
00018 #include <stdarg.h>
00019 #include "imupic32mcj.h"
00020
00021 /* Application's state machine enum */
00022 typedef enum {
00023     APP_STATE_CAN_RECEIVE,
00024     APP_STATE_CAN_TRANSMIT,
00025     APP_STATE_CAN_IDLE,
00026     APP_STATE_CAN_USER_INPUT,
00027     APP_STATE_CAN_XFER_SUCCESSFUL,
00028     APP_STATE_CAN_XFER_ERROR
00029 } APP_STATES;
00030
00031 typedef enum {
00032     CAN_RECEIVE,
00033     CAN_TRANSMIT_N,
00034     CAN_TRANSMIT_FD,
00035     CAN_IDLE,
00036     CAN_USER_INPUT,
00037     CAN_XFER_SUCCESSFUL,
00038     CAN_XFER_ERROR
00039 } CANFD_STATES;
00040
00041 typedef enum {
00042     CAN_NULL,
00043     CAN_IMU_DATA,
00044     CAN_IMU_INFO,
00045     CAN_IMU_RAW,
00046     CAN_IMU_GET,
00047     CAN_IMU_SET,
00048     CAN_STAT,
00049     CAN_MISC,
00050     CAN_FFT_LO,
00051     CAN_FFT_HI,
00052 } CANFD_MESSAGE;
00053
00054 #ifdef __32MZ1025W104132__
00055 #define CANDEV2
00056 #endif
00057
00058 // #define DEBUG_FILTER
00059
00060 void APP_CAN_Callback(uintptr_t);
00061 int canfd_state(CANFD_STATES, void *);
00062 uint32_t canfd_num_tx(void);
00063 uint32_t canfd_num_rx(void);
00064 uint32_t canfd_num_stall(void);
00065
00066 extern volatile bool tx_msg_ready;
00067 extern volatile bool rx_msg_ready;
00068
00069 #ifdef __cplusplus
00070 }
00071 #endif

```

```

00072
00073 #endif  /* CANFD_H */
00074

```

4.14 cmd_scanner.h

```

00001 /*
00002  * File:   cmd_scanner.h
00003  * Author: root
00004  *
00005  * Created on February 2, 2023, 8:23 AM
00006  */
00007
00008 #ifndef CMD_SCANNER_H
00009 #define CMD_SCANNER_H
00010
00011 #ifdef __cplusplus
00012 extern "C" {
00013 #endif
00014
00015 #include <stddef.h>           // Defines NULL
00016 #include <stdbool.h>         // Defines true
00017 #include <stdlib.h>          // Defines EXIT_FAILURE
00018 #include <stdarg.h>
00019 #include <proc/p32mk0512mcj048.h>
00020 #include "definitions.h"     // SYS function prototypes
00021 #include "imu.h"
00022
00023 #define CMD_DRIVER "V1.701"
00024 #define CMD_ALIAS "COMMAND"
00025
00026 /*
00027  * serial port command parser
00028  */
00029 #define CLI_CMD_BUFFER_SIZE 128
00030
00031 typedef struct _t_cmd {
00032     const char *cmd;
00033     void (*fh)(void*);
00034 } t_cmd;
00035
00036 typedef struct _t_cli_ctx {
00037     t_cmd *cmds;
00038     char cmd[CLI_CMD_BUFFER_SIZE];
00039     uint8_t cpos;
00040 } t_cli_ctx;
00041
00042 #define CLI_IO_INPUT(__data) \
00043     linux_getc(__data)
00044
00045 #define CLI_IO_OUTPUT(__data, __len) \
00046     linux_putc(__data, __len)
00047
00048 #define KEY_CODE_BACKSPACE 0x7f
00049 #define KEY_CODE_DELETE 0x7e
00050 #define KEY_CODE_ENTER '\r'
00051 #define KEY_CODE_ESCAPE 0x1b
00052
00053 #define POSINC(__x) (((__x) < (CLI_CMD_BUFFER_SIZE - 1)) ? (__x + 1) : (__x))
00054 #define POSDEC(__x) ((__x) ? ((__x) - 1) : 0)
00055
00056 typedef enum _t_cmd_status {
00057     E_CMD_OK = 0,
00058     E_CMD_NOT_FOUND,
00059     E_CMD_TOO_SHORT,
00060     E_CMD_EMPTY
00061 } t_cmd_status;
00062
00063 /*
00064  * command parser functions
00065  */
00066 void scmd_init(void);
00067 void cli_read(t_cli_ctx *);
00068 void dump_uart(void);
00069
00070 #ifdef __cplusplus
00071 }
00072 #endif
00073
00074 #endif /* CMD_SCANNER_H */
00075

```


4.15 definitions.h

```

00001 /*****
00002  System Definitions
00003
00004  File Name:
00005      definitions.h
00006
00007  Summary:
00008      project system definitions.
00009
00010  Description:
00011      This file contains the system-wide prototypes and definitions for a project.
00012
00013  *****/
00014
00015 //DOM-IGNORE-BEGIN
00016 /*****
00017  * Copyright (C) 2018 Microchip Technology Inc. and its subsidiaries.
00018  *
00019  * Subject to your compliance with these terms, you may use Microchip software
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00036  * THAT YOU HAVE PAID DIRECTLY TO MICROCHIP FOR THIS SOFTWARE.
00037  *****/
00038 //DOM-IGNORE-END
00039
00040 #ifndef DEFINITIONS_H
00041 #define DEFINITIONS_H
00042
00043 // ****
00044 // ****
00045 // Section: Included Files
00046 // ****
00047 // ****
00048 #include <stdint.h>
00049 #include <stddef.h>
00050 #include <stdbool.h>
00051 #include <stdio.h>
00052 #include "peripheral/canfd/plib_canfd1.h"
00053 #include "peripheral/clk/plib_clk.h"
00054 #include "peripheral/gpio/plib_gpio.h"
00055 #include "peripheral/evic/plib_evic.h"
00056 #include "peripheral/dmac/plib_dmac.h"
00057 #include "peripheral/coretimer/plib_coretimer.h"
00058 #include "peripheral/tmr/plib_tmr6.h"
00059 #include "peripheral/adchs/plib_adchs.h"
00060 #include "peripheral/qei/plib_qei2.h"
00061 #include "peripheral/uart/plib_uart1.h"
00062 #include "peripheral/tmr/plib_tmr4.h"
00063 #include "peripheral/uart/plib_uart2.h"
00064 #include "peripheral/tmr/plib_tmr5.h"
00065 #include "peripheral/tmr/plib_tmr2.h"
00066 #include "peripheral/tmr/plib_tmr3.h"
00067 #include "peripheral/mcpwm/plib_mcpwm.h"
00068 #include "peripheral/spi/spi_master/plib_spi2_master.h"
00069 #include "peripheral/spi/spi_master/plib_spi1_master.h"
00070 #include "peripheral/rtcc/plib_rtcc.h"
00071 #include "peripheral/tmr/plib_tmr9.h"
00072
00073 // DOM-IGNORE-BEGIN
00074 #ifdef __cplusplus // Provide C++ Compatibility
00075 extern "C" {
00076
00077
00078 #endif
00079 // DOM-IGNORE-END
00080
00081 /* CPU clock frequency */
00082 #define CPU_CLOCK_FREQUENCY 120000000
00083
00084 // ****

```

```

00085 // *****
00086 // Section: System Functions
00087 // *****
00088 // *****
00089
00090 // *****
00091 /* System Initialization Function
00092
00093     Function:
00094         void SYS_Initialize( void *data )
00095
00096     Summary:
00097         Function that initializes all modules in the system.
00098
00099     Description:
00100         This function initializes all modules in the system, including any drivers,
00101         services, middleware, and applications.
00102
00103     Precondition:
00104         None.
00105
00106     Parameters:
00107         data          - Pointer to the data structure containing any data
00108                        necessary to initialize the module. This pointer may
00109                        be null if no data is required and default initialization
00110                        is to be used.
00111
00112     Returns:
00113         None.
00114
00115     Example:
00116         <code>
00117         SYS_Initialize ( NULL );
00118
00119         while ( true )
00120         {
00121             SYS_Tasks ( );
00122         }
00123         </code>
00124
00125     Remarks:
00126         This function will only be called once, after system reset.
00127 */
00128
00129 void SYS_Initialize( void *data );
00130
00131 /* Nullify SYS_Tasks() if only PLIBs are used. */
00132 #define     SYS_Tasks()
00133
00134 // *****
00135 // *****
00136 // Section: extern declarations
00137 // *****
00138 // *****
00139
00140
00141
00142
00143 //DOM-IGNORE-BEGIN
00144 #ifdef __cplusplus
00145 }
00146 #endif
00147 //DOM-IGNORE-END
00148
00149 #endif /* DEFINITIONS_H */
00150 /*****
00151 End of File
00152 */
00153

```

4.16 device.h

```

00001 /*****
00002     Device Header File
00003
00004     Company:
00005         Microchip Technology Inc.
00006
00007     File Name:
00008         device.h
00009
00010     Summary:
00011         This file includes the selected device from within the project.

```

```

00012     The device will provide access to respective device packs.
00013
00014     Description:
00015     None
00016
00017     *****/
00018
00019     // DOM-IGNORE-BEGIN
00020     /*****
00021     * Copyright (C) 2018 Microchip Technology Inc. and its subsidiaries.
00022     *
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00040     * THAT YOU HAVE PAID DIRECTLY TO MICROCHIP FOR THIS SOFTWARE.
00041     *****/
00042     // DOM-IGNORE-END
00043
00044     #ifndef DEVICE_H
00045     #define DEVICE_H
00046
00047     #include <xc.h>
00048     #include <sys/attribs.h>
00049     #include "toolchain_specifics.h"
00050
00051     #endif //DEVICE_H

```

4.17 interrupts.h

```

00001     /*****
00002     System Interrupts File
00003
00004     Company:
00005     Microchip Technology Inc.
00006
00007     File Name:
00008     interrupt.h
00009
00010     Summary:
00011     Interrupt vectors mapping
00012
00013     Description:
00014     This file contains declarations of device vectors used by Harmony 3
00015     *****/
00016
00017     // DOM-IGNORE-BEGIN
00018     /*****
00019     * Copyright (C) 2018 Microchip Technology Inc. and its subsidiaries.
00020     *
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00039     *****/
00040     // DOM-IGNORE-END

```

```

00041
00042 #ifndef INTERRUPTS_H
00043 #define INTERRUPTS_H
00044
00045 // *****
00046 // *****
00047 // Section: Included Files
00048 // *****
00049 // *****
00050 #include <stdint.h>
00051
00052
00053
00054 // *****
00055 // *****
00056 // Section: Handler Routines
00057 // *****
00058 // *****
00059
00060
00061
00062 #endif // INTERRUPTS_H

```

4.18 plib_adchs.h

```

00001 /*****
00002  ADCHS Peripheral Library Interface Header File
00003
00004  Company
00005      Microchip Technology Inc.
00006
00007  File Name
00008      plib_adchs.h
00009
00010  Summary
00011      ADCHS peripheral library interface.
00012
00013  Description
00014      This file defines the interface to the ADCHS peripheral library. This
00015      library provides access to and control of the associated peripheral
00016      instance.
00017
00018  *****/
00019
00020 // DOM-IGNORE-BEGIN
00021 /*****
00022  * Copyright (C) 2019 Microchip Technology Inc. and its subsidiaries.
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00041  * THAT YOU HAVE PAID DIRECTLY TO MICROCHIP FOR THIS SOFTWARE.
00042  *****/
00043 // DOM-IGNORE-END
00044
00045 #ifndef PLIB_ADCHS_H    // Guards against multiple inclusion
00046 #define PLIB_ADCHS_H
00047
00048
00049 // *****
00050 // *****
00051 // Section: Included Files
00052 // *****
00053 // *****
00054
00055 /* This section lists the other files that are included in this file.
00056 */
00057
00058 #include "plib_adchs_common.h"

```

```

00059
00060 // DOM-IGNORE-BEGIN
00061 #ifdef __cplusplus // Provide C++ Compatibility
00062 extern "C" {
00063
00064 #endif
00065 // DOM-IGNORE-END
00066
00067 // *****
00068 // *****
00069 // *****
00070 // *****
00071 // Section: Data Types
00072 // *****
00073 // *****
00074 /* The following data type definitions are used by the functions in this
00075    interface and should be considered part it.
00076 */
00077
00078 // *****
00079 // *****
00080 // *****
00081 // Section: Interface Routines
00082 // *****
00083 // *****
00084 /* The following functions make up the methods (set of possible operations) of
00085    this interface.
00086 */
00087
00088 void ADCHS_Initialize (void);
00089
00090 void ADCHS_ModulesEnable (ADCHS_MODULE_MASK moduleMask);
00091 void ADCHS_ModulesDisable (ADCHS_MODULE_MASK moduleMask);
00092
00093 void ADCHS_GlobalEdgeConversionStart(void);
00094 void ADCHS_GlobalLevelConversionStart(void);
00095 void ADCHS_GlobalLevelConversionStop(void);
00096 void ADCHS_ChannelConversionStart(ADCHS_CHANNEL_NUM channel);
00097
00098 void ADCHS_ChannelResultInterruptEnable (ADCHS_CHANNEL_NUM channel);
00099 void ADCHS_ChannelResultInterruptDisable (ADCHS_CHANNEL_NUM channel);
00100 void ADCHS_ChannelEarlyInterruptEnable (ADCHS_CHANNEL_NUM channel);
00101 void ADCHS_ChannelEarlyInterruptDisable (ADCHS_CHANNEL_NUM channel);
00102
00103 bool ADCHS_ChannelResultIsReady(ADCHS_CHANNEL_NUM channel);
00104 uint16_t ADCHS_ChannelResultGet(ADCHS_CHANNEL_NUM channel);
00105
00106
00107
00108 bool ADCHS_EOSStatusGet(void);
00109
00110
00111 // *****
00112 // *****
00113 // DOM-IGNORE-BEGIN
00114 #ifdef __cplusplus // Provide C++ Compatibility
00115
00116 }
00117 #endif
00118 // DOM-IGNORE-END
00119
00120 #endif //PLIB_ADCHS_H
00121

```

4.19 plib_adchs_common.h

```

00001 /*****
00002    ADCHS Peripheral Library Interface Header File
00003
00004    Company
00005        Microchip Technology Inc.
00006
00007    File Name
00008        plib_adchs_common.h
00009
00010    Summary
00011        Commonly needed stuff for the ADCHS peripheral libraries interfaces.
00012
00013    Description
00014        This file defines several items commonly needed by the interfaces to
00015        the ADCHS peripheral libraries.
00016
00017 *****/

```

```

00018
00019 // DOM-IGNORE-BEGIN
00020 /*****
00021 * Copyright (C) 2018 Microchip Technology Inc. and its subsidiaries.
00022 *
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00040 * THAT YOU HAVE PAID DIRECTLY TO MICROCHIP FOR THIS SOFTWARE.
00041 *****/
00042 // DOM-IGNORE-END
00043
00044 #ifndef PLIB_ADCHS_COMMON_H // Guards against multiple inclusion
00045 #define PLIB_ADCHS_COMMON_H
00046
00047
00048 // ****
00049 // ****
00050 // Section: Included Files
00051 // ****
00052 // ****
00053
00054 /* This section lists the other files that are included in this file.
00055 */
00056
00057 #include <stddef.h>
00058 #include <stdbool.h>
00059
00060
00061 // DOM-IGNORE-BEGIN
00062 #ifdef __cplusplus // Provide C++ Compatibility
00063
00064 extern "C" {
00065
00066 #endif
00067
00068 // DOM-IGNORE-END
00069
00070
00071
00072
00073
00074 // ****
00075 // ****
00076 // Section: Data Types
00077 // ****
00078 // ****
00079
00080 // ****
00081
00082 typedef enum
00083 {
00084     ADCHS_MODULE0_MASK = (1U < 0U),
00085     ADCHS_MODULE1_MASK = (1U < 1U),
00086     ADCHS_MODULE2_MASK = (1U < 2U),
00087     ADCHS_MODULE3_MASK = (1U < 3U),
00088     ADCHS_MODULE4_MASK = (1U < 4U),
00089     ADCHS_MODULE5_MASK = (1U < 5U),
00090     ADCHS_MODULE7_MASK = (1U < 7U)
00091 }ADCHS_MODULE_MASK;
00092
00093
00094 typedef enum
00095 {
00096     ADCHS_CH0 = 0U,
00097     ADCHS_CH1 = 1U,
00098     ADCHS_CH2 = 2U,
00099     ADCHS_CH3 = 3U,
00100     ADCHS_CH4 = 4U,
00101     ADCHS_CH5 = 5U,
00102     ADCHS_CH6 = 6U,
00103     ADCHS_CH7 = 7U,
00104     ADCHS_CH8 = 8U,

```

```

00105     ADCHS_CH9 = 9U,
00106     ADCHS_CH10 = 10U,
00107     ADCHS_CH11 = 11U,
00108     ADCHS_CH24 = 24U,
00109     ADCHS_CH25 = 25U,
00110     ADCHS_CH26 = 26U,
00111     ADCHS_CH27 = 27U,
00112     ADCHS_CH48 = 48U,
00113     ADCHS_CH49 = 49U,
00114     ADCHS_CH50 = 50U,
00115     ADCHS_CH53 = 53U,
00116 }ADCHS_CHANNEL_NUM;
00117
00118
00119
00120 // *****
00121
00122 typedef void (*ADCHS_CALLBACK)(ADCHS_CHANNEL_NUM channel, uintptr_t context);
00123
00124 typedef void (*ADCHS_EOS_CALLBACK)(uintptr_t context);
00125
00126
00127
00128
00129 // *****
00130
00131 typedef struct
00132 {
00133     ADCHS_CALLBACK callback_fn;
00134     uintptr_t context;
00135 }ADCHS_CALLBACK_OBJECT;
00136
00137 typedef struct
00138 {
00139     ADCHS_EOS_CALLBACK callback_fn;
00140     uintptr_t context;
00141 }ADCHS_EOS_CALLBACK_OBJECT;
00142
00143
00144
00145
00146 // DOM-IGNORE-BEGIN
00147 #ifdef __cplusplus // Provide C++ Compatibility
00148 }
00149 }
00150
00151 #endif
00152 // DOM-IGNORE-END
00153
00154 #endif //PLIB_ADCHS_COMMON_H
00155

```

4.20 plib_canfd1.h

```

00001 /*****
00002     CANFD Peripheral Library Interface Header File
00003
00004     Company:
00005         Microchip Technology Inc.
00006
00007     File Name:
00008         plib_canfd1.h
00009
00010     Summary:
00011         CANFD PLIB interface declarations.
00012
00013     Description:
00014         The CANFD plib provides a simple interface to manage the CANFD modules on
00015         Microchip microcontrollers. This file defines the interface declarations
00016         for the CANFD plib.
00017
00018     Remarks:
00019         None.
00020
00021     *****/
00022 //DOM-IGNORE-BEGIN
00023 /*****
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```

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00044 *****/
00045 //DOM-IGNORE-END
00046
00047 #ifndef PLIB_CANFD1_H
00048 #define PLIB_CANFD1_H
00049
00050 // *****
00051 // *****
00052 // Section: Included Files
00053 // *****
00054 // *****
00055
00056 /*
00057  * This section lists the other files that are included in this file.
00058  */
00059 #include <stdbool.h>
00060 #include <string.h>
00061 #include "device.h"
00062 #include "plib_canfd_common.h"
00063
00064 // DOM-IGNORE-BEGIN
00065 #ifdef __cplusplus // Provide C++ Compatibility
00066     extern "C" {
00067 #endif
00068 // DOM-IGNORE-END
00069
00070 // *****
00071 // *****
00072 // Section: Interface Routines
00073 // *****
00074 // *****
00075 void CAN1_Initialize(void);
00076 bool CAN1_MessageTransmit(uint32_t id, uint8_t length, uint8_t* data, uint8_t fifoQueueNum, CANFD_MODE
mode, CANFD_MSG_TX_ATTRIBUTE msgAttr);
00077 bool CAN1_MessageReceive(uint32_t *id, uint8_t *length, uint8_t *data, uint32_t *timestamp, uint8_t
fifoNum, CANFD_MSG_RX_ATTRIBUTE *msgAttr);
00078 void CAN1_MessageAbort(uint8_t fifoQueueNum);
00079 void CAN1_MessageAcceptanceFilterSet(uint8_t filterNum, uint32_t id);
00080 uint32_t CAN1_MessageAcceptanceFilterGet(uint8_t filterNum);
00081 void CAN1_MessageAcceptanceFilterMaskSet(uint8_t acceptanceFilterMaskNum, uint32_t id);
00082 uint32_t CAN1_MessageAcceptanceFilterMaskGet(uint8_t acceptanceFilterMaskNum);
00083 bool CAN1_TransmitEventFIFOElementGet(uint32_t *id, uint32_t *sequence, uint32_t *timestamp);
00084 CANFD_ERROR CAN1_ErrorGet(void);
00085 void CAN1_ErrorCountGet(uint8_t *txErrorCount, uint8_t *rxErrorCount);
00086 bool CAN1_InterruptGet(uint8_t fifoQueueNum, CANFD_FIFO_INTERRUPT_FLAG_MASK fifoInterruptFlagMask);
00087 bool CAN1_TxFIFOQueueIsFull(uint8_t fifoQueueNum);
00088 bool CAN1_AutoTRResponseSet(uint32_t id, uint8_t length, uint8_t* data, uint8_t fifoNum);
00089 // DOM-IGNORE-BEGIN
00090 #ifdef __cplusplus // Provide C++ Compatibility
00091     }
00092 #endif
00093 // DOM-IGNORE-END
00094
00095 #endif // PLIB_CANFD1_H

```

4.21 plib_canfd_common.h

```

00001 /*****
00002  CANFD Peripheral Library Interface Header File
00003
00004  Company
00005  Microchip Technology Inc.
00006
00007  File Name
00008  plib_canfd_common.h
00009
00010  Summary
00011  CANFD peripheral library interface.
00012

```



```

00013 Description
00014     This file defines the interface to the CANFD peripheral library. This
00015     library provides access to and control of the associated peripheral
00016     instance.
00017
00018 Remarks:
00019
00020 *****/
00021
00022 // DOM-IGNORE-BEGIN
00023 /*****
00024  * Copyright (C) 2019 Microchip Technology Inc. and its subsidiaries.
00025  *
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00044  *****/
00045 // DOM-IGNORE-END
00046
00047 #ifndef PLIB_CANFD_COMMON_H
00048 #define PLIB_CANFD_COMMON_H
00049
00050 // DOM-IGNORE-BEGIN
00051 #ifdef __cplusplus // Provide C++ Compatibility
00052
00053     extern "C" {
00054
00055 #endif
00056 // DOM-IGNORE-END
00057
00058 // *****/
00059 // *****/
00060 // Section: Data Types
00061 // *****/
00062 // *****/
00063
00064 // *****/
00065 /* CANFD Mode
00066
00067 Summary:
00068     CANFD Mode for Classic CAN and CAN FD.
00069
00070 Description:
00071     This data type defines CANFD mode Classic CAN, CAN FD without BRS(Bit rate switching)
00072     and CAN FD with BRS.
00073
00074 Remarks:
00075     None.
00076 */
00077 typedef enum
00078 {
00079     CANFD_MODE_NORMAL = 0,
00080     CANFD_MODE_FD_WITHOUT_BRS,
00081     CANFD_MODE_FD_WITH_BRS
00082 } CANFD_MODE;
00083
00084 // *****/
00085 /* CANFD Tx Message Attribute
00086
00087 Summary:
00088     CANFD Tx Message Attribute for Tx FIFO and Tx Queue.
00089
00090 Description:
00091     This data type defines CANFD Message TX Attribute for Data Frame and Remote Frame.
00092     Only One attribute need to be passed as parameter value while invoking
00093     message transmit function.
00094
00095 Remarks:
00096     None.
00097 */
00098 typedef enum
00099 {

```

```

00100     CANFD_MSG_TX_DATA_FRAME = 0,
00101     CANFD_MSG_TX_REMOTE_FRAME
00102 } CANFD_MSG_TX_ATTRIBUTE;
00103
00104 // *****
00105 /* CANFD Message Object Type RX Attribute
00106
00107     Summary:
00108         CANFD Message RX Attribute for Data Frame and Remote Frame.
00109
00110     Description:
00111         This data type defines CANFD Message RX Attribute for Data Frame and Remote Frame.
00112
00113     Remarks:
00114         None.
00115 */
00116 typedef enum
00117 {
00118     CANFD_MSG_RX_DATA_FRAME = 0,
00119     CANFD_MSG_RX_REMOTE_FRAME
00120 } CANFD_MSG_RX_ATTRIBUTE;
00121
00122 // *****
00123 /* CANFD FIFO Interrupt Status Flag Mask
00124
00125     Summary:
00126         CANFD FIFO Interrupt Status Flag Mask.
00127
00128     Description:
00129         This data type defines the CANFD FIFO Interrupt Status Flag Mask.
00130
00131     Remarks:
00132         None.
00133 */
00134 typedef enum
00135 {
00136     /* Receive FIFO Not Empty / Transmit FIFO Not Full Interrupt Mask */
00137     CANFD_FIFO_INTERRUPT_TFNRFNIF_MASK = 0x1,
00138
00139     /* Transmit FIFO Half Empty / Receive FIFO Half Full Interrupt Mask */
00140     CANFD_FIFO_INTERRUPT_TFHRFHIF_MASK = 0x2,
00141
00142     /* Transmit FIFO Empty / Receive FIFO Full Interrupt Mask */
00143     CANFD_FIFO_INTERRUPT_TFERFFIF_MASK = 0x4,
00144
00145     /* Receive FIFO Overflow Interrupt Mask */
00146     CANFD_FIFO_INTERRUPT_RXOVIF_MASK = 0x8,
00147
00148     /* Transmit Attempts Exhausted Interrupt Pending Mask */
00149     CANFD_FIFO_INTERRUPT_TXATIF_MASK = 0x10
00150 } CANFD_FIFO_INTERRUPT_FLAG_MASK;
00151
00152 // *****
00153 /* CANFD Transfer Error
00154
00155     Summary:
00156         CANFD Transfer Error data type.
00157
00158     Description:
00159         This data type defines the CANFD Transfer Error.
00160
00161     Remarks:
00162         None.
00163 */
00164 typedef enum
00165 {
00166     CANFD_ERROR_NONE = 0x0,
00167     CANFD_ERROR_TX_RX_WARNING_STATE = 0x10000,
00168     CANFD_ERROR_RX_WARNING_STATE = 0x20000,
00169     CANFD_ERROR_TX_WARNING_STATE = 0x40000,
00170     CANFD_ERROR_RX_BUS_PASSIVE_STATE = 0x80000,
00171     CANFD_ERROR_TX_BUS_PASSIVE_STATE = 0x100000,
00172     CANFD_ERROR_TX_BUS_OFF_STATE = 0x200000
00173 } CANFD_ERROR;
00174
00175 // *****
00176 /* CANFD Callback
00177
00178     Summary:
00179         CANFD Callback Function Pointer.
00180
00181     Description:
00182         This data type defines the CANFD Callback Function Pointer.
00183
00184     Remarks:
00185         None.
00186 */

```

```

00187 typedef void (*CANFD_CALLBACK) (uintptr_t contextHandle);
00188
00189 // *****
00190 /* CANFD RX Message
00191
00192     Summary:
00193         CANFD RX Message Buffer structure.
00194
00195     Description:
00196         This data structure stores RX Message.
00197
00198     Remarks:
00199         None.
00200 */
00201 typedef struct
00202 {
00203     /* Rx Message ID */
00204     uint32_t *id;
00205     /* Rx Message buffer */
00206     uint8_t *buffer;
00207     /* Rx Message size */
00208     uint8_t *size;
00209     /* Rx Message timestamp */
00210     uint32_t *timestamp;
00211     /* Rx Message attribute */
00212     CANFD_MSG_RX_ATTRIBUTE *msgAttr;
00213 } CANFD_RX_MSG;
00214
00215 // *****
00216 /* CANFD Receive Message Object
00217
00218     Summary:
00219         CANFD Receive Message Object structure.
00220
00221     Description:
00222         This data structure defines the CANFD Rx Message Object format.
00223
00224     Remarks:
00225         None.
00226 */
00227 typedef struct
00228 {
00229     /* CANFD Rx message - SID[10:0] and EID[28:11] */
00230     uint32_t r0;
00231
00232     /* CANFD Rx message - DLC[3:0], IDE[4], RTR[5], BRS[6], FDF[7]
00233                                     ESI[8] and FILHIT[15:11] */
00234     uint32_t r1;
00235
00236     /* CANFD Rx message - RXMSGTS and Receive Buffer Data Bytes */
00237     uint8_t data[];
00238 } CANFD_RX_MSG_OBJECT;
00239
00240 // *****
00241 /* CANFD Trasmit Message Object
00242
00243     Summary:
00244         CANFD Transmit Message Object structure.
00245
00246     Description:
00247         This data structure defines the CANFD Tx Message Object format.
00248
00249     Remarks:
00250         None.
00251 */
00252 typedef struct
00253 {
00254     /* CANFD Tx message - SID[10:0] and EID[28:11] */
00255     uint32_t t0;
00256
00257     /* CANFD Tx message - DLC[3:0], IDE[4], RTR[5], BRS[6], FDF[7]
00258                                     ESI[8] and SEQ[31:9] */
00259     uint32_t t1;
00260
00261     /* CANFD Tx message - Transmit Buffer Data Bytes */
00262     uint8_t data[];
00263 } CANFD_TX_MSG_OBJECT;
00264
00265 // *****
00266 /* CANFD Trasmit Event FIFO Element
00267
00268     Summary:
00269         CANFD Trasmit Event FIFO Element structure.
00270
00271     Description:
00272         This data structure defines the CANFD Tx Event FIFO Element format.
00273

```

```

00274     Remarks:
00275     None.
00276 */
00277 typedef struct
00278 {
00279     /* CANFD Tx Event FIFO - SID[10:0] and EID[28:11] */
00280     uint32_t te0;
00281
00282     /* CANFD Tx Event FIFO - DLC[3:0], IDE[4], RTR[5], BRS[6], FDF[7]
00283     ESI[8] and SEQ[31:9] */
00284     uint32_t tel;
00285
00286     /* CANFD Tx Event FIFO - Transmit Message Time Stamp */
00287     uint8_t timestamp[];
00288 } CANFD_TX_EVENT_FIFO_ELEMENT;
00289
00290 // *****
00291 /* CANFD Callback Object
00292
00293     Summary:
00294     CANFD transfer event callback structure.
00295
00296     Description:
00297     This data structure stores transfer event callback and it's context.
00298
00299     Remarks:
00300     None.
00301 */
00302 typedef struct
00303 {
00304     /* Transfer Event Callback */
00305     CANFD_CALLBACK callback;
00306
00307     /* Transfer Event Callback Context */
00308     uintptr_t context;
00309 } CANFD_CALLBACK_OBJ;
00310
00311 // *****
00312 /* CANFD PLib Instance Object
00313
00314     Summary:
00315     CANFD PLib Object structure.
00316
00317     Description:
00318     This data structure defines the CANFD PLib Instance Object.
00319
00320     Remarks:
00321     None.
00322 */
00323 typedef struct
00324 {
00325     /* CANFD Error Status */
00326     uint32_t errorStatus;
00327 } CANFD_OBJ;
00328
00329 // DOM-IGNORE-BEGIN
00330 #ifdef __cplusplus // Provide C++ Compatibility
00331 }
00332 #endif
00333 // DOM-IGNORE-END
00334
00335 #endif //PLIB_CANFD_COMMON_H

```

4.22 plib_clk.h

```

00001 /*****
00002     SYS CLK Static Functions for Clock System Service
00003
00004     Company:
00005     Microchip Technology Inc.
00006
00007     File Name:
00008     plib_clk.h
00009
00010     Summary:
00011     SYS CLK static function interface for the Clock System Service.
00012
00013     Description:
00014     The Clock System Service provides a simple interface to manage the
00015     oscillators on Microchip microcontrollers. This file defines the static
00016     implementation for the Clock System Service.
00017
00018     Remarks:

```

```

00019     Static functions incorporate all system clock configuration settings as
00020     determined by the user via the Microchip Harmony Configurator GUI.
00021     It provides static version of the routines, eliminating the need for an
00022     object ID or object handle.
00023
00024     Static single-open interfaces also eliminate the need for the open handle.
00025
00026     *****/
00027
00028 /*****
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00048  * THAT YOU HAVE PAID DIRECTLY TO MICROCHIP FOR THIS SOFTWARE.
00049  *****/
00050
00051 #ifndef PLIB_CLK_H
00052 #define PLIB_CLK_H
00053
00054 // ****
00055 // ****
00056 // Section: Included Files
00057 // ****
00058 // ****
00059
00060 #include <stddef.h>
00061 #include <stdbool.h>
00062
00063 // DOM-IGNORE-BEGIN
00064 #ifdef __cplusplus // Provide C++ Compatibility
00065     extern "C" {
00066
00067 #endif
00068 // DOM-IGNORE-END
00069
00070
00071 // ****
00072 // ****
00073 // Section: CLK Module System Interface Routines
00074 // ****
00075 // ****
00076
00077 // ****
00078 /* Function:
00079     void CLK_Initialize( void )
00080
00081     Summary:
00082         Initializes hardware of the System Clock and Peripheral Clock.
00083
00084     Description:
00085         This function initializes the hardware of System Clock and Peripheral Clocks.
00086
00087     Precondition:
00088         None.
00089
00090     Parameters:
00091         None.
00092
00093     Returns:
00094         None.
00095
00096     Example:
00097         <code>
00098         //Example 1: Do not alter the configuration bit settings
00099         CLK_Initialize ( );
00100
00101         </code>
00102
00103     Remarks:
00104         None.
00105 */

```

```

00106
00107 void CLK_Initialize( void );
00108
00109 // DOM-IGNORE-BEGIN
00110 #ifdef __cplusplus // Provide C++ Compatibility
00111
00112     }
00113
00114 #endif
00115 // DOM-IGNORE-END
00116
00117 #endif //PLIB_CLK_H
00118

```

4.23 plib_coretimer.h

```

00001 /*****
00002  Interface definition of Core Timer PLIB.
00003
00004  Company:
00005      Microchip Technology Inc.
00006
00007  File Name:
00008      plib_coretimer.h
00009
00010  Summary:
00011      Interface definition of the Core Timer Plib .
00012
00013  Description:
00014      This file defines the interface for the Core Timer Plib.
00015  *****/
00016
00017 /*****
00018  * Copyright (C) 2019 Microchip Technology Inc. and its subsidiaries.
00019  *
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00038  *****/
00039
00040 #ifndef PLIB_CORETIMER_H // Guards against multiple inclusion
00041 #define PLIB_CORETIMER_H
00042
00043 #include <stdint.h>
00044 #include <stdbool.h>
00045
00046 #ifdef __cplusplus // Provide C++ Compatibility
00047     extern "C" {
00048 #endif
00049
00050 #define CORE_TIMER_FREQUENCY      60000000
00051
00052 #define CORE_TIMER_INTERRUPT_PERIOD_VALUE    0xea60
00053 #define CORE_TIMER_INTERRUPT_PERIOD_IN_US    1000
00054
00055 typedef void (*CORETIMER_CALLBACK)(uint32_t status, uintptr_t context);
00056
00057 typedef struct
00058 {
00059     CORETIMER_CALLBACK    callback;
00060     uintptr_t             context;
00061     volatile uint32_t     tickCounter;
00062     uint32_t              period;
00063 } CORETIMER_OBJECT ;
00064
00065 void CORETIMER_Initialize(void);
00066 void CORETIMER_CallbackSet ( CORETIMER_CALLBACK callback, uintptr_t context );
00067 uint32_t CORETIMER_FrequencyGet (void);

```

```

00068 void CORETIMER_PeriodSet (uint32_t period);
00069 void CORETIMER_Start(void);
00070 void CORETIMER_Stop(void);
00071
00072
00073
00074
00075 void CORETIMER_DelayMs (uint32_t delay_ms);
00076 void CORETIMER_DelayUs (uint32_t delay_us);
00077
00078 #ifndef __cplusplus // Provide C++ Compatibility
00079 }
00080 #endif
00081
00082 #endif

```

4.24 plib_dmac.h

```

00001 /*****
00002  DMAC Peripheral Library Interface Header File
00003
00004  Company:
00005      Microchip Technology Inc.
00006
00007  File Name:
00008      plib_dmac.h
00009
00010  Summary:
00011      DMAC peripheral library interface.
00012
00013  Description:
00014      This file defines the interface to the DMAC peripheral library. This
00015      library provides access to and control of the DMAC controller.
00016
00017  Remarks:
00018      None.
00019
00020 *****/
00021
00022 // DOM-IGNORE-BEGIN
00023 /*****
00024  * Copyright (C) 2019 Microchip Technology Inc. and its subsidiaries.
00025  *
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00043  * THAT YOU HAVE PAID DIRECTLY TO MICROCHIP FOR THIS SOFTWARE.
00044 *****/
00045 // DOM-IGNORE-END
00046
00047 #ifndef PLIB_DMACH_H // Guards against multiple inclusion
00048 #define PLIB_DMACH_H
00049
00050 // ****
00051 // ****
00052 // Section: Included Files
00053 // ****
00054 // ****
00055
00056 #include <device.h>
00057 #include <string.h>
00058 #include <stdbool.h>
00059 #include <sys/kmem.h>
00060
00061 // DOM-IGNORE-BEGIN
00062 #ifndef __cplusplus // Provide C++ Compatibility
00063
00064     extern "C" {
00065

```

```

00066 #endif
00067 // DOM-IGNORE-END
00068
00069
00070 // *****
00071 // *****
00072 // Section: type definitions
00073 // *****
00074 // *****
00075
00076 // *****
00077 /* DMA error type
00078
00079     Summary:
00080         Identifies the available DMA operating modes.
00081
00082     Description:
00083         This data type Identifies if a transfer had an address error or not.
00084
00085
00086     Remarks:
00087         The identification of an error transaction takes place in the DMA ISR.
00088 */
00089 typedef enum
00090 {
00091     /* Data was transferred successfully. */
00092     DMAC_ERROR_NONE /*DOM-IGNORE-BEGIN*/ = 1, /* DOM-IGNORE-END*/
00093
00094     /* DMA address error. */
00095     DMAC_ERROR_ADDRESS_ERROR /*DOM-IGNORE-BEGIN*/ = 2 /* DOM-IGNORE-END*/
00096 } DMAC_ERROR;
00097
00098 // *****
00099 /* DMA transfer event
00100
00101
00102     Summary:
00103         Identifies the status of the transfer event.
00104
00105     Description:
00106         Used to report back, via registered callback, the status of a transaction.
00107
00108     Remarks:
00109         None
00110 */
00111 typedef enum
00112 {
00113     /* No events yet. */
00114     DMAC_TRANSFER_EVENT_NONE = 0,
00115
00116     /* Data was transferred successfully. */
00117     DMAC_TRANSFER_EVENT_COMPLETE = 1,
00118
00119     /* Error while processing the request */
00120     DMAC_TRANSFER_EVENT_ERROR = 2,
00121
00122     /* Half Data is transferred */
00123     DMAC_TRANSFER_EVENT_HALF_COMPLETE = 4
00124 } DMAC_TRANSFER_EVENT;
00125
00126 // *****
00127 /* DMA Data Pattern Size
00128
00129
00130     Summary:
00131         Identifies the pattern size for data matching
00132
00133     Description:
00134         This data type Identifies size of pattern data which will be matched with
00135         transmitted data.
00136
00137     Remarks:
00138         None.
00139 */
00140 typedef enum
00141 {
00142     /* pattern size is of 1 byte */
00143     DMAC_DATA_PATTERN_SIZE_1_BYTE,
00144
00145     /* pattern size is of 2 bytes */
00146     DMAC_DATA_PATTERN_SIZE_2_BYTE
00147 } DMAC_DATA_PATTERN_SIZE;
00148
00149 typedef void (*DMAC_CHANNEL_CALLBACK) (DMAC_TRANSFER_EVENT status, uintptr_t contextHandle);
00150
00151 // *****

```



```

00153 /* DMA channel object
00154
00155 Summary:
00156     Fundamental data object for a DMA channel.
00157
00158 Description:
00159     Used by DMA logic to register/use a DMA callback, report back error information
00160     from the ISR handling a transfer event.
00161
00162 Remarks:
00163     None
00164 */
00165 typedef struct
00166 {
00167     bool inUse;
00168
00169     /* Indicates the error information for
00170     the last DMA operation on this channel */
00171     DMAC_ERROR errorInfo;
00172
00173     /* Call back function for this DMA channel */
00174     DMAC_CHANNEL_CALLBACK pEventCallBack;
00175
00176     /* Client data(Event Context) that will be returned at callback */
00177     uintptr_t hClientArg;
00178 } DMAC_CHANNEL_OBJECT;
00179
00180 // *****
00181 /* DMA channel
00182
00183 Summary:
00184     Fundamental data object that represents DMA channel number.
00185
00186 Description:
00187     None
00188
00189 Remarks:
00190     None
00191 */
00192 typedef enum
00193 {
00194     DMAC_CHANNEL_0 = 0x0,
00195     DMAC_CHANNEL_1 = 0x1,
00196     DMAC_CHANNEL_2 = 0x2,
00197     DMAC_CHANNEL_3 = 0x3,
00198     DMAC_CHANNEL_4 = 0x4,
00199     DMAC_CHANNEL_5 = 0x5,
00200     DMAC_CHANNEL_6 = 0x6,
00201     DMAC_CHANNEL_7 = 0x7,
00202     DMAC_NUMBER_OF_CHANNELS = 0x8
00203 } DMAC_CHANNEL;
00204
00205 // *****
00206 /* DMA CRC Setup
00207
00208 Summary:
00209     Fundamental data object that represents DMA CRC setup parameters.
00210
00211 Description:
00212     None
00213
00214 Remarks:
00215     None
00216 */
00217 typedef struct
00218 {
00219     /* DCRCCON[CRCAPP]: The DMA transfers data from the source into the CRC engine and
00220     * writes the calculated CRC value to the destination when enabled
00221     */
00222     bool append_mode;
00223
00224     /* DCRCCON[BITO]: The input data is bit reversed (reflected input) when enabled */
00225     bool reverse_crc_input;
00226
00227     /* DCRCCON[PLEN]: Determines the length of the polynomial Example: 16, 32 */
00228     uint8_t polynomial_length;
00229 }

```

```

00240     /* DCRDXOR: Polynomial for calculating the CRC */
00241     uint32_t polynomial;
00242
00243     /* DCRCDATA: Input Non direct Seed for calculating the CRC */
00244     uint32_t non_direct_seed;
00245
00246     /* The calculated CRC is bit reversed (reflected output) before final XOR */
00247     bool reverse_crc_output;
00248
00249     /* The XOR value used to generate final CRC output */
00250     uint32_t final_xor_value;
00251 } DMAC_CRC_SETUP;
00252
00253 // *****
00254 // *****
00255 // Section: DMAC API
00256 // *****
00257 // *****
00258 // *****
00259 /* Function:
00260     void DMAC_ChannelCallbackRegister
00261
00262     Summary:
00263         Callback function registration function
00264
00265     Description:
00266         Registers the callback function (and context pointer, if used) for a given DMA interrupt.
00267
00268     Parameters:
00269         DMAC_CHANNEL channel - DMA channel this callback pertains to
00270         const DMAC_CHANNEL_CALLBACK eventHandler - pointer to callback function
00271         const uintptr_t contextHandle - pointer of context information callback is to use (set to NULL if
not used)
00272
00273     Returns:
00274         void
00275
00276     Example:
00277         <code>
00278             DMAC_ChannelCallbackRegister(DMAC_CHANNEL_0, DMAC_Callback, 0);
00279         </code>
00280 */
00281 void DMAC_ChannelCallbackRegister(DMAC_CHANNEL channel, const DMAC_CHANNEL_CALLBACK eventHandler,
const uintptr_t contextHandle );
00282
00283 // *****
00284 /* Function:
00285     bool DMAC_ChannelTransfer
00286
00287     Summary:
00288         DMA channel transfer function
00289
00290     Description:
00291         Sets up a DMA transfer, and starts the transfer if user specified a
00292         software-initiated transfer in Harmony.
00293
00294     Parameters:
00295         DMAC_CHANNEL channel - DMA channel to use for this transfer
00296         const void *srcAddr - pointer to source data
00297         size_t srcSize - Size of the source
00298         const void *destAddr - pointer to where data is to be moved to
00299         size_t destSize - Size of the destination
00300         size_t cellSize - Size of the cell
00301
00302     Returns:
00303         false, if DMA already is busy / true, if DMA is not busy before calling function
00304
00305     Example:
00306         <code>
00307             DMAC_ChannelCallbackRegister(DMAC_CHANNEL_0, DMAC_Callback, 0);
00308             DMAC_ChannelTransfer(DMAC_CHANNEL_0, srcAddr, srcSize, destAddr, destSize, cellSize);
00309         </code>
00310 */
00311 bool DMAC_ChannelTransfer( DMAC_CHANNEL channel, const void *srcAddr, size_t srcSize, const void
*destAddr, size_t destSize, size_t cellSize);
00312
00313 bool DMAC_ChainTransferSetup( DMAC_CHANNEL channel, const void *srcAddr, size_t srcSize, const void
*destAddr, size_t destSize, size_t cellSize);
00314
00315 void DMAC_ChannelPatternMatchSetup(DMAC_CHANNEL channel, DMAC_DATA_PATTERN_SIZE patternSize, uint16_t
patternMatchData);
00316 void DMAC_ChannelPatternMatchDisable(DMAC_CHANNEL channel);
00317 // *****
00318 /* Function:
00319     void DMAC_ChannelDisable (DMAC_CHANNEL channel)
00320
00321     Summary:

```

```

00322     This function disables the DMA channel.
00323
00324     Description:
00325         Disables the DMA channel specified.
00326
00327     Parameters:
00328         DMAC_CHANNEL channel - the particular channel to be disabled
00329
00330     Returns:
00331         void
00332
00333     Example:
00334         <code>
00335         DMAC_ChannelDisable (DMAC_CHANNEL_0);
00336         </code>
00337 */
00338 void DMAC_ChannelDisable (DMAC_CHANNEL channel);
00339
00340 // *****
00341 /* Function:
00342     bool DMAC_ChannelIsBusy (DMAC_CHANNEL channel)
00343
00344     Summary:
00345         Reads the busy status of a channel.
00346
00347     Description:
00348         Reads the busy status of a channel and returns status to caller.
00349
00350     Parameters:
00351         DMAC_CHANNEL channel - the particular channel to be interrogated
00352
00353     Returns:
00354         true - channel is busy
00355         false - channel is not busy
00356
00357     Example:
00358         <code>
00359         bool returnVal;
00360         returnVal = DMAC_ChannelIsBusy(DMAC_CHANNEL_0);
00361         while( returnVal )
00362             ;
00363         </code>
00364 */
00365 bool DMAC_ChannelIsBusy (DMAC_CHANNEL channel);
00366
00367 // *****
00368 /* Function:
00369     void DMAC_ChannelCRCSetup
00370
00371     Summary:
00372         DMA Channel CRC setup and enable function
00373
00374     Description:
00375         Sets up the DMA CRC engine for a particular channel and enables it.
00376         CRC can be enabled for only one channel at a time.
00377         Application needs to call this API with proper setup parameters every time
00378         before starting any DMA transfer.
00379
00380     Note:
00381         - A non direct seed should be used while setting up the DMA CRC
00382
00383         - The source buffer used for the DMA transfer should be appended with
00384           additional zero bits based on the CRC to be generated as shown in example
00385           - For 16 Bit CRC - Two bytes of 0's needs to be appended
00386           - For 32 Bit CRC - Four bytes of 0's needs to be appended
00387
00388         - Currently LFSR CRC type is only supported
00389
00390     Parameters:
00391         - DMAC_CHANNEL channel      : DMA channel this callback pertains to
00392         - DMAC_CRC_SETUP crcSetup   : parameter holding the crc setup information
00393
00394     Returns:
00395         void
00396
00397     Example:
00398         <code>
00399         const uint8_t srcBuffer[13] = {'1', '2', '3', '4', '5', '6', '7', '8', '9', 0, 0, 0, 0};
00400         uint8_t CACHE_ALIGN dstBuffer[13] = {0};
00401
00402         DMAC_CRC_SETUP crcSetup = {0};
00403
00404         crcSetup.reverse_data_input = true;
00405         crcSetup.polynomial_length = 32;
00406         crcSetup.polynomial        = 0x04C11DB7;
00407         crcSetup.non_direct_seed   = 0x46AF6449;
00408         crcSetup.reverse_crc_output = true;

```

```

00409     crcSetup.final_xor_value    = 0xFFFFFFFF;
00410
00411     DMAC_ChannelCRCSetup(DMAC_CHANNEL_0, crcSetup);
00412
00413     DMAC_ChannelTransfer(DMAC_CHANNEL_0, &srcBuffer, 13, &dstBuffer, 13, 13);
00414 </code>
00415 */
00416 void DMAC_ChannelCRCSetup( DMAC_CHANNEL channel, DMAC_CRC_SETUP CRCSetup );
00417
00418 // *****
00419 /* Function:
00420     void DMAC_CRCDisable
00421
00422     Summary:
00423         DMA CRC disable function
00424
00425     Description:
00426         Disables CRC generation for the DMA transfers
00427
00428     Parameters:
00429         None
00430
00431     Returns:
00432         void
00433
00434     Example:
00435     <code>
00436         DMAC_CRCDisable();
00437     </code>
00438 */
00439 void DMAC_CRCDisable( void );
00440
00441 // *****
00442 /* Function:
00443     uint32_t DMAC_CRCRead
00444
00445     Summary:
00446         DMA CRC read function
00447
00448     Description:
00449         Reads the generated DMA CRC value. It performs crc reverse and final xor
00450         operation based on setup paramters during DMAC_ChannelCRCSetup()
00451
00452         Note: Once Read is done, DMAC_ChannelCRCSetup() has to be called
00453         again to setup the seed before performing DMA transfer for CRC generation.
00454
00455     Parameters:
00456         None
00457
00458     Returns:
00459         - crc: Generated crc value
00460
00461     Example:
00462     <code>
00463         DMAC_CRC_SETUP crcSetup = {0};
00464         transfer_completed = false;
00465         uint32_t crc = 0;
00466
00467         crcSetup.reverse_data_input = true;
00468         crcSetup.polynomial_length = 32;
00469         crcSetup.polynomial        = 0x04C11DB7;
00470         crcSetup.non_direct_seed   = 0x46AF6449;
00471         crcSetup.reverse_crc_output = true;
00472         crcSetup.final_xor_value   = 0xFFFFFFFF;
00473
00474         DMAC_ChannelCRCSetup(DMAC_CHANNEL_0, crcSetup);
00475
00476         DMAC_ChannelTransfer(...);
00477
00478         if (transfer_completed == true)
00479         {
00480             crc = DMAC_CRCRead();
00481         }
00482     </code>
00483 */
00484 uint32_t DMAC_CRCRead( void );
00485
00486 // *****
00487 /* Function:
00488     void DMAC_Initialize( void )
00489
00490     Summary:
00491         This function initializes the DMAC controller of the device.
00492
00493     Description:
00494         Sets up a DMA controller for subsequent transfer activity.
00495

```

```

00496
00497     Parameters:
00498         none
00499
00500     Returns:
00501         void
00502
00503     Example:
00504         <code>
00505         DMAC_Initialize();
00506         </code>
00507 */
00508 void DMAC_Initialize( void );
00509
00510
00511 DMAC_TRANSFER_EVENT DMAC_ChannelTransferStatusGet( DMAC_CHANNEL channel);
00512 // DOM-IGNORE-BEGIN
00513 #ifdef __cplusplus // Provide C++ Compatibility
00514     }
00515 }
00516
00517 #endif
00518 // DOM-IGNORE-END
00519
00520 #endif //PLIB_DMACH

```

4.25 plib_evic.h

```

00001 /*****
00002  EVIC PLIB Header
00003
00004  Company:
00005      Microchip Technology Inc.
00006
00007  File Name:
00008      plib_evic.h
00009
00010  Summary:
00011      PIC32MZ Interrupt Module PLIB Header File
00012
00013  Description:
00014      None
00015
00016 *****/
00017
00018 // DOM-IGNORE-BEGIN
00019 /*****
00020  * Copyright (C) 2019 Microchip Technology Inc. and its subsidiaries.
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00040 *****/
00041 // DOM-IGNORE-END
00042
00043 #ifndef PLIB_EVIC_H
00044 #define PLIB_EVIC_H
00045
00046 // ****
00047 // ****
00048 // Section: Included Files
00049 // ****
00050 // ****
00051 #include <device.h>
00052 #include <stdint.h>
00053 #include <stdbool.h>
00054 #include <device.h>
00055

```

```
00056 // DOM-IGNORE-BEGIN
00057 #ifdef __cplusplus // Provide C++ Compatibility
00058
00059     extern "C" {
00060
00061 #endif
00062 // DOM-IGNORE-END
00063
00064
00065
00066 // *****
00067 // *****
00068 // Section: Data Types
00069 // *****
00070 // *****
00071
00072 typedef enum
00073 {
00074     INT_SOURCE_CORE_TIMER = _CORE_TIMER_VECTOR,
00075
00076     INT_SOURCE_CORE_SOFTWARE_0 = _CORE_SOFTWARE_0_VECTOR,
00077
00078     INT_SOURCE_CORE_SOFTWARE_1 = _CORE_SOFTWARE_1_VECTOR,
00079
00080     INT_SOURCE_EXTERNAL_0 = _EXTERNAL_0_VECTOR,
00081
00082     INT_SOURCE_TIMER_1 = _TIMER_1_VECTOR,
00083
00084     INT_SOURCE_INPUT_CAPTURE_1_ERROR = _INPUT_CAPTURE_1_ERROR_VECTOR,
00085
00086     INT_SOURCE_INPUT_CAPTURE_1 = _INPUT_CAPTURE_1_VECTOR,
00087
00088     INT_SOURCE_OUTPUT_COMPARE_1 = _OUTPUT_COMPARE_1_VECTOR,
00089
00090     INT_SOURCE_EXTERNAL_1 = _EXTERNAL_1_VECTOR,
00091
00092     INT_SOURCE_TIMER_2 = _TIMER_2_VECTOR,
00093
00094     INT_SOURCE_INPUT_CAPTURE_2_ERROR = _INPUT_CAPTURE_2_ERROR_VECTOR,
00095
00096     INT_SOURCE_INPUT_CAPTURE_2 = _INPUT_CAPTURE_2_VECTOR,
00097
00098     INT_SOURCE_OUTPUT_COMPARE_2 = _OUTPUT_COMPARE_2_VECTOR,
00099
00100     INT_SOURCE_EXTERNAL_2 = _EXTERNAL_2_VECTOR,
00101
00102     INT_SOURCE_TIMER_3 = _TIMER_3_VECTOR,
00103
00104     INT_SOURCE_INPUT_CAPTURE_3_ERROR = _INPUT_CAPTURE_3_ERROR_VECTOR,
00105
00106     INT_SOURCE_INPUT_CAPTURE_3 = _INPUT_CAPTURE_3_VECTOR,
00107
00108     INT_SOURCE_OUTPUT_COMPARE_3 = _OUTPUT_COMPARE_3_VECTOR,
00109
00110     INT_SOURCE_EXTERNAL_3 = _EXTERNAL_3_VECTOR,
00111
00112     INT_SOURCE_TIMER_4 = _TIMER_4_VECTOR,
00113
00114     INT_SOURCE_INPUT_CAPTURE_4_ERROR = _INPUT_CAPTURE_4_ERROR_VECTOR,
00115
00116     INT_SOURCE_INPUT_CAPTURE_4 = _INPUT_CAPTURE_4_VECTOR,
00117
00118     INT_SOURCE_OUTPUT_COMPARE_4 = _OUTPUT_COMPARE_4_VECTOR,
00119
00120     INT_SOURCE_EXTERNAL_4 = _EXTERNAL_4_VECTOR,
00121
00122     INT_SOURCE_TIMER_5 = _TIMER_5_VECTOR,
00123
00124     INT_SOURCE_INPUT_CAPTURE_5_ERROR = _INPUT_CAPTURE_5_ERROR_VECTOR,
00125
00126     INT_SOURCE_INPUT_CAPTURE_5 = _INPUT_CAPTURE_5_VECTOR,
00127
00128     INT_SOURCE_OUTPUT_COMPARE_5 = _OUTPUT_COMPARE_5_VECTOR,
00129
00130     INT_SOURCE_RTCC = _RTCC_VECTOR,
00131
00132     INT_SOURCE_FLASH_CONTROL = _FLASH_CONTROL_VECTOR,
00133
00134     INT_SOURCE_COMPARATOR_1 = _COMPARATOR_1_VECTOR,
00135
00136     INT_SOURCE_COMPARATOR_2 = _COMPARATOR_2_VECTOR,
00137
00138     INT_SOURCE_SPI1_FAULT = _SPI1_FAULT_VECTOR,
00139
00140     INT_SOURCE_SPI1_RX = _SPI1_RX_VECTOR,
00141
00142     INT_SOURCE_SPI1_TX = _SPI1_TX_VECTOR,
```

```
00143
00144     INT_SOURCE_UART1_FAULT = _UART1_FAULT_VECTOR,
00145
00146     INT_SOURCE_UART1_RX = _UART1_RX_VECTOR,
00147
00148     INT_SOURCE_UART1_TX = _UART1_TX_VECTOR,
00149
00150     INT_SOURCE_I2C1_BUS = _I2C1_BUS_VECTOR,
00151
00152     INT_SOURCE_I2C1_SLAVE = _I2C1_SLAVE_VECTOR,
00153
00154     INT_SOURCE_I2C1_MASTER = _I2C1_MASTER_VECTOR,
00155
00156     INT_SOURCE_CHANGE_NOTICE_A = _CHANGE_NOTICE_A_VECTOR,
00157
00158     INT_SOURCE_CHANGE_NOTICE_B = _CHANGE_NOTICE_B_VECTOR,
00159
00160     INT_SOURCE_CHANGE_NOTICE_C = _CHANGE_NOTICE_C_VECTOR,
00161
00162     INT_SOURCE_CHANGE_NOTICE_D = _CHANGE_NOTICE_D_VECTOR,
00163
00164     INT_SOURCE_CHANGE_NOTICE_E = _CHANGE_NOTICE_E_VECTOR,
00165
00166     INT_SOURCE_CHANGE_NOTICE_F = _CHANGE_NOTICE_F_VECTOR,
00167
00168     INT_SOURCE_CHANGE_NOTICE_G = _CHANGE_NOTICE_G_VECTOR,
00169
00170     INT_SOURCE_SPI2_FAULT = _SPI2_FAULT_VECTOR,
00171
00172     INT_SOURCE_SPI2_RX = _SPI2_RX_VECTOR,
00173
00174     INT_SOURCE_SPI2_TX = _SPI2_TX_VECTOR,
00175
00176     INT_SOURCE_UART2_FAULT = _UART2_FAULT_VECTOR,
00177
00178     INT_SOURCE_UART2_RX = _UART2_RX_VECTOR,
00179
00180     INT_SOURCE_UART2_TX = _UART2_TX_VECTOR,
00181
00182     INT_SOURCE_I2C2_BUS = _I2C2_BUS_VECTOR,
00183
00184     INT_SOURCE_I2C2_SLAVE = _I2C2_SLAVE_VECTOR,
00185
00186     INT_SOURCE_I2C2_MASTER = _I2C2_MASTER_VECTOR,
00187
00188     INT_SOURCE_CTMU = _CTMU_VECTOR,
00189
00190     INT_SOURCE_DMA0 = _DMA0_VECTOR,
00191
00192     INT_SOURCE_DMA1 = _DMA1_VECTOR,
00193
00194     INT_SOURCE_DMA2 = _DMA2_VECTOR,
00195
00196     INT_SOURCE_DMA3 = _DMA3_VECTOR,
00197
00198     INT_SOURCE_TIMER_6 = _TIMER_6_VECTOR,
00199
00200     INT_SOURCE_INPUT_CAPTURE_6_ERROR = _INPUT_CAPTURE_6_ERROR_VECTOR,
00201
00202     INT_SOURCE_INPUT_CAPTURE_6 = _INPUT_CAPTURE_6_VECTOR,
00203
00204     INT_SOURCE_OUTPUT_COMPARE_6 = _OUTPUT_COMPARE_6_VECTOR,
00205
00206     INT_SOURCE_TIMER_7 = _TIMER_7_VECTOR,
00207
00208     INT_SOURCE_INPUT_CAPTURE_7_ERROR = _INPUT_CAPTURE_7_ERROR_VECTOR,
00209
00210     INT_SOURCE_INPUT_CAPTURE_7 = _INPUT_CAPTURE_7_VECTOR,
00211
00212     INT_SOURCE_OUTPUT_COMPARE_7 = _OUTPUT_COMPARE_7_VECTOR,
00213
00214     INT_SOURCE_TIMER_8 = _TIMER_8_VECTOR,
00215
00216     INT_SOURCE_INPUT_CAPTURE_8_ERROR = _INPUT_CAPTURE_8_ERROR_VECTOR,
00217
00218     INT_SOURCE_INPUT_CAPTURE_8 = _INPUT_CAPTURE_8_VECTOR,
00219
00220     INT_SOURCE_OUTPUT_COMPARE_8 = _OUTPUT_COMPARE_8_VECTOR,
00221
00222     INT_SOURCE_TIMER_9 = _TIMER_9_VECTOR,
00223
00224     INT_SOURCE_INPUT_CAPTURE_9_ERROR = _INPUT_CAPTURE_9_ERROR_VECTOR,
00225
00226     INT_SOURCE_INPUT_CAPTURE_9 = _INPUT_CAPTURE_9_VECTOR,
00227
00228     INT_SOURCE_OUTPUT_COMPARE_9 = _OUTPUT_COMPARE_9_VECTOR,
00229
```

```
00230     INT_SOURCE_ADC = _ADC_VECTOR,
00231
00232     INT_SOURCE_ADC_DC1 = _ADC_DC1_VECTOR,
00233
00234     INT_SOURCE_ADC_DC2 = _ADC_DC2_VECTOR,
00235
00236     INT_SOURCE_ADC_DF1 = _ADC_DF1_VECTOR,
00237
00238     INT_SOURCE_ADC_DF2 = _ADC_DF2_VECTOR,
00239
00240     INT_SOURCE_ADC_DF3 = _ADC_DF3_VECTOR,
00241
00242     INT_SOURCE_ADC_DF4 = _ADC_DF4_VECTOR,
00243
00244     INT_SOURCE_ADC_FAULT = _ADC_FAULT_VECTOR,
00245
00246     INT_SOURCE_ADC_EOS = _ADC_EOS_VECTOR,
00247
00248     INT_SOURCE_ADC_ARDY = _ADC_ARDY_VECTOR,
00249
00250     INT_SOURCE_ADC_URDY = _ADC_URDY_VECTOR,
00251
00252     INT_SOURCE_ADC_DMA = _ADC_DMA_VECTOR,
00253
00254     INT_SOURCE_ADC_EARLY = _ADC_EARLY_VECTOR,
00255
00256     INT_SOURCE_ADC_DATA0 = _ADC_DATA0_VECTOR,
00257
00258     INT_SOURCE_ADC_DATA1 = _ADC_DATA1_VECTOR,
00259
00260     INT_SOURCE_ADC_DATA2 = _ADC_DATA2_VECTOR,
00261
00262     INT_SOURCE_ADC_DATA3 = _ADC_DATA3_VECTOR,
00263
00264     INT_SOURCE_ADC_DATA4 = _ADC_DATA4_VECTOR,
00265
00266     INT_SOURCE_ADC_DATA5 = _ADC_DATA5_VECTOR,
00267
00268     INT_SOURCE_ADC_DATA6 = _ADC_DATA6_VECTOR,
00269
00270     INT_SOURCE_ADC_DATA7 = _ADC_DATA7_VECTOR,
00271
00272     INT_SOURCE_ADC_DATA8 = _ADC_DATA8_VECTOR,
00273
00274     INT_SOURCE_ADC_DATA9 = _ADC_DATA9_VECTOR,
00275
00276     INT_SOURCE_ADC_DATA10 = _ADC_DATA10_VECTOR,
00277
00278     INT_SOURCE_ADC_DATA11 = _ADC_DATA11_VECTOR,
00279
00280     INT_SOURCE_ADC_DATA12 = _ADC_DATA12_VECTOR,
00281
00282     INT_SOURCE_ADC_DATA13 = _ADC_DATA13_VECTOR,
00283
00284     INT_SOURCE_ADC_DATA14 = _ADC_DATA14_VECTOR,
00285
00286     INT_SOURCE_ADC_DATA15 = _ADC_DATA15_VECTOR,
00287
00288     INT_SOURCE_ADC_DATA16 = _ADC_DATA16_VECTOR,
00289
00290     INT_SOURCE_ADC_DATA17 = _ADC_DATA17_VECTOR,
00291
00292     INT_SOURCE_ADC_DATA18 = _ADC_DATA18_VECTOR,
00293
00294     INT_SOURCE_ADC_DATA19 = _ADC_DATA19_VECTOR,
00295
00296     INT_SOURCE_ADC_DATA24 = _ADC_DATA24_VECTOR,
00297
00298     INT_SOURCE_ADC_DATA25 = _ADC_DATA25_VECTOR,
00299
00300     INT_SOURCE_ADC_DATA26 = _ADC_DATA26_VECTOR,
00301
00302     INT_SOURCE_ADC_DATA27 = _ADC_DATA27_VECTOR,
00303
00304     INT_SOURCE_ADC_DATA40 = _ADC_DATA40_VECTOR,
00305
00306     INT_SOURCE_ADC_DATA41 = _ADC_DATA41_VECTOR,
00307
00308     INT_SOURCE_ADC_DATA46 = _ADC_DATA46_VECTOR,
00309
00310     INT_SOURCE_ADC_DATA47 = _ADC_DATA47_VECTOR,
00311
00312     INT_SOURCE_ADC_DATA48 = _ADC_DATA48_VECTOR,
00313
00314     INT_SOURCE_ADC_DATA49 = _ADC_DATA49_VECTOR,
00315
00316     INT_SOURCE_ADC_DATA50 = _ADC_DATA50_VECTOR,
```



```

00317
00318     INT_SOURCE_ADC_DATA51 = _ADC_DATA51_VECTOR,
00319
00320     INT_SOURCE_ADC_DATA52 = _ADC_DATA52_VECTOR,
00321
00322     INT_SOURCE_ADC_DATA53 = _ADC_DATA53_VECTOR,
00323
00324     INT_SOURCE_COMPARATOR_3 = _COMPARATOR_3_VECTOR,
00325
00326     INT_SOURCE_COMPARATOR_4 = _COMPARATOR_4_VECTOR,
00327
00328     INT_SOURCE_COMPARATOR_5 = _COMPARATOR_5_VECTOR,
00329
00330     INT_SOURCE_CAN1 = _CAN1_VECTOR,
00331
00332     INT_SOURCE_QEI1 = _QEI1_VECTOR,
00333
00334     INT_SOURCE_QEI2 = _QEI2_VECTOR,
00335
00336     INT_SOURCE_PWM_PRI = _PWM_PRI_VECTOR,
00337
00338     INT_SOURCE_PWM_SEC = _PWM_SEC_VECTOR,
00339
00340     INT_SOURCE_PWM1 = _PWM1_VECTOR,
00341
00342     INT_SOURCE_PWM2 = _PWM2_VECTOR,
00343
00344     INT_SOURCE_PWM3 = _PWM3_VECTOR,
00345
00346     INT_SOURCE_PWM4 = _PWM4_VECTOR,
00347
00348     INT_SOURCE_PWM5 = _PWM5_VECTOR,
00349
00350     INT_SOURCE_PWM6 = _PWM6_VECTOR,
00351
00352     INT_SOURCE_DMA4 = _DMA4_VECTOR,
00353
00354     INT_SOURCE_DMA5 = _DMA5_VECTOR,
00355
00356     INT_SOURCE_DMA6 = _DMA6_VECTOR,
00357
00358     INT_SOURCE_DMA7 = _DMA7_VECTOR,
00359
00360     INT_SOURCE_QEI3 = _QEI3_VECTOR,
00361
00362     INT_SOURCE_SYSTEM_BUS_PROTECTION = _SYSTEM_BUS_PROTECTION_VECTOR,
00363
00364     INT_SOURCE_PWM7 = _PWM7_VECTOR,
00365
00366     INT_SOURCE_PWM8 = _PWM8_VECTOR,
00367
00368     INT_SOURCE_PWM9 = _PWM9_VECTOR,
00369
00370     INT_SOURCE_ADC_DC3 = _ADC_DC3_VECTOR,
00371
00372     INT_SOURCE_ADC_DC4 = _ADC_DC4_VECTOR,
00373
00374     INT_SOURCE_PCACHE = _PCACHE_VECTOR,
00375
00376     INT_SOURCE_CLC1 = _CLC1_VECTOR,
00377
00378     INT_SOURCE_CLC2 = _CLC2_VECTOR,
00379
00380     INT_SOURCE_CLC3 = _CLC3_VECTOR,
00381
00382     INT_SOURCE_CLC4 = _CLC4_VECTOR,
00383
00384     INT_SOURCE_CORE_PERF_COUNT = _CORE_PERF_COUNT_VECTOR,
00385
00386     INT_SOURCE_CORE_FAST_DEBUG_CHAN = _CORE_FAST_DEBUG_CHAN_VECTOR,
00387
00388 } INT_SOURCE;
00389
00390 typedef enum
00391 {
00392     EXTERNAL_INT_2 = _IEC0_INT2IE_MASK,
00393
00394
00395
00396 } EXTERNAL_INT_PIN;
00397
00398
00399 typedef void (*EXTERNAL_INT_PIN_CALLBACK) (EXTERNAL_INT_PIN pin, uintptr_t context);
00400
00401 // *****
00402 // *****
00403 // Section: Interface Routines

```

```

00404 // *****
00405 // *****
00406
00407 void EVIC_Initialize ( void );
00408
00409 void EVIC_SourceEnable( INT_SOURCE source );
00410
00411 void EVIC_SourceDisable( INT_SOURCE source );
00412
00413 bool EVIC_SourceIsEnabled( INT_SOURCE source );
00414
00415 bool EVIC_SourceStatusGet( INT_SOURCE source );
00416
00417 void EVIC_SourceStatusSet( INT_SOURCE source );
00418
00419 void EVIC_SourceStatusClear( INT_SOURCE source );
00420
00421 void EVIC_INT_Enable( void );
00422
00423 bool EVIC_INT_Disable( void );
00424
00425 void EVIC_INT_Restore( bool state );
00426
00427 bool EVIC_ExternalInterruptCallbackRegister(
00428     EXTERNAL_INT_PIN extIntPin,
00429     const EXTERNAL_INT_PIN_CALLBACK callback,
00430     uintptr_t context
00431 );
00432
00433 void EVIC_ExternalInterruptEnable( EXTERNAL_INT_PIN extIntPin );
00434
00435 void EVIC_ExternalInterruptDisable( EXTERNAL_INT_PIN extIntPin );
00436
00437 typedef struct {
00438
00439     /* Callback for event on target pin*/
00440     EXTERNAL_INT_PIN_CALLBACK    callback;
00441
00442     /* Callback Context */
00443     uintptr_t                    context;
00444 } EXT_INT_PIN_CALLBACK_OBJ;
00445
00446 // DOM-IGNORE-BEGIN
00447 #ifdef __cplusplus // Provide C++ Compatibility
00448
00449 }
00450
00451 #endif
00452 // DOM-IGNORE-END
00453 #endif // PLIB_EVIC_H

```

4.26 plib_gpio.h

```

00001 /*****
00002  GPIO PLIB
00003
00004  Company:
00005      Microchip Technology Inc.
00006
00007  File Name:
00008      plib_gpio.h UUUUUUUUU
00009
00010  Summary:
00011      GPIO PLIB Header File
00012
00013  Description:
00014      This library provides an interface to control and interact with Parallel
00015      Input/Output controller (GPIO) module.
00016
00017  *****/
00018
00019 /*****
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00021  *
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00039 * THAT YOU HAVE PAID DIRECTLY TO MICROCHIP FOR THIS SOFTWARE.
00040 *****/
00041
00042 #ifndef PLIB_GPIO_H
00043 #define PLIB_GPIO_H
00044
00045 #include <device.h>
00046 #include <stdint.h>
00047 #include <stdbool.h>
00048 #include <stddef.h>
00049
00050 // DOM-IGNORE-BEGIN
00051 #ifdef __cplusplus // Provide C++ Compatibility
00052
00053     extern "C" {
00054
00055 #endif
00056 // DOM-IGNORE-END
00057
00058 // ****
00059 // ****
00060 // Section: Data types and constants
00061 // ****
00062 // ****
00063
00064
00065 /*** Macros for SW4 pin ***/
00066 #define SW4_Set() (LATASET = (1U<7))
00067 #define SW4_Clear() (LATACLAR = (1U<7))
00068 #define SW4_Toggle() (LATAINV= (1U<7))
00069 #define SW4_OutputEnable() (TRISACLR = (1U<7))
00070 #define SW4_InputEnable() (TRISASET = (1U<7))
00071 #define SW4_Get() ((PORTA >> 7) & 0x1U)
00072 #define SW4_PIN GPIO_PIN_RA7
00073 #define SW4_InterruptEnable() (CNENASET = (1U<7))
00074 #define SW4_InterruptDisable() (CNENACLR = (1U<7))
00075
00076 /*** Macros for BUZZER pin ***/
00077 #define BUZZER_Get() ((PORTB >> 15) & 0x1U)
00078 #define BUZZER_PIN GPIO_PIN_RB15
00079
00080 /*** Macros for EXT_SW2 pin ***/
00081 #define EXT_SW2_Set() (LATASET = (1U<12))
00082 #define EXT_SW2_Clear() (LATACLAR = (1U<12))
00083 #define EXT_SW2_Toggle() (LATAINV= (1U<12))
00084 #define EXT_SW2_OutputEnable() (TRISACLR = (1U<12))
00085 #define EXT_SW2_InputEnable() (TRISASET = (1U<12))
00086 #define EXT_SW2_Get() ((PORTA >> 12) & 0x1U)
00087 #define EXT_SW2_PIN GPIO_PIN_RA12
00088
00089 /*** Macros for DIS_CS pin ***/
00090 #define DIS_CS_Set() (LATASET = (1U<11))
00091 #define DIS_CS_Clear() (LATACLAR = (1U<11))
00092 #define DIS_CS_Toggle() (LATAINV= (1U<11))
00093 #define DIS_CS_OutputEnable() (TRISACLR = (1U<11))
00094 #define DIS_CS_InputEnable() (TRISASET = (1U<11))
00095 #define DIS_CS_Get() ((PORTA >> 11) & 0x1U)
00096 #define DIS_CS_PIN GPIO_PIN_RA11
00097
00098 /*** Macros for TP3 pin ***/
00099 #define TP3_Set() (LATBSET = (1U<0))
00100 #define TP3_Clear() (LATBCLR = (1U<0))
00101 #define TP3_Toggle() (LATBINV= (1U<0))
00102 #define TP3_OutputEnable() (TRISBCLR = (1U<0))
00103 #define TP3_InputEnable() (TRISBSET = (1U<0))
00104 #define TP3_Get() ((PORTB >> 0) & 0x1U)
00105 #define TP3_PIN GPIO_PIN_RB0
00106
00107 /*** Macros for TP2 pin ***/
00108 #define TP2_Set() (LATCSET = (1U<2))
00109 #define TP2_Clear() (LATCCLR = (1U<2))
00110 #define TP2_Toggle() (LATCINV= (1U<2))
00111 #define TP2_OutputEnable() (TRISCCLR = (1U<2))
00112 #define TP2_InputEnable() (TRISCSET = (1U<2))
00113 #define TP2_Get() ((PORTC >> 2) & 0x1U)
00114 #define TP2_PIN GPIO_PIN_RC2
00115

```

```

00116 /*** Macros for TP1 pin ***/
00117 #define TP1_Set() (LATCSET = (1U<<11))
00118 #define TP1_Clear() (LATCCLR = (1U<<11))
00119 #define TP1_Toggle() (LATCINV= (1U<<11))
00120 #define TP1_OutputEnable() (TRISCLR = (1U<<11))
00121 #define TP1_InputEnable() (TRISSET = (1U<<11))
00122 #define TP1_Get() ((PORTC >> 11) & 0x1U)
00123 #define TP1_PIN GPIO_PIN_RC11
00124
00125 /*** Macros for LED_GREEN pin ***/
00126 #define LED_GREEN_Set() (LATASET = (1U<<4))
00127 #define LED_GREEN_Clear() (LATACLR = (1U<<4))
00128 #define LED_GREEN_Toggle() (LATAINV= (1U<<4))
00129 #define LED_GREEN_OutputEnable() (TRISACL = (1U<<4))
00130 #define LED_GREEN_InputEnable() (TRISASET = (1U<<4))
00131 #define LED_GREEN_Get() ((PORTA >> 4) & 0x1U)
00132 #define LED_GREEN_PIN GPIO_PIN_RA4
00133
00134 /*** Macros for LED_RED pin ***/
00135 #define LED_RED_Set() (LATCSET = (1U<<15))
00136 #define LED_RED_Clear() (LATCCLR = (1U<<15))
00137 #define LED_RED_Toggle() (LATCINV= (1U<<15))
00138 #define LED_RED_OutputEnable() (TRISCLR = (1U<<15))
00139 #define LED_RED_InputEnable() (TRISSET = (1U<<15))
00140 #define LED_RED_Get() ((PORTC >> 15) & 0x1U)
00141 #define LED_RED_PIN GPIO_PIN_RC15
00142
00143 /*** Macros for SW2 pin ***/
00144 #define SW2_Set() (LATDSET = (1U<<8))
00145 #define SW2_Clear() (LATDCLR = (1U<<8))
00146 #define SW2_Toggle() (LATDINV= (1U<<8))
00147 #define SW2_OutputEnable() (TRISDCLR = (1U<<8))
00148 #define SW2_InputEnable() (TRISDSET = (1U<<8))
00149 #define SW2_Get() ((PORTD >> 8) & 0x1U)
00150 #define SW2_PIN GPIO_PIN_RD8
00151 #define SW2_InterruptEnable() (CNENDSET = (1U<<8))
00152 #define SW2_InterruptDisable() (CNENDCLR = (1U<<8))
00153
00154 /*** Macros for DIS_MODE pin ***/
00155 #define DIS_MODE_Set() (LATBSET = (1U<<6))
00156 #define DIS_MODE_Clear() (LATBCLR = (1U<<6))
00157 #define DIS_MODE_Toggle() (LATBINV= (1U<<6))
00158 #define DIS_MODE_OutputEnable() (TRISBCLR = (1U<<6))
00159 #define DIS_MODE_InputEnable() (TRISBSET = (1U<<6))
00160 #define DIS_MODE_Get() ((PORTB >> 6) & 0x1U)
00161 #define DIS_MODE_PIN GPIO_PIN_RB6
00162
00163 /*** Macros for SW3 pin ***/
00164 #define SW3_Set() (LATBSET = (1U<<8))
00165 #define SW3_Clear() (LATBCLR = (1U<<8))
00166 #define SW3_Toggle() (LATBINV= (1U<<8))
00167 #define SW3_OutputEnable() (TRISBCLR = (1U<<8))
00168 #define SW3_InputEnable() (TRISBSET = (1U<<8))
00169 #define SW3_Get() ((PORTB >> 8) & 0x1U)
00170 #define SW3_PIN GPIO_PIN_RB8
00171
00172 /*** Macros for DIS_RESET pin ***/
00173 #define DIS_RESET_Set() (LATBSET = (1U<<9))
00174 #define DIS_RESET_Clear() (LATBCLR = (1U<<9))
00175 #define DIS_RESET_Toggle() (LATBINV= (1U<<9))
00176 #define DIS_RESET_OutputEnable() (TRISBCLR = (1U<<9))
00177 #define DIS_RESET_InputEnable() (TRISBSET = (1U<<9))
00178 #define DIS_RESET_Get() ((PORTB >> 9) & 0x1U)
00179 #define DIS_RESET_PIN GPIO_PIN_RB9
00180
00181 /*** Macros for IMU_CS pin ***/
00182 #define IMU_CS_Set() (LATCSET = (1U<<9))
00183 #define IMU_CS_Clear() (LATCCLR = (1U<<9))
00184 #define IMU_CS_Toggle() (LATCINV= (1U<<9))
00185 #define IMU_CS_OutputEnable() (TRISCLR = (1U<<9))
00186 #define IMU_CS_InputEnable() (TRISSET = (1U<<9))
00187 #define IMU_CS_Get() ((PORTC >> 9) & 0x1U)
00188 #define IMU_CS_PIN GPIO_PIN_RC9
00189
00190 /*** Macros for MISC_0 pin ***/
00191 #define MISC_0_Set() (LATBSET = (1U<<11))
00192 #define MISC_0_Clear() (LATBCLR = (1U<<11))
00193 #define MISC_0_Toggle() (LATBINV= (1U<<11))
00194 #define MISC_0_OutputEnable() (TRISBCLR = (1U<<11))
00195 #define MISC_0_InputEnable() (TRISBSET = (1U<<11))
00196 #define MISC_0_Get() ((PORTB >> 11) & 0x1U)
00197 #define MISC_0_PIN GPIO_PIN_RB11
00198
00199 /*** Macros for DIS_BLANK pin ***/
00200 #define DIS_BLANK_Set() (LATBSET = (1U<<12))
00201 #define DIS_BLANK_Clear() (LATBCLR = (1U<<12))
00202 #define DIS_BLANK_Toggle() (LATBINV= (1U<<12))

```

```

00203 #define DIS_BLANK_OutputEnable()      (TRISBCLR = (1U<<12))
00204 #define DIS_BLANK_InputEnable()       (TRISBSET = (1U<<12))
00205 #define DIS_BLANK_Get()                ((PORTB >> 12) & 0x1U)
00206 #define DIS_BLANK_PIN                  GPIO_PIN_RB12
00207
00208 /*** Macros for SW5 pin ***/
00209 #define SW5_Set()                      (LATASET = (1U<<10))
00210 #define SW5_Clear()                   (LATACLR = (1U<<10))
00211 #define SW5_Toggle()                  (LATAINV= (1U<<10))
00212 #define SW5_OutputEnable()            (TRISACLR = (1U<<10))
00213 #define SW5_InputEnable()             (TRISASET = (1U<<10))
00214 #define SW5_Get()                    ((PORTA >> 10) & 0x1U)
00215 #define SW5_PIN                       GPIO_PIN_RA10
00216 #define SW5_InterruptEnable()         (CNENASET = (1U<<10))
00217 #define SW5_InterruptDisable()        (CNENACLR = (1U<<10))
00218
00219
00220 // *****
00221 /* GPIO Port
00222
00223     Summary:
00224         Identifies the available GPIO Ports.
00225
00226     Description:
00227         This enumeration identifies the available GPIO Ports.
00228
00229     Remarks:
00230         The caller should not rely on the specific numbers assigned to any of
00231         these values as they may change from one processor to the next.
00232
00233         Not all ports are available on all devices. Refer to the specific
00234         device data sheet to determine which ports are supported.
00235 */
00236
00237 #define GPIO_PORT_A    (0)
00238 #define GPIO_PORT_B    (1)
00239 #define GPIO_PORT_C    (2)
00240 #define GPIO_PORT_D    (3)
00241 #define GPIO_PORT_E    (4)
00242 typedef uint32_t GPIO_PORT;
00243
00244 typedef enum
00245 {
00246     GPIO_INTERRUPT_ON_MISMATCH,
00247     GPIO_INTERRUPT_ON_RISING_EDGE,
00248     GPIO_INTERRUPT_ON_FALLING_EDGE,
00249     GPIO_INTERRUPT_ON_BOTH_EDGES,
00250 }GPIO_INTERRUPT_STYLE;
00251
00252 // *****
00253 /* GPIO Port Pins
00254
00255     Summary:
00256         Identifies the available GPIO port pins.
00257
00258     Description:
00259         This enumeration identifies the available GPIO port pins.
00260
00261     Remarks:
00262         The caller should not rely on the specific numbers assigned to any of
00263         these values as they may change from one processor to the next.
00264
00265         Not all pins are available on all devices. Refer to the specific
00266         device data sheet to determine which pins are supported.
00267 */
00268
00269
00270 #define GPIO_PIN_RA0    (0U)
00271 #define GPIO_PIN_RA1    (1U)
00272 #define GPIO_PIN_RA4    (4U)
00273 #define GPIO_PIN_RA7    (7U)
00274 #define GPIO_PIN_RA8    (8U)
00275 #define GPIO_PIN_RA10   (10U)
00276 #define GPIO_PIN_RA11   (11U)
00277 #define GPIO_PIN_RA12   (12U)
00278 #define GPIO_PIN_RB0    (16U)
00279 #define GPIO_PIN_RB1    (17U)
00280 #define GPIO_PIN_RB2    (18U)
00281 #define GPIO_PIN_RB3    (19U)
00282 #define GPIO_PIN_RB4    (20U)
00283 #define GPIO_PIN_RB5    (21U)
00284 #define GPIO_PIN_RB6    (22U)
00285 #define GPIO_PIN_RB7    (23U)
00286 #define GPIO_PIN_RB8    (24U)
00287 #define GPIO_PIN_RB9    (25U)
00288 #define GPIO_PIN_RB10   (26U)
00289 #define GPIO_PIN_RB11   (27U)

```

```

00290 #define      GPIO_PIN_RB12   (28U)
00291 #define      GPIO_PIN_RB13   (29U)
00292 #define      GPIO_PIN_RB14   (30U)
00293 #define      GPIO_PIN_RB15   (31U)
00294 #define      GPIO_PIN_RC0    (32U)
00295 #define      GPIO_PIN_RC1    (33U)
00296 #define      GPIO_PIN_RC2    (34U)
00297 #define      GPIO_PIN_RC6    (38U)
00298 #define      GPIO_PIN_RC7    (39U)
00299 #define      GPIO_PIN_RC8    (40U)
00300 #define      GPIO_PIN_RC9    (41U)
00301 #define      GPIO_PIN_RC10   (42U)
00302 #define      GPIO_PIN_RC11   (43U)
00303 #define      GPIO_PIN_RC12   (44U)
00304 #define      GPIO_PIN_RC13   (45U)
00305 #define      GPIO_PIN_RC15   (47U)
00306 #define      GPIO_PIN_RD8    (56U)
00307
00308     /* This element should not be used in any of the GPIO APIs.
00309        It will be used by other modules or application to denote that none of the GPIO Pin is used */
00310 #define      GPIO_PIN_NONE    (-1)
00311
00312 typedef uint32_t GPIO_PIN;
00313
00314 typedef void (*GPIO_PIN_CALLBACK) ( GPIO_PIN pin, uintptr_t context);
00315
00316 void GPIO_Initialize(void);
00317
00318 // *****
00319 // *****
00320 // Section: GPIO Functions which operates on multiple pins of a port
00321 // *****
00322 // *****
00323
00324 uint32_t GPIO_PortRead(GPIO_PORT port);
00325
00326 void GPIO_PortWrite(GPIO_PORT port, uint32_t mask, uint32_t value);
00327
00328 uint32_t GPIO_PortLatchRead ( GPIO_PORT port );
00329
00330 void GPIO_PortSet(GPIO_PORT port, uint32_t mask);
00331
00332 void GPIO_PortClear(GPIO_PORT port, uint32_t mask);
00333
00334 void GPIO_PortToggle(GPIO_PORT port, uint32_t mask);
00335
00336 void GPIO_PortInputEnable(GPIO_PORT port, uint32_t mask);
00337
00338 void GPIO_PortOutputEnable(GPIO_PORT port, uint32_t mask);
00339
00340 void GPIO_PortInterruptEnable(GPIO_PORT port, uint32_t mask);
00341
00342 void GPIO_PortInterruptDisable(GPIO_PORT port, uint32_t mask);
00343
00344 // *****
00345 // *****
00346 // Section: Local Data types and Prototypes
00347 // *****
00348 // *****
00349
00350 typedef struct {
00351
00352     /* target pin */
00353     GPIO_PIN                pin;
00354
00355     /* Callback for event on target pin*/
00356     GPIO_PIN_CALLBACK        callback;
00357
00358     /* Callback Context */
00359     uintptr_t                context;
00360 } GPIO_PIN_CALLBACK_OBJ;
00361
00362 // *****
00363 // *****
00364 // Section: GPIO Functions which operates on one pin at a time
00365 // *****
00366 // *****
00367
00368 static inline void GPIO_PinWrite(GPIO_PIN pin, bool value)
00369 {
00370     uint32_t xvalue = (uint32_t)value;
00371     GPIO_PortWrite((pin>>4U), (uint32_t)(0x1U) << (pin & 0xFU), (xvalue) << (pin & 0xFU));
00372 }
00373
00374 static inline bool GPIO_PinRead(GPIO_PIN pin)
00375 {
00376

```

```

00377     return (((GPIO_PortRead((GPIO_PORT)(pin»4U))) » (pin & 0xFU)) & 0x1U) != 0U);
00378 }
00379
00380 static inline bool GPIO_PinLatchRead(GPIO_PIN pin)
00381 {
00382     return (((GPIO_PortLatchRead((GPIO_PORT)(pin»4U))) » (pin & 0xFU)) & 0x1U) != 0U);
00383 }
00384
00385 static inline void GPIO_PinToggle(GPIO_PIN pin)
00386 {
00387     GPIO_PortToggle((pin»4U), (uint32_t)0x1U « (pin & 0xFU));
00388 }
00389
00390 static inline void GPIO_PinSet(GPIO_PIN pin)
00391 {
00392     GPIO_PortSet((pin»4U), (uint32_t)0x1U « (pin & 0xFU));
00393 }
00394
00395 static inline void GPIO_PinClear(GPIO_PIN pin)
00396 {
00397     GPIO_PortClear((pin»4U), (uint32_t)0x1U « (pin & 0xFU));
00398 }
00399
00400 static inline void GPIO_PinInputEnable(GPIO_PIN pin)
00401 {
00402     GPIO_PortInputEnable((pin»4U), (uint32_t)0x1U « (pin & 0xFU));
00403 }
00404
00405 static inline void GPIO_PinOutputEnable(GPIO_PIN pin)
00406 {
00407     GPIO_PortOutputEnable((pin»4U), (uint32_t)0x1U « (pin & 0xFU));
00408 }
00409
00410 #define GPIO_PinInterruptEnable(pin)        GPIO_PinIntEnable(pin, GPIO_INTERRUPT_ON_MISMATCH)
00411 #define GPIO_PinInterruptDisable(pin)       GPIO_PinIntDisable(pin)
00412
00413 void GPIO_PinIntEnable(GPIO_PIN pin, GPIO_INTERRUPT_STYLE style);
00414 void GPIO_PinIntDisable(GPIO_PIN pin);
00415
00416 bool GPIO_PinInterruptCallbackRegister(
00417     GPIO_PIN pin,
00418     const GPIO_PIN_CALLBACK callBack,
00419     uintptr_t context
00420 );
00421
00422 // DOM-IGNORE-BEGIN
00423 #ifdef __cplusplus // Provide C++ Compatibility
00424     }
00425 #endif
00426 // DOM-IGNORE-END
00427 #endif
00428 // DOM-IGNORE-END
00429 #endif // PLIB_GPIO_H

```

4.27 plib_mcpwm.h

```

00001 /*****
00002  Motor Control PWM (MCPWM) Peripheral Library Interface Header File
00003
00004  Company:
00005  Microchip Technology Inc.
00006
00007  File Name:
00008  plib_mcpwm.h
00009
00010  Summary:
00011  MCPWM PLIB Header File
00012
00013  Description:
00014  None
00015
00016 *****/
00017
00018 /*****
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00038 * THAT YOU HAVE PAID DIRECTLY TO MICROCHIP FOR THIS SOFTWARE.
00039 *****/
00040
00041 #ifndef _PLIB_MCPWM_H
00042 #define _PLIB_MCPWM_H
00043
00044 #include <stddef.h>
00045 #include <stdbool.h>
00046 #include <stdint.h>
00047 #include "device.h"
00048 #include "plib_mcpwm_common.h"
00049
00050 // DOM-IGNORE-BEGIN
00051 #ifdef __cplusplus // Provide C++ Compatibility
00052     extern "C" {
00053 #endif
00054 // DOM-IGNORE-END
00055
00056
00057 // *****
00058 // Section: Interface
00059 // *****
00060 // *****
00061
00062 // ***** MCPWM API *****/
00063 // *****
00064
00065 void MCPWM_Initialize (void);
00066
00067 void MCPWM_Start (void);
00068
00069 void MCPWM_Stop (void);
00070
00071 void MCPWM_PrimaryPeriodSet (uint16_t period);
00072
00073 uint16_t MCPWM_PrimaryPeriodGet (void);
00074
00075 void MCPWM_SecondaryPeriodSet (uint16_t period);
00076
00077 uint16_t MCPWM_SecondaryPeriodGet (void);
00078
00079 void MCPWM_ChannelPrimaryDutySet (MCPWM_CH_NUM channel, uint16_t duty);
00080
00081 void MCPWM_ChannelSecondaryDutySet (MCPWM_CH_NUM channel, uint16_t duty);
00082
00083 void MCPWM_ChannelDeadTimeSet (MCPWM_CH_NUM channel, uint16_t high_deadtime, uint16_t low_deadtime);
00084
00085 void MCPWM_ChannelPrimaryTriggerSet (MCPWM_CH_NUM channel, uint16_t trigger);
00086
00087 void MCPWM_ChannelSecondaryTriggerSet (MCPWM_CH_NUM channel, uint16_t trigger);
00088
00089 void MCPWM_ChannelLeadingEdgeBlankingDelaySet (MCPWM_CH_NUM channel, uint16_t delay);
00090
00091 void MCPWM_ChannelPinsOverrideEnable (MCPWM_CH_NUM channel);
00092
00093 void MCPWM_ChannelPinsOverrideDisable (MCPWM_CH_NUM channel);
00094
00095 void MCPWM_ChannelPinsOwnershipEnable (MCPWM_CH_NUM channel);
00096
00097 void MCPWM_ChannelPinsOwnershipDisable (MCPWM_CH_NUM channel);
00098
00099
00100
00101
00102
00103 // DOM-IGNORE-BEGIN
00104 #ifdef __cplusplus // Provide C++ Compatibility
00105     }
00106 #endif
00107
00108 // DOM-IGNORE-END
00109 #endif // _PLIB_MCPWM_H
```


4.28 plib_mcpwm_common.h

```

00001 /*****
00002  Motor Control PWM (MCPWM) Peripheral Library Interface Header File
00003
00004  Company
00005      Microchip Technology Inc.
00006
00007  File Name
00008      plib_mcpwm_common.h
00009
00010  Summary
00011      Data Type definition of the MCPWM Peripheral Interface Plib.
00012
00013  Description
00014      This file defines the Data Types for the MCPWM Plib.
00015
00016  Remarks:
00017      None.
00018
00019 *****/
00020 // DOM-IGNORE-BEGIN
00021 /*****
00022  * Copyright (C) 2019 Microchip Technology Inc. and its subsidiaries.
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00041  * THAT YOU HAVE PAID DIRECTLY TO MICROCHIP FOR THIS SOFTWARE.
00042  *****/
00043 // DOM-IGNORE-END
00044
00045 #ifndef PLIB_MCPWM_COMMON_H    // Guards against multiple inclusion
00046 #define PLIB_MCPWM_COMMON_H
00047
00048 // DOM-IGNORE-BEGIN
00049 #ifdef __cplusplus // Provide C++ Compatibility
00050
00051     extern "C" {
00052
00053 #endif
00054 // DOM-IGNORE-END
00055
00056 // ****
00057 // ****
00058 // Section: Data Types
00059 // ****
00060 // ****
00061
00062 /* The following data type definitions are used by the functions in this
00063    interface.
00064 */
00065
00066 // ****
00067
00068
00069 typedef enum
00070 {
00071     MCPWM_CH_1 ,
00072     MCPWM_CH_2 ,
00073     MCPWM_CH_3 ,
00074     MCPWM_CH_4 ,
00075     MCPWM_CH_5 ,
00076     MCPWM_CH_6 ,
00077 } MCPWM_CH_NUM;
00078
00079 typedef enum
00080 {
00081     MCPWM_NONE = 0,
00082     MCPWM_PWMH = _PWMCON1_PWMHIF_MASK,
00083     MCPWM_PWML = _PWMCON1_PWMLIF_MASK,
00084     MCPWM_TRG = _PWMCON1_TRGIF_MASK,

```

```

00085     MCPWM_CL = _PWMCON1_CLIF_MASK,
00086     MCPWM_FLT = _PWMCON1_FLTIF_MASK,
00087     MCPWM_STATUS_MASK = _PWMCON1_PWMHIF_MASK | _PWMCON1_PWMLIF_MASK | _PWMCON1_TRGIF_MASK |
    _PWMCON1_CLIF_MASK | _PWMCON1_FLTIF_MASK
00088 }MCPWM_CH_STATUS;
00089
00090 typedef void (*MCPWM_CH_CALLBACK) (MCPWM_CH_STATUS status, uintptr_t context);
00091
00092 typedef void (*MCPWM_CALLBACK) (uintptr_t context);
00093
00094 // *****
00095 // *****
00096 // Section: Local: **** Do Not Use ****
00097 // *****
00098 // *****
00099
00100 typedef struct
00101 {
00102     MCPWM_CH_CALLBACK callback;
00103     uintptr_t context;
00104 } MCPWM_CH_OBJECT ;
00105
00106 typedef struct
00107 {
00108     MCPWM_CALLBACK callback;
00109     uintptr_t context;
00110 } MCPWM_OBJECT ;
00111
00112 // DOM-IGNORE-BEGIN
00113 #ifdef __cplusplus // Provide C++ Compatibility
00114 }
00115 #endif
00116 // DOM-IGNORE-END
00117 #endif
00118 // DOM-IGNORE-END
00119 #endif // PLIB_ACC_COMMON_H

```

4.29 plib_qei2.h

```

00001 /*****
00002 Quadrature Encoder Interface (QEI) Peripheral Library Interface Header File
00003
00004 Company:
00005     Microchip Technology Inc.
00006
00007 File Name:
00008     plib_qei2.h
00009
00010 Summary:
00011     QEI PLIB Header File
00012
00013 Description:
00014     None
00015
00016 *****/
00017
00018 /*****
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00039 *****/
00040
00041 #ifndef _PLIB_QEI2_H
00042 #define _PLIB_QEI2_H
00043
00044 #include <stddef.h>

```

```

00045 #include <stdbool.h>
00046 #include <stdint.h>
00047 #include "device.h"
00048 #include "plib_qei_common.h"
00049
00050 // DOM-IGNORE-BEGIN
00051 #ifdef __cplusplus // Provide C++ Compatibility
00052     extern "C" {
00053 #endif
00054 // DOM-IGNORE-END
00055
00056
00057 // *****
00058 // Section: Interface
00059 // *****
00060 // *****
00061
00062 // ***** QEI2 API *****/
00063 // *****
00064
00065 void QEI2_Initialize (void);
00066
00067 void QEI2_Start(void);
00068
00069 void QEI2_Stop(void);
00070
00071 __STATIC_INLINE uint32_t QEI2_PositionGet(void)
00072 {
00073     return (POS2CNT);
00074 }
00075
00076 __STATIC_INLINE uint32_t QEI2_VelocityGet(void)
00077 {
00078     return (VEL2CNT);
00079 }
00080
00081 __STATIC_INLINE uint32_t QEI2_RevolutionsGet(void)
00082 {
00083     return (INDX2CNT);
00084 }
00085
00086 uint32_t QEI2_PulseIntervalGet(void);
00087
00088 void QEI2_PositionWindowSet(uint32_t high_threshold, uint32_t low_threshold);
00089
00090 void QEI2_PositionCountSet(uint32_t position_count);
00091
00092 void QEI2_VelocityCountSet(uint32_t velocity_count);
00093
00094
00095
00096 // DOM-IGNORE-BEGIN
00097 #ifdef __cplusplus // Provide C++ Compatibility
00098     }
00099 #endif
00100
00101 // DOM-IGNORE-END
00102 #endif // _PLIB_QEI2_H

```

4.30 plib_qei_common.h

```

00001 /*****
00002  Quadrature Encoder Interface (QEI) Peripheral Library Interface Header File
00003
00004  Company
00005      Microchip Technology Inc.
00006
00007  File Name
00008      plib_qei_common.h
00009
00010  Summary
00011      Data Type definition of the QEI Peripheral Interface Plib.
00012
00013  Description
00014      This file defines the Data Types for the QEI Plib.
00015
00016  Remarks:
00017      None.
00018
00019 *****/
00020 // DOM-IGNORE-BEGIN
00021 /*****
00022  * Copyright (C) 2019 Microchip Technology Inc. and its subsidiaries.

```

```

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00042 *****/
00043 // DOM-IGNORE-END
00044
00045 #ifndef PLIB_QEI_COMMON_H // Guards against multiple inclusion
00046 #define PLIB_QEI_COMMON_H
00047
00048 // DOM-IGNORE-BEGIN
00049 #ifdef __cplusplus // Provide C++ Compatibility
00050
00051     extern "C" {
00052
00053 #endif
00054 // DOM-IGNORE-END
00055
00056 // *****
00057 // *****
00058 // Section: Data Types
00059 // *****
00060 // *****
00061
00062 /* The following data type definitions are used by the functions in this
00063 interface.
00064 */
00065
00066 // *****
00067
00068
00069 typedef enum
00070 {
00071     QEI_NONE = 0,
00072     QEI_INDEX = _QEI1STAT_IDXIEN_MASK,
00073     QEI_HOME = _QEI1STAT_HOMIEN_MASK,
00074     QEI_VELOCITY_OVERFLOW = _QEI1STAT_VELOVIEN_MASK,
00075     QEI_POSITION_OVERFLOW = _QEI1STAT_POSOVIEN_MASK,
00076     QEI_POS_INIT_COMPLETE = _QEI1STAT_PCIIEN_MASK,
00077     QEI_POS_LESS_EQ = _QEI1STAT_PCLEQIEN_MASK,
00078     QEI_POS_HIGH_EQ = _QEI1STAT_PCHEQIEN_MASK
00079 }QEI_INTERRUPT;
00080
00081 typedef enum
00082 {
00083     QEI_STATUS_NONE = 0,
00084     QEI_INDEX_STATUS = _QEI1STAT_IDXIRQ_MASK,
00085     QEI_HOME_STATUS = _QEI1STAT_HOMIRQ_MASK,
00086     QEI_VELOCITY_OVERFLOW_STATUS = _QEI1STAT_VELOVIRO_MASK,
00087     QEI_POSITION_OVERFLOW_STATUS = _QEI1STAT_POSOVIRO_MASK,
00088     QEI_POS_INIT_COMPLETE_STATUS = _QEI1STAT_PCIIRO_MASK,
00089     QEI_POS_LESS_EQ_STATUS = _QEI1STAT_PCLEQIRO_MASK,
00090     QEI_POS_HIGH_EQ_STATUS = _QEI1STAT_PCHEQIRO_MASK,
00091     QEI_STATUS_MASK = QEI_INDEX_STATUS | QEI_HOME_STATUS | QEI_VELOCITY_OVERFLOW_STATUS |
    QEI_POSITION_OVERFLOW_STATUS
00092     | QEI_POSITION_OVERFLOW_STATUS | QEI_POS_INIT_COMPLETE_STATUS |
    QEI_POS_LESS_EQ_STATUS | QEI_POS_HIGH_EQ_STATUS
00093 }QEI_STATUS;
00094
00095 typedef void (*QEI_CALLBACK) (QEI_STATUS status, uintptr_t context);
00096
00097
00098 // *****
00099 // *****
00100 // Section: Local: **** Do Not Use ****
00101 // *****
00102 // *****
00103
00104 typedef struct
00105 {
00106     QEI_CALLBACK callback;
00107     uintptr_t context;

```

```

00108
00109 } QEI_CH_OBJECT ;
00110
00111
00112
00113 // DOM-IGNORE-BEGIN
00114 #ifdef __cplusplus // Provide C++ Compatibility
00115 }
00116 #endif
00117 // DOM-IGNORE-END
00118 #endif // PLIB_ACC_COMMON_H

```

4.31 plib_rtcc.h

```

00001 /*****
00002   Real Time Counter (RTCC) PLIB
00003
00004   Company:
00005     Microchip Technology Inc.
00006
00007   File Name:
00008     plib_rtcc.h
00009
00010   Summary:
00011     RTCC PLIB Header file
00012
00013   Description:
00014     This file defines the interface to the RTC peripheral library. This
00015     library provides access to and control of the associated peripheral
00016     instance.
00017
00018   *****/
00019
00020 // DOM-IGNORE-BEGIN
00021 /*****
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00042   *****/
00043 // DOM-IGNORE-END
00044
00045 #ifndef PLIB_RTCC_H
00046 #define PLIB_RTCC_H
00047
00048 // ****
00049 // ****
00050 // Section: Included Files
00051 // ****
00052 // ****
00053
00054 #include <stddef.h>
00055 #include <stdbool.h>
00056 #include <stdint.h>
00057 #include "device.h"
00058 #include <time.h>
00059
00060 // DOM-IGNORE-BEGIN
00061 #ifdef __cplusplus // Provide C++ Compatibility
00062
00063     extern "C" {
00064
00065 #endif
00066 // DOM-IGNORE-END
00067
00068 // ****
00069 // ****

```

```

00070 // Section: Data Types
00071 // *****
00072 // *****
00073
00074 typedef enum
00075 {
00076     RTCC_ALARM_MASK_HALF_SECOND = 0x00,    // Every half-second
00077     RTCC_ALARM_MASK_SECOND = 0x01,         // Every second
00078     RTCC_ALARM_MASK_10_SECONDS = 0x02,     // Every 10 seconds
00079     RTCC_ALARM_MASK_SS = 0x03,             // Every minute
00080     RTCC_ALARM_MASK_10_MINUTES = 0x04,     // Every 10 minutes
00081     RTCC_ALARM_MASK_HOUR = 0x05,          // Every hour
00082     RTCC_ALARM_MASK_HHMISS = 0x06,        // Once a day
00083     RTCC_ALARM_MASK_WEEK = 0x07,          // Once a week
00084     RTCC_ALARM_MASK_MONTH = 0x08,         // Once a month
00085     RTCC_ALARM_MASK_YEAR = 0x09,          // Once a year
00086     RTCC_ALARM_MASK_OFF = 0xFF            // Disabled
00087
00088 } RTCC_ALARM_MASK;
00089
00090 typedef enum
00091 {
00092     RTCC_INT_ALARM = 0x40000000
00093 } RTCC_INT_MASK;
00094
00095 typedef void (*RTCC_CALLBACK)( uintptr_t context );
00096
00097 // *****
00098 // *****
00099 // *****
00100 // Section: Interface Routines
00101 // *****
00102 // *****
00103
00104 void RTCC_Initialize( void );
00105
00106 bool RTCC_TimeSet( struct tm *Time );
00107
00108 void RTCC_TimeGet( struct tm *Time );
00109
00110 bool RTCC_AlarmSet( struct tm *alarmTime, RTCC_ALARM_MASK alarmFreq );
00111
00112 void RTCC_CallbackRegister( RTCC_CALLBACK callback, uintptr_t context );
00113
00114 void RTCC_InterruptEnable( RTCC_INT_MASK interrupt );
00115
00116 void RTCC_InterruptDisable( RTCC_INT_MASK interrupt );
00117
00118 // DOM-IGNORE-BEGIN
00119 #ifdef __cplusplus // Provide C++ Compatibility
00120     }
00121 #endif
00122 // DOM-IGNORE-END
00123 #endif
00124 // DOM-IGNORE-END
00125 #endif // PLIB_RTCC_H

```

4.32 plib_spi1_master.h

```

00001 /*****
00002  SPI PLIB
00003
00004  Company:
00005      Microchip Technology Inc.
00006
00007  File Name:
00008      plib_spi1_master.h
00009
00010  Summary:
00011      SPI1 Master PLIB Header File
00012
00013  Description:
00014      This file has prototype of all the interfaces provided for particular
00015      SPI peripheral.
00016
00017  *****/
00018
00019 /*****
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00040 *****/
00041
00042 #ifndef PLIB_SPI1_MASTER_H
00043 #define PLIB_SPI1_MASTER_H
00044
00045 #include "device.h"
00046 #include "plib_spi_master_common.h"
00047
00048 /* Provide C++ Compatibility */
00049 #ifdef __cplusplus
00050     extern "C" {
00051 #endif
00052
00053
00054
00055
00056 /***** SPI1 Interface *****/
00057
00058 void SPI1_Initialize ( void );
00059
00060 bool SPI1_WriteRead (void* pTransmitData, size_t txSize, void* pReceiveData, size_t rxSize);
00061
00062 bool SPI1_Write(void* pTransmitData, size_t txSize);
00063
00064 bool SPI1_Read(void* pReceiveData, size_t rxSize);
00065
00066 bool SPI1_TransferSetup (SPI_TRANSFER_SETUP *setup, uint32_t spiSourceClock);
00067
00068 bool SPI1_IsTransmitterBusy (void);
00069
00070 bool SPI1_IsBusy(void);
00071
00072 void SPI1_CallbackRegister(SPI_CALLBACK callback, uintptr_t context);
00073
00074 /* Provide C++ Compatibility */
00075 #ifdef __cplusplus
00076     }
00077 #endif
00078
00079 #endif
00080
00081 #endif // PLIB_SPI1_MASTER_H

```

4.33 plib_spi2_master.h

```

00001 /*****
00002 SPI PLIB
00003
00004 Company:
00005     Microchip Technology Inc.
00006
00007 File Name:
00008     plib_spi2_master.h
00009
00010 Summary:
00011     SPI2 Master PLIB Header File
00012
00013 Description:
00014     This file has prototype of all the interfaces provided for particular
00015     SPI peripheral.
00016
00017 *****/
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00019 /*****
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00040 *****/
00041
00042 #ifndef PLIB_SPI2_MASTER_H
00043 #define PLIB_SPI2_MASTER_H
00044
00045 #include "device.h"
00046 #include "plib_spi_master_common.h"
00047
00048 /* Provide C++ Compatibility */
00049 #ifdef __cplusplus
00050     extern "C" {
00051
00052
00053 #endif
00054
00055
00056 /***** SPI2 Interface *****/
00057
00058 void SPI2_Initialize ( void );
00059
00060 bool SPI2_WriteRead (void* pTransmitData, size_t txSize, void* pReceiveData, size_t rxSize);
00061
00062 bool SPI2_Write(void* pTransmitData, size_t txSize);
00063
00064 bool SPI2_Read(void* pReceiveData, size_t rxSize);
00065
00066 bool SPI2_TransferSetup (SPI_TRANSFER_SETUP *setup, uint32_t spiSourceClock);
00067
00068 bool SPI2_IsTransmitterBusy (void);
00069
00070
00071 /* Provide C++ Compatibility */
00072 #ifdef __cplusplus
00073     }
00074 #endif
00075
00076 #endif
00077
00078 #endif // PLIB_SPI2_MASTER_H

```

4.34 plib_spi_master_common.h

```

00001 /*****
00002  SPI PLIB
00003
00004  Company:
00005    Microchip Technology Inc.
00006
00007  File Name:
00008    plib_spi_master_common.h
00009
00010  Summary:
00011    SPI PLIB Master Common Header File
00012
00013  Description:
00014    This file has prototype of all the interfaces which are common for all the
00015    SPI peripherals.
00016 *****/
00017
00018
00019 /*****
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00040 *****/
00041
00042 #ifndef PLIB_SPI_MASTER_COMMON_H
00043 #define PLIB_SPI_MASTER_COMMON_H
00044
00045 #include <stdint.h>
00046 #include <stddef.h>
00047 #include <stdbool.h>
00048
00049
00050 /* Provide C++ Compatibility */
00051 #ifdef __cplusplus
00052     extern "C" {
00053
00054
00055 #endif
00056
00057 /***** SPI{SPI_INDEX?string} Interface *****/
00058
00059 typedef enum
00060 {
00061     SPI_CLOCK_PHASE_TRAILING_EDGE = 0x00000000,
00062     SPI_CLOCK_PHASE_LEADING_EDGE = 0x00000100,
00063
00064     /* Force the compiler to reserve 32-bit space for each enum value */
00065     SPI_CLOCK_PHASE_INVALID = 0xFFFFFFFF
00066 }SPI_CLOCK_PHASE;
00067
00068 typedef enum
00069 {
00070     SPI_CLOCK_POLARITY_IDLE_LOW = 0x00000000,
00071     SPI_CLOCK_POLARITY_IDLE_HIGH = 0x00000040,
00072
00073     /* Force the compiler to reserve 32-bit space for each enum value */
00074     SPI_CLOCK_POLARITY_INVALID = 0xFFFFFFFF
00075 }SPI_CLOCK_POLARITY;
00076
00077 typedef enum
00078 {
00079     SPI_DATA_BITS_8 = 0x00000000,
00080     SPI_DATA_BITS_16 = 0x00000400,
00081     SPI_DATA_BITS_32 = 0x00000800,
00082
00083     /* Force the compiler to reserve 32-bit space for each enum value */
00084     SPI_DATA_BITS_INVALID = 0xFFFFFFFF
00085 }SPI_DATA_BITS;
00086
00087 typedef struct
00088 {
00089     uint32_t    clockFrequency;
00090     SPI_CLOCK_PHASE clockPhase;
00091     SPI_CLOCK_POLARITY clockPolarity;
00092     SPI_DATA_BITS dataBits;
00093 }SPI_TRANSFER_SETUP;
00094
00095 typedef void (*SPI_CALLBACK) (uintptr_t context);
00096
00097 // ****
00098 // ****
00099 // Section: Local: **** Do Not Use ****
00100 // ****
00101 // ****
00102
00103 typedef struct
00104 {
00105     void* txBuffer;
00106     void* rxBuffer;
00107     size_t txSize;

```

```

00112     size_t                rxSize;
00113     size_t                dummySize;
00114     size_t                rxCount;
00115     size_t                txCount;
00116     bool                  transferIsBusy;
00117     SPI_CALLBACK          callback;
00118     uintptr_t             context;
00119
00120 } SPI_OBJECT ;
00121
00122 /* Provide C++ Compatibility */
00123 #ifdef __cplusplus
00124     }
00125 #endif
00126
00127 #endif
00128
00129 #endif // PLIB_SPI_MASTER_COMMON_H

```

4.35 plib_tmr2.h

```

00001 /*****
00002  Data Type definition of Timer PLIB
00003
00004  Company:
00005      Microchip Technology Inc.
00006
00007  File Name:
00008      plib_tmr2.h
00009
00010  Summary:
00011      Data Type definition of the Timer Peripheral Interface Plib.
00012
00013  Description:
00014      This file defines the Data Types for the Timer Plib.
00015
00016  Remarks:
00017      None.
00018
00019 *****/
00020
00021 /*****
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00042  *****/
00043
00044 #ifndef PLIB_TMR2_H
00045 #define PLIB_TMR2_H
00046
00047 #include <stddef.h>
00048 #include <stdint.h>
00049 #include "device.h"
00050 #include "plib_tmr_common.h"
00051
00052 // DOM-IGNORE-BEGIN
00053 #ifdef __cplusplus // Provide C++ Compatibility
00054     extern "C" {
00055 #endif
00056
00057 #endif
00058 // DOM-IGNORE-END
00059
00060 // *****
00061 // *****
00062 // Section: Data Types

```

```

00063 // *****
00064 // *****
00065
00066 // *****
00067 // *****
00068 // Section: Interface Routines
00069 // *****
00070 // *****
00071
00072 // *****
00073 void TMR2_Initialize(void);
00074
00075 void TMR2_Start(void);
00076
00077 void TMR2_Stop(void);
00078
00079 void TMR2_PeriodSet(uint32_t);
00080
00081 uint32_t TMR2_PeriodGet(void);
00082
00083 uint32_t TMR2_CounterGet(void);
00084
00085 uint32_t TMR2_FrequencyGet(void);
00086
00087 void TMR2_InterruptEnable(void);
00088
00089 void TMR2_InterruptDisable(void);
00090
00091 void TMR2_CallbackRegister( TMR_CALLBACK callback_fn, uintptr_t context );
00092
00093 // DOM-IGNORE-BEGIN
00094 #ifdef __cplusplus // Provide C++ Compatibility
00095     }
00096 #endif
00097 // DOM-IGNORE-END
00098
00099 #endif /* PLIB_TMR2_H */

```

4.36 plib_tmr3.h

```

00001 /*****
00002  Data Type definition of Timer PLIB
00003
00004  Company:
00005      Microchip Technology Inc.
00006
00007  File Name:
00008      plib_tmr3.h
00009
00010  Summary:
00011      Data Type definition of the Timer Peripheral Interface Plib.
00012
00013  Description:
00014      This file defines the Data Types for the Timer Plib.
00015
00016  Remarks:
00017      None.
00018
00019 *****/
00020
00021 /*****
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00042  *****/

```

```

00043
00044 #ifndef PLIB_TMR3_H
00045 #define PLIB_TMR3_H
00046
00047 #include <stddef.h>
00048 #include <stdint.h>
00049 #include "device.h"
00050 #include "plib_tmr_common.h"
00051
00052 // DOM-IGNORE-BEGIN
00053 #ifdef __cplusplus // Provide C++ Compatibility
00054
00055     extern "C" {
00056
00057 #endif
00058 // DOM-IGNORE-END
00059
00060 // *****
00061 // *****
00062 // Section: Data Types
00063 // *****
00064 // *****
00065
00066 // *****
00067 // *****
00068 // Section: Interface Routines
00069 // *****
00070 // *****
00071
00072 // *****
00073 void TMR3_Initialize(void);
00074
00075 void TMR3_Start(void);
00076
00077 void TMR3_Stop(void);
00078
00079 void TMR3_PeriodSet(uint32_t);
00080
00081 uint32_t TMR3_PeriodGet(void);
00082
00083 uint32_t TMR3_CounterGet(void);
00084
00085 uint32_t TMR3_FrequencyGet(void);
00086
00087
00088 // DOM-IGNORE-BEGIN
00089 #ifdef __cplusplus // Provide C++ Compatibility
00090
00091     }
00092 #endif
00093 // DOM-IGNORE-END
00094
00095 #endif /* PLIB_TMR3_H */

```

4.37 plib_tmr4.h

```

00001 /*****
00002  Data Type definition of Timer PLIB
00003
00004  Company:
00005    Microchip Technology Inc.
00006
00007  File Name:
00008    plib_tmr4.h
00009
00010  Summary:
00011    Data Type definition of the Timer Peripheral Interface Plib.
00012
00013  Description:
00014    This file defines the Data Types for the Timer Plib.
00015
00016  Remarks:
00017    None.
00018
00019 *****/
00020
00021 /*****
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00041 * THAT YOU HAVE PAID DIRECTLY TO MICROCHIP FOR THIS SOFTWARE.
00042 *****/
00043
00044 #ifndef PLIB_TMR4_H
00045 #define PLIB_TMR4_H
00046
00047 #include <stddef.h>
00048 #include <stdint.h>
00049 #include "device.h"
00050 #include "plib_tmr_common.h"
00051
00052 // DOM-IGNORE-BEGIN
00053 #ifdef __cplusplus // Provide C++ Compatibility
00054
00055     extern "C" {
00056
00057 #endif
00058 // DOM-IGNORE-END
00059
00060 // *****
00061 // *****
00062 // Section: Data Types
00063 // *****
00064 // *****
00065
00066 // *****
00067 // *****
00068 // Section: Interface Routines
00069 // *****
00070 // *****
00071
00072 // *****
00073 void TMR4_Initialize(void);
00074
00075 void TMR4_Start(void);
00076
00077 void TMR4_Stop(void);
00078
00079 void TMR4_PeriodSet(uint32_t);
00080
00081 uint32_t TMR4_PeriodGet(void);
00082
00083 uint32_t TMR4_CounterGet(void);
00084
00085 uint32_t TMR4_FrequencyGet(void);
00086
00087 void TMR4_InterruptEnable(void);
00088
00089 void TMR4_InterruptDisable(void);
00090
00091 void TMR4_CallbackRegister( TMR_CALLBACK callback_fn, uintptr_t context );
00092
00093 // DOM-IGNORE-BEGIN
00094 #ifdef __cplusplus // Provide C++ Compatibility
00095
00096     }
00097 #endif
00098 // DOM-IGNORE-END
00099
00100 #endif /* PLIB_TMR4_H */

```

4.38 plib_tmr5.h

```

00001 /*****
00002     Data Type definition of Timer PLIB
00003
00004     Company:
00005         Microchip Technology Inc.
00006
00007     File Name:

```

```
00008     plib_tmr5.h
00009
00010     Summary:
00011         Data Type definition of the Timer Peripheral Interface Plib.
00012
00013     Description:
00014         This file defines the Data Types for the Timer Plib.
00015
00016     Remarks:
00017         None.
00018
00019     *****/
00020
00021 /*****/
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00042 *****/
00043
00044 #ifndef PLIB_TMR5_H
00045 #define PLIB_TMR5_H
00046
00047 #include <stddef.h>
00048 #include <stdint.h>
00049 #include "device.h"
00050 #include "plib_tmr_common.h"
00051
00052 // DOM-IGNORE-BEGIN
00053 #ifdef __cplusplus // Provide C++ Compatibility
00054     extern "C" {
00055
00056
00057 #endif
00058 // DOM-IGNORE-END
00059
00060 // *****/
00061 // *****/
00062 // Section: Data Types
00063 // *****/
00064 // *****/
00065
00066 // *****/
00067 // *****/
00068 // Section: Interface Routines
00069 // *****/
00070 // *****/
00071
00072 // *****/
00073 void TMR5_Initialize(void);
00074
00075 void TMR5_Start(void);
00076
00077 void TMR5_Stop(void);
00078
00079 void TMR5_PeriodSet(uint32_t);
00080
00081 uint32_t TMR5_PeriodGet(void);
00082
00083 uint32_t TMR5_CounterGet(void);
00084
00085 uint32_t TMR5_FrequencyGet(void);
00086
00087 void TMR5_InterruptEnable(void);
00088
00089 void TMR5_InterruptDisable(void);
00090
00091 void TMR5_CallbackRegister( TMR_CALLBACK callback_fn, uintptr_t context );
00092
00093 // DOM-IGNORE-BEGIN
00094 #ifdef __cplusplus // Provide C++ Compatibility
```

```

00095
00096     }
00097 #endif
00098 // DOM-IGNORE-END
00099
00100 #endif /* PLIB_TMR5_H */

```

4.39 plib_tmr6.h

```

00001 /*****
00002  Data Type definition of Timer PLIB
00003
00004  Company:
00005      Microchip Technology Inc.
00006
00007  File Name:
00008      plib_tmr6.h
00009
00010  Summary:
00011      Data Type definition of the Timer Peripheral Interface Plib.
00012
00013  Description:
00014      This file defines the Data Types for the Timer Plib.
00015
00016  Remarks:
00017      None.
00018
00019 *****/
00020
00021 /*****
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00042  *****/
00043
00044 #ifndef PLIB_TMR6_H
00045 #define PLIB_TMR6_H
00046
00047 #include <stddef.h>
00048 #include <stdint.h>
00049 #include "device.h"
00050 #include "plib_tmr_common.h"
00051
00052 // DOM-IGNORE-BEGIN
00053 #ifdef __cplusplus // Provide C++ Compatibility
00054     extern "C" {
00055
00056 #endif
00057 #endif
00058 // DOM-IGNORE-END
00059
00060 // ****
00061 // ****
00062 // Section: Data Types
00063 // ****
00064 // ****
00065
00066 // ****
00067 // ****
00068 // Section: Interface Routines
00069 // ****
00070 // ****
00071
00072 // ****
00073 void TMR6_Initialize(void);
00074

```

```

00075 void TMR6_Start(void);
00076
00077 void TMR6_Stop(void);
00078
00079 void TMR6_PeriodSet(uint16_t);
00080
00081 uint16_t TMR6_PeriodGet(void);
00082
00083 uint16_t TMR6_CounterGet(void);
00084
00085 uint32_t TMR6_FrequencyGet(void);
00086
00087 void TMR6_InterruptEnable(void);
00088
00089 void TMR6_InterruptDisable(void);
00090
00091 void TMR6_CallbackRegister( TMR_CALLBACK callback_fn, uintptr_t context );
00092
00093 // DOM-IGNORE-BEGIN
00094 #ifdef __cplusplus // Provide C++ Compatibility
00095     }
00096 #endif
00097 // DOM-IGNORE-END
00098
00099 #endif /* PLIB_TMR6_H */

```

4.40 plib_tmr9.h

```

00001 /*****
00002  Data Type definition of Timer PLIB
00003
00004  Company:
00005      Microchip Technology Inc.
00006
00007  File Name:
00008      plib_tmr9.h
00009
00010  Summary:
00011      Data Type definition of the Timer Peripheral Interface Plib.
00012
00013  Description:
00014      This file defines the Data Types for the Timer Plib.
00015
00016  Remarks:
00017      None.
00018
00019 *****/
00020
00021 /*****
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00042  *****/
00043
00044 #ifndef PLIB_TMR9_H
00045 #define PLIB_TMR9_H
00046
00047 #include <stdint.h>
00048 #include <stdint.h>
00049 #include "device.h"
00050 #include "plib_tmr_common.h"
00051
00052 // DOM-IGNORE-BEGIN
00053 #ifdef __cplusplus // Provide C++ Compatibility
00054

```



```

00055     extern "C" {
00056
00057 #endif
00058 // DOM-IGNORE-END
00059
00060 // *****
00061 // *****
00062 // Section: Data Types
00063 // *****
00064 // *****
00065
00066 // *****
00067 // *****
00068 // Section: Interface Routines
00069 // *****
00070 // *****
00071
00072 // *****
00073 void TMR9_Initialize(void);
00074
00075 void TMR9_Start(void);
00076
00077 void TMR9_Stop(void);
00078
00079 void TMR9_PeriodSet(uint32_t);
00080
00081 uint32_t TMR9_PeriodGet(void);
00082
00083 uint32_t TMR9_CounterGet(void);
00084
00085 uint32_t TMR9_FrequencyGet(void);
00086
00087
00088 // DOM-IGNORE-BEGIN
00089 #ifdef __cplusplus // Provide C++ Compatibility
00090     }
00091 #endif
00092 // DOM-IGNORE-END
00093
00094 #endif /* PLIB_TMR9_H */

```

4.41 plib_tmr_common.h

```

00001 /*****
00002  TMR Peripheral Library Interface Header File
00003
00004  Company
00005    Microchip Technology Inc.
00006
00007  File Name
00008    plib_tmr_common.h
00009
00010  Summary
00011    TMR peripheral library interface.
00012
00013  Description
00014    This file defines the interface to the TC peripheral library. This
00015    library provides access to and control of the associated peripheral
00016    instance.
00017
00018  *****/
00019
00020 // DOM-IGNORE-BEGIN
00021 /*****
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```

```

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00041 * THAT YOU HAVE PAID DIRECTLY TO MICROCHIP FOR THIS SOFTWARE.
00042 *****/
00043 // DOM-IGNORE-END
00044
00045 #ifndef PLIB_TMR_COMMON_H    // Guards against multiple inclusion
00046 #define PLIB_TMR_COMMON_H
00047
00048
00049 // ****
00050 // ****
00051 // Section: Included Files
00052 // ****
00053 // ****
00054
00055 /* This section lists the other files that are included in this file.
00056 */
00057 #include <stddef.h>
00058
00059 // DOM-IGNORE-BEGIN
00060 #ifdef __cplusplus // Provide C++ Compatibility
00061
00062 extern "C" {
00063
00064 #endif
00065
00066 // DOM-IGNORE-END
00067
00068 // ****
00069 // ****
00070 // Section: Data Types
00071 // ****
00072 // ****
00073 /* The following data type definitions are used by the functions in this
00074    interface and should be considered part of it.
00075 */
00076
00077
00078 // ****
00079 /* TMR_CALLBACK
00080
00081    Summary:
00082        Use to register a callback with the TMR.
00083
00084    Description:
00085        When a match is asserted, a callback can be activated.
00086        Use TMR_CALLBACK as the function pointer to register the callback
00087        with the match.
00088
00089    Remarks:
00090        The callback should look like:
00091        void callback(handle, context);
00092        Make sure the return value and parameters of the callback are correct.
00093 */
00094
00095 typedef void (*TMR_CALLBACK)(uint32_t status, uintptr_t context);
00096
00097 // ****
00098
00099 typedef struct
00100 {
00101     /*TMR callback function happens on Period match*/
00102     TMR_CALLBACK callback_fn;
00103     /* - Client data (Event Context) that will be passed to callback */
00104     uintptr_t context;
00105 }TMR_TIMER_OBJECT;
00106
00107
00108 // DOM-IGNORE-BEGIN
00109 #ifdef __cplusplus // Provide C++ Compatibility
00110
00111 }
00112
00113 #endif
00114 // DOM-IGNORE-END
00115
00116 #endif // _PLIB_TMR_COMMON_H
00117

```

4.42 plib_uart1.h

```

00001 /*****
00002     UART1 PLIB

```

```

00003
00004     Company:
00005         Microchip Technology Inc.
00006
00007     File Name:
00008         plib_uart1.h
00009
00010     Summary:
00011         UART1 PLIB Header File
00012
00013     Description:
00014         None
00015
00016     *****/
00017
00018 /*****
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00039  *****/
00040
00041 #ifndef PLIB_UART1_H
00042 #define PLIB_UART1_H
00043
00044 #include <stddef.h>
00045 #include <stdbool.h>
00046 #include <stdint.h>
00047 #include "device.h"
00048 #include "plib_uart_common.h"
00049
00050 // DOM-IGNORE-BEGIN
00051 #ifdef __cplusplus // Provide C++ Compatibility
00052
00053     extern "C" {
00054
00055 #endif
00056 // DOM-IGNORE-END
00057
00058 // *****
00059 // *****
00060 // Section: Interface
00061 // *****
00062 // *****
00063
00064 #define UART1_FrequencyGet()      (uint32_t) (60000000UL)
00065
00066 /***** UART1 API *****/
00067
00068 void UART1_Initialize( void );
00069
00070 bool UART1_SerialSetup( UART_SERIAL_SETUP *setup, uint32_t srcClkFreq );
00071
00072 bool UART1_Write( void *buffer, const size_t size );
00073
00074 bool UART1_Read( void *buffer, const size_t size );
00075
00076 UART_ERROR UART1_ErrorGet( void );
00077
00078 bool UART1_AutoBaudQuery( void );
00079
00080 void UART1_AutoBaudSet( bool enable );
00081
00082 int UART1_ReadByte( void );
00083
00084 bool UART1_ReceiverIsReady( void );
00085
00086 void UART1_WriteByte( int data );
00087
00088 bool UART1_TransmitterIsReady( void );
00089

```

```

00090
00091 bool UART1_TransmitComplete( void );
00092
00093 // DOM-IGNORE-BEGIN
00094 #ifdef __cplusplus // Provide C++ Compatibility
00095
00096     }
00097
00098 #endif
00099 // DOM-IGNORE-END
00100
00101 #endif // PLIB_UART1_H

```

4.43 plib_uart2.h

```

00001 /*****
00002     UART2 PLIB
00003
00004     Company:
00005         Microchip Technology Inc.
00006
00007     File Name:
00008         plib_uart2.h
00009
00010     Summary:
00011         UART2 PLIB Header File
00012
00013     Description:
00014         None
00015
00016 *****/
00017
00018 /*****
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00039 *****/
00040
00041 #ifndef PLIB_UART2_H
00042 #define PLIB_UART2_H
00043
00044 #include <stddef.h>
00045 #include <stdbool.h>
00046 #include <stdint.h>
00047 #include "device.h"
00048 #include "plib_uart_common.h"
00049
00050 // DOM-IGNORE-BEGIN
00051 #ifdef __cplusplus // Provide C++ Compatibility
00052
00053     extern "C" {
00054
00055 #endif
00056 // DOM-IGNORE-END
00057
00058 // ****
00059 // ****
00060 // Section: Interface
00061 // ****
00062 // ****
00063
00064 #define UART2_FrequencyGet()    (uint32_t)(60000000UL)
00065
00066 /***** UART2 API *****/
00067
00068 void UART2_Initialize( void );

```

```

00069
00070 bool UART2_SerialSetup( UART_SERIAL_SETUP *setup, uint32_t srcClkFreq );
00071
00072 UART_ERROR UART2_ErrorGet( void );
00073
00074 bool UART2_AutoBaudQuery( void );
00075
00076 void UART2_AutoBaudSet( bool enable );
00077
00078 size_t UART2_Write(uint8_t* pWrBuffer, const size_t size );
00079
00080 size_t UART2_WriteCountGet(void);
00081
00082 size_t UART2_WriteFreeBufferCountGet(void);
00083
00084 size_t UART2_WriteBufferSizeGet(void);
00085
00086 bool UART2_TransmitComplete(void);
00087
00088 bool UART2_WriteNotificationEnable(bool isEnabled, bool isPersistent);
00089
00090 void UART2_WriteThresholdSet(uint32_t nBytesThreshold);
00091
00092 void UART2_WriteCallbackRegister( UART_RING_BUFFER_CALLBACK callback, uintptr_t context);
00093
00094 size_t UART2_Read(uint8_t* pRdBuffer, const size_t size);
00095
00096 size_t UART2_ReadCountGet(void);
00097
00098 size_t UART2_ReadFreeBufferCountGet(void);
00099
00100 size_t UART2_ReadBufferSizeGet(void);
00101
00102 bool UART2_ReadNotificationEnable(bool isEnabled, bool isPersistent);
00103
00104 void UART2_ReadThresholdSet(uint32_t nBytesThreshold);
00105
00106 void UART2_ReadCallbackRegister( UART_RING_BUFFER_CALLBACK callback, uintptr_t context);
00107
00108 // DOM-IGNORE-BEGIN
00109 #ifdef __cplusplus // Provide C++ Compatibility
00110     }
00111 #endif
00112 // DOM-IGNORE-END
00113 #endif
00114 // DOM-IGNORE-END
00115 #endif
00116 #endif // PLIB_UART2_H

```

4.44 plib_uart_common.h

```

00001 /*****
00002  UART PLIB
00003
00004  Company:
00005      Microchip Technology Inc.
00006
00007  File Name:
00008      plib_uart_common.h
00009
00010  Summary:
00011      UART PLIB Common Header File
00012
00013  Description:
00014      This file has prototype of all the interfaces which are common for all the
00015      UART peripherals.
00016
00017  *****/
00018
00019 /*****
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00038 * ANY WAY RELATED TO THIS SOFTWARE WILL NOT EXCEED THE AMOUNT OF FEES, IF ANY,
00039 * THAT YOU HAVE PAID DIRECTLY TO MICROCHIP FOR THIS SOFTWARE.
00040 *****/
00041
00042 #ifndef PLIB_UART_COMMON_H
00043 #define PLIB_UART_COMMON_H
00044
00045 #include <stdint.h>
00046 #include <stddef.h>
00047 #include <stdbool.h>
00048
00049
00050 /* Provide C++ Compatibility */
00051 #ifdef __cplusplus
00052     extern "C" {
00053
00054 #endif
00055
00056 #define UART_RXFIFO_DEPTH          9
00057
00058 typedef enum
00059 {
00060     UART_PARITY_NONE = 0x00,
00061     UART_PARITY_EVEN = 0x02,
00062     UART_PARITY_ODD = 0x04,
00063
00064     /* Force the compiler to reserve 32-bit space for each enum */
00065     UART_PARITY_INVALID = 0xFFFFFFFF
00066 } UART_PARITY;
00067
00068 typedef enum
00069 {
00070     UART_DATA_8_BIT = 0x00,
00071     UART_DATA_9_BIT = 0x06,
00072
00073     /* Force the compiler to reserve 32-bit memory for each enum */
00074     UART_DATA_INVALID = 0xFFFFFFFF
00075 } UART_DATA;
00076
00077 typedef enum
00078 {
00079     UART_STOP_1_BIT = 0x00,
00080     UART_STOP_2_BIT = 0x01,
00081
00082     /* Force the compiler to reserve 32-bit memory for each enum */
00083     UART_STOP_INVALID = 0xFFFFFFFF
00084 } UART_STOP;
00085
00086 typedef struct
00087 {
00088     uint32_t baudRate;
00089     UART_PARITY parity;
00090     UART_DATA dataWidth;
00091     UART_STOP stopBits;
00092 } UART_SERIAL_SETUP;
00093
00094 // ****
00095 /* UART Errors
00096 */
00097
00098 typedef enum
00099 {
00100     UART_ERROR_NONE = 0,
00101     UART_ERROR_OVERRUN = 0x02,
00102     UART_ERROR_FRAMING = 0x04,
00103     UART_ERROR_PARITY = 0x08
00104 } UART_ERROR;
00105
00106

```

```

00120 typedef void (* UART_CALLBACK)( uintptr_t context );
00121
00122 // *****
00123 // *****
00124 // Section: Local: **** Do Not Use ****
00125 // *****
00126 // *****
00127
00128 typedef struct
00129 {
00130     uint8_t *          txBuffer;
00131     size_t             txSize;
00132     size_t             txProcessedSize;
00133     UART_CALLBACK      txCallback;
00134     uintptr_t          txContext;
00135     bool               txBusyStatus;
00136
00137     uint8_t *          rxBuffer;
00138     size_t             rxSize;
00139     size_t             rxProcessedSize;
00140     UART_CALLBACK      rxCallback;
00141     uintptr_t          rxContext;
00142     bool               rxBusyStatus;
00143
00144     volatile UART_ERROR errors;
00145 } UART_OBJECT ;
00146
00147 typedef enum
00148 {
00149     /* Threshold number of bytes are available in the receive ring buffer */
00150     UART_EVENT_READ_THRESHOLD_REACHED = 0,
00151
00152     /* Receive ring buffer is full. Application must read the data out to avoid missing data on the
00153     next RX interrupt. */
00154     UART_EVENT_READ_BUFFER_FULL,
00155
00156     /* USART error. Application must call the UARTx_ErrorGet API to get the type of error and clear
00157     the error. */
00158     UART_EVENT_READ_ERROR,
00159
00160     /* Threshold number of free space is available in the transmit ring buffer */
00161     UART_EVENT_WRITE_THRESHOLD_REACHED,
00162 } UART_EVENT;
00163
00164 typedef void (* UART_RING_BUFFER_CALLBACK)( UART_EVENT event, uintptr_t context );
00165
00166 // *****
00167 // *****
00168 // Section: Local: **** Do Not Use ****
00169 // *****
00170
00171 typedef struct
00172 {
00173     UART_RING_BUFFER_CALLBACK wrCallback;
00174
00175     uintptr_t wrContext;
00176
00177     volatile uint32_t wrInIndex;
00178
00179     volatile uint32_t wrOutIndex;
00180
00181     uint32_t wrThreshold;
00182
00183     bool isWrNotificationEnabled;
00184
00185     bool isWrNotifyPersistently;
00186
00187     uint32_t wrBufferSize;
00188
00189     UART_RING_BUFFER_CALLBACK rdCallback;
00190
00191     uintptr_t rdContext;
00192
00193     volatile uint32_t rdInIndex;
00194
00195     volatile uint32_t rdOutIndex;
00196
00197     uint32_t rdBufferSize;
00198
00199     bool isRdNotificationEnabled;
00200
00201     uint32_t rdThreshold;
00202
00203     bool isRdNotifyPersistently;
00204

```

```

00205     volatile UART_ERROR                errors;
00206
00207 } UART_RING_BUFFER_OBJECT;
00208
00209 /* Provide C++ Compatibility */
00210 #ifdef __cplusplus
00211     }
00212 }
00213
00214 #endif
00215
00216 #endif // PLIB_UART_COMMON_H
00217
00218 /*****
00219  End of File
00220 */

```

4.45 toolchain_specifics.h

```

00001 /*****
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00003  *
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00020  * ANY WAY RELATED TO THIS SOFTWARE WILL NOT EXCEED THE AMOUNT OF FEES, IF ANY,
00021  * THAT YOU HAVE PAID DIRECTLY TO MICROCHIP FOR THIS SOFTWARE.
00022  *****/
00023
00024 #ifndef TOOLCHAIN_SPECIFICS_H
00025 #define TOOLCHAIN_SPECIFICS_H
00026
00027 #include <sys/types.h>
00028
00029 #define NO_INIT                __attribute__((section(".no_init")))
00030 #define SECTION(a)             __attribute__((section(a)))
00031
00032 #ifndef __ASM
00033     #define __ASM                __asm__
00034 #endif
00035 #ifndef __INLINE
00036     #define __INLINE            __inline__
00037 #endif
00038 #ifndef __STATIC_INLINE
00039     #define __STATIC_INLINE      static __inline__
00040 #endif
00041 #ifndef __STATIC_FORCEINLINE
00042     #define __STATIC_FORCEINLINE __attribute__((always_inline)) static __inline__
00043 #endif
00044 #ifndef __NO_RETURN
00045     #define __NO_RETURN          __attribute__((noreturn))
00046 #endif
00047 #ifndef __USED
00048     #define __USED              __attribute__((used))
00049 #endif
00050 #ifndef __WEAK
00051     #define __WEAK              __attribute__((weak))
00052 #endif
00053 #ifndef __PACKED
00054     #define __PACKED            __attribute__((packed, aligned(1)))
00055 #endif
00056 #ifndef __PACKED_STRUCT
00057     #define __PACKED_STRUCT      struct __attribute__((packed, aligned(1)))
00058 #endif
00059 #ifndef __PACKED_UNION
00060     #define __PACKED_UNION      union __attribute__((packed, aligned(1)))
00061 #endif
00062 #ifndef __COHERENT
00063     #define __COHERENT          __attribute__((coherent))
00064 #endif

```



```

00065 #ifndef __ALIGNED
00066     #define __ALIGNED(x)          __attribute__((aligned(x)))
00067 #endif
00068 #ifndef __RESTRICT
00069     #define __RESTRICT            __restrict__
00070 #endif
00071
00072 #define CACHE_LINE_SIZE          (4u)
00073 #define CACHE_ALIGN
00074
00075 #define CACHE_ALIGNED_SIZE_GET(size)      (size + ((size % CACHE_LINE_SIZE)? (CACHE_LINE_SIZE - (size %
    CACHE_LINE_SIZE)) : 0))
00076
00077 #ifndef FORMAT_ATTRIBUTE
00078     #define FORMAT_ATTRIBUTE(archetype, string_index, first_to_check) __attribute__ ((format
    (archetype, string_index, first_to_check)))
00079 #endif
00080
00081 #endif // end of header
00082

```

4.46 do_fft.h

```

00001 /*
00002  * File:    do_fft.h
00003  * Author:  root
00004  *
00005  * Created on November 6, 2022, 1:53 PM
00006  */
00007
00008 #ifndef DO_FFT_H
00009 #define DO_FFT_H
00010
00011 #ifdef __cplusplus
00012 extern "C" {
00013 #endif
00014
00015 #include <xc.h>
00016 #include <p32xxxx.h>
00017 #include <stdint.h>
00018 #include <stdbool.h>
00019 #include <stdio.h>
00020 #include <stdlib.h>
00021
00022 #include "fft.h"
00023
00024 #define _XTAL_FREQ 120000000UL
00025
00026 #define DO_FFT_DRIVER    "V1.002"
00027 #define DO_FFT_ALIAS     "DO_FFT "
00028
00029 #define fft_sample_rate 100.0
00030 #define fft_cutoff_freq 0.5
00031
00032 void do_fft(bool);
00033 void __delay_ms_fft(uint32_t);
00034 void __delay_us_fft(uint32_t);
00035 void do_fft_version(void);
00036
00037 double do_fft_dc_x(double input);
00038 double do_fft_dc_y(double input);
00039 double do_fft_dc_z(double input);
00040
00041 extern uint8_t inB[N_FFT];
00042
00043 #ifdef __cplusplus
00044 }
00045 #endif
00046
00047 #endif /* DO_FFT_H */
00048

```

4.47 fft.h

```

00001 /*
00002  * File:    fft.h
00003  * Author:  root
00004  *

```

```

00005  * Created on November 6, 2022, 1:51 PM
00006  */
00007
00008 #ifndef FFT_H
00009 #define FFT_H
00010
00011 #ifdef __cplusplus
00012 extern "C" {
00013 #endif
00014
00015 #include <stdint.h>
00016 #include <stdbool.h>
00017 #include <stdio.h>
00018 #include <stdlib.h>
00019 #include <math.h>
00020 #include "imupic32mcj.h"
00021
00022 #define FFT_DRIVER "V1.160"
00023 #define FFT_ALIAS "FFT"
00024
00025 #define N_FFT 256
00026 #define PI2N 2 * M_PI / N_FFT
00027 #define FFT_COUNT 256
00028
00029 #ifndef BNO086
00030 #ifdef BMA400 // Bosch devices, gain and noise product
00031 #define FFT_GAIN 5.0 // low cost, high noise
00032 #else
00033 #define FFT_GAIN 40.0 // Murata imu
00034 #endif
00035 #else
00036 #define FFT_GAIN 5.0 // BNO gain
00037 #endif
00038
00039     extern uint8_t inB[], fft_buffer[];
00040     extern double fft_gain;
00041
00042     void initFFT(void);
00043     void windowFFT(uint8_t *source);
00044     void FFT(void);
00045     void powerScale(uint8_t *dest);
00046     void fft_version(void);
00047
00048 #ifdef __cplusplus
00049 }
00050 #endif
00051
00052 #endif /* FFT_H */
00053

```

4.48 gfx.h

```

00001 /*
00002  * File:   gfx.h
00003  * Author: root
00004  *
00005  * Created on October 30, 2020, 9:35 AM
00006  */
00007
00008 #ifndef GFX_H
00009 #define GFX_H
00010
00011 #ifdef __cplusplus
00012 extern "C" {
00013 #endif
00014
00015 #include <math.h>
00016 #include "../firmware/lcd_drv/OledGrph.h"
00017
00018     // #define GFX_BOX
00019 #define SHOW_LA
00020 #define fft_max_pixel 64
00021 #define fft_resolution_shift 1
00022 #define fft_scale 2
00023
00024     static const double sine_res = 3600.0;
00025     static const double sinea = 0.0;
00026     static const double sineb = 1200.0;
00027     static const double sinec = 2400.0;
00028
00029     extern int32_t xa, ya, za;
00030     extern volatile double q0, q1, q2, q3;
00031

```

```

00032     void line_rot(uint32_t, uint32_t, uint32_t, uint32_t);
00033     void vector_graph(void);
00034     void vector_graph_fs(void);
00035     void LA_gfx(bool, bool, uint32_t);
00036     void fft_draw(uint8_t, uint8_t);
00037
00038 #ifdef __cplusplus
00039 }
00040 #endif
00041
00042 #endif /* GFX_H */
00043

```

4.49 hid.h

```

00001 /*
00002  * TMR3 debounce timer
00003  * TMR4 screen blank timer
00004  */
00005
00006 #ifndef HID_H
00007 #define HID_H
00008
00009 #ifdef __cplusplus
00010 extern "C" {
00011 #endif
00012
00013 #include "buzzer.h"
00014 #include "gfx.h"
00015
00016 #define HID_DRIVER "V1.015"
00017 #define HID_ALIAS "HID"
00018
00019 #define FFT_WAKE 1500
00020 #define HID_LONG_BLANK 14062499U * 10U
00021
00022 #define DBOUNCE_COUNTS 70000
00023 /*
00024  * silent mode animation
00025  */
00026 #define SDOT_ON 70
00027 #define SDOT_OFF 140
00028
00029 #define PROX_T_H 8.5f
00030 #define PROX_T_L 1.0f
00031
00032 typedef enum {
00033     H_init,
00034     H_home,
00035     H_run,
00036     H_blank,
00037     H_zero_blank,
00038     H_abort,
00039 } H_STATE;
00040
00041 typedef struct H_type {
00042     H_STATE hid_state;
00043     volatile bool dis_alt, show_la, la_mod, dis_unblank, dis_reset, silent;
00044 } H_type;
00045
00046 extern H_type H;
00047
00048 extern void hid_init(H_STATE);
00049
00050 #ifdef __cplusplus
00051 }
00052 #endif
00053
00054 #endif /* HID_H */
00055

```

4.50 host.h

```

00001 /*
00002  * File: host.h
00003  * Author: nsasp
00004  *
00005  * Created on December 29, 2022, 9:14 PM

```

```

00006  */
00007
00008 #ifndef HOST_H
00009 #define HOST_H
00010
00011 #ifdef __cplusplus
00012 extern "C" {
00013 #endif
00014
00015 #include <stddef.h>           // Defines NULL
00016 #include <stdbool.h>         // Defines true
00017 #include <stdlib.h>          // Defines EXIT_FAILURE
00018 #include <stdarg.h>
00019 #include <proc/p32mk0512mcj048.h>
00020 #include "definitions.h"      // SYS function prototypes
00021 #include "imu.h"
00022 #include "imupic32mcj.h"
00023 #include "../firmware/lcd_drv/lcd_drv.h"
00024 #include "remote_cmd.h"
00025
00026 #define HOST_DRIVER "V1.701"
00027 #define HOST_ALIAS "HOST"
00028
00029 #define HOST_MAGIC 0x1957
00030 #define HOST_MAGIC_ID 0x101957
00031 #define HOST_SECRET 0xBD6FDC7BC925CD3E // 64-bit random number for unlocking machine control functions
00032
00033 #define host_lcd_update 100
00034
00035 #ifdef XPRJ_nsensor
00036 #define HOST_BOARD
00037 #define INT_BOARD
00038 #define debounce_delay 7000 // a few seconds in host mode
00039 #else
00040 #define debounce_delay 3000000 // a few seconds in sensor mode
00041 #endif
00042
00043 #ifdef XPRJ_mcj
00044 #define SENSOR_BOARD
00045 #define BLOCK_BOARD
00046 #endif
00047
00048 #ifdef XPRJ_bma400
00049 #define SENSOR_BOARD
00050 #define BLOCK_BOARD
00051 #endif
00052
00053 #ifdef XPRJ_mcj_remote
00054 #define SENSOR_BOARD
00055 #define BLOCK_BOARD
00056 #endif
00057
00058 /*
00059  * these routines can corrupt the display buffer with ISR level modifications of memory
00060  */
00061 //#define DEBUG_can_callback
00062 //#define DEBUG_can_callback1
00063
00064 #define avg_samples 10000.0
00065 #define uart_wait 32300
00066 #define host_xmit_wait 10000
00067
00068 /*
00069  * CAN-FD vibration sensor host to network state machine
00070  */
00071 int host_sm(void);
00072 bool TP1_check(void);
00073 void send_from_host(uint32_t);
00074
00075 #ifdef __cplusplus
00076 }
00077 #endif
00078
00079 #endif /* HOST_H */
00080

```

4.51 imu.h

```

00001 /*
00002  * File:   imu.h
00003  * Author: root
00004  *
00005  * Created on September 19, 2022, 8:50 AM

```

```

00006  */
00007
00008 #ifndef IMU_H
00009 #define IMU_H
00010
00011 #ifdef __cplusplus
00012 extern "C" {
00013 #endif
00014
00015 #include <stddef.h>           // Defines NULL
00016 #include <stdbool.h>         // Defines true
00017 #include <stdlib.h>          // Defines EXIT_FAILURE
00018 #include <math.h>
00019 #include "imupic32mcj.h"
00020 #include "bno086_reg.h"
00021 #include "quat.h"
00022
00023 #define IMU_DRIVER "V1.715"
00024 #ifdef BNO086
00025 #define IMU_ALIAS "BNO086 IMU"
00026 #else
00027 #define IMU_ALIAS "IMU"
00028 #endif
00029
00030 #define IMU_ID_DELAY 400
00031
00032 #define IMU_DATA_RAW_LEN 30
00033 #define IMU_DATA_BUFFER_INDEX 1
00034
00035 typedef struct _sSensorData_t {
00036     const uint16_t id;
00037     sh2_Quaternion_t fusion;
00038     double x;
00039     double y;
00040     double z;
00041     float xa;
00042     float ya;
00043     float za;
00044     float xerr;
00045     float yerr;
00046     float zerr;
00047     uint32_t sensortime;
00048     float sensortemp;
00049     uint8_t buffer[64]; // can-fd frame buffer space
00050 } sSensorData_t;
00051
00052 /*
00053  * function pointer templates structure
00054  * for the device I/O routines and data
00055  */
00056 typedef const struct _op_t {
00057     void (*info_ptr)(void);
00058     void (*imu_set_spimode)(void *);
00059     bool (*imu_getid)(void *);
00060     bool (*imu_getdata)(void *);
00061 } op_t;
00062
00063 enum device_type {
00064     IMU_BMA490L = 0, // IMU chip model
00065     IMU_SCA3300,
00066     IMU_SCL3300,
00067     IMU_BMA400,
00068     IMU_BNO086,
00069     IMU_NONE,
00070     IMU_LAST,
00071 };
00072
00073 /*
00074  * IMU data structure for driver
00075  */
00076 typedef struct _imu_cmd_t {
00077     const uint16_t id;
00078     uint32_t board_serial_id;
00079     enum device_type device;
00080     const uint8_t cs, spi_bytes, acc_range_scl;
00081     uint8_t acc_range;
00082     uint32_t log_timeout, rs, ss;
00083     volatile bool online, run, update, features, crc_error, angles;
00084     uint64_t host_serial_id;
00085     bool locked, warn, down;
00086 #ifdef BNO086
00087     uint8_t rbuf[512], tbuf[512];
00088 #else
00089     uint8_t rbuf[512], tbuf[512];
00090 #endif
00091     uint32_t rbuf32[2], tbuf32[2];
00092     uint16_t serial1, serial2;

```

```

00093         op_t op;
00094         volatile bool init_good;
00095     } imu_cmd_t;
00096
00097     enum hcmd_type { // IMU sensor chip commands from the host
00098         CMD_UNLOCK = 0,
00099         CMD_LOCK,
00100         CMD_SAFE,
00101         CMD_SPIN_DOWN,
00102         CMD_WARN_ON,
00103         CMD_WARN_OFF,
00104         CMD_ACK,
00105         CMD_IDLE,
00106         CMD_ACTIVE,
00107         CMD_LAST,
00108     };
00109
00110     /*
00111     * IMU data structure for host messages
00112     */
00113     typedef struct _imu_host_t {
00114         uint16_t id;
00115         uint64_t host_serial_id;
00116         uint8_t cmd;
00117         uint32_t cmd_data[4];
00118         uint64_t secret;
00119         uint8_t buf[64];
00120     } imu_host_t;
00121
00122     struct sca3300_data {
00123         const uint16_t id;
00124
00125         struct {
00126             int16_t channels[9];
00127             uint16_t ret_status;
00128             uint32_t ts;
00129         } scan;
00130         uint8_t rs;
00131         uint8_t ss;
00132     };
00133
00134     enum accel_g {
00135         range_2g = 0,
00136         range_4g,
00137         range_8g,
00138         range_16g,
00139         range_scale,
00140         range_15g,
00141         range_3g,
00142         range_6g,
00143         range_15gl,
00144         range_12g,
00145         range_24g,
00146         range_incl,
00147         range_inc2,
00148         range_bno,
00149     };
00150
00151     enum sca3300_scan_indexes {
00152         SCA3300_ACC_X = 0,
00153         SCA3300_ACC_Y,
00154         SCA3300_ACC_Z,
00155         SCA3300_TEMP,
00156         SCA3300_TIMESTAMP,
00157         SCL3300_ANG_X,
00158         SCL3300_ANG_Y,
00159         SCL3300_ANG_Z,
00160     };
00161
00162     #define GRAVITY_EARTH (9.80665f)
00163     /*
00164     * device earth gravity range calibration scalars
00165     */
00166     #define BMA400_ACCEL_MG_LSB_2G 0.000980665f
00167     #define BMA490_ACCEL_MG_LSB_2G 0.000061035f
00168     #define BMA490_ACCEL_MG_LSB_4G 0.000122070f
00169     #define BMA490_ACCEL_MG_LSB_8G 0.000244141f
00170     #define BMA490_ACCEL_MG_LSB_16G 0.000488281f
00171     #define IMU_ACCEL_MG_SCALE 1.000000000f
00172     #define IMU_BNO_ACCEL_MG_SCALE 0.9840000000f
00173     #define IMU_BNO_ACCEL_MG_SCALE_0 0.9840000000f
00174     #define IMU_BNO_ACCEL_MG_SCALE_1 0.9220000000f
00175     #define SCA3300_ACCEL_MG_LSB_15G 0.000207000f
00176     #define SCA3300_ACCEL_MG_LSB_3G 0.000395000f
00177     #define SCA3300_ACCEL_MG_LSB_6G 0.000765000f
00178     #define SCL3300_ACCEL_MG_LSB_12G 0.000167400f
00179     #define SCL3300_ACCEL_MG_LSB_24G 0.000333800f

```

```

00181 #define SCL3300_INC1    0.000083700f
00182 #define SCL3300_INC2    0.000083700f
00183
00184 #define ANGLE_RES1    16384.0f
00185 #define ANGLE_RES2    90.0f
00186 #define TEMPERATURE_RES    18.9f
00187 #define TEMPERATURE_OFFSET    -273.0f
00188
00189     typedef struct sFFTDData_t {
00190         uint16_t id;
00191         uint8_t buffer[64]; // can-fd frame buffer space
00192     } sFFTDData_t;
00193
00194     double get_imu_scale(imu_cmd_t *);
00195     void getAllData(sSensorData_t *, imu_cmd_t *);
00196     const uint8_t * imu_string(imu_cmd_t *);
00197
00198     extern char imu_buffer[256];
00199
00200 #ifdef __cplusplus
00201 }
00202 #endif
00203
00204 #endif /* IMU_H */
00205

```

4.52 imupic32mcj.h

```

00001 /*
00002  * File:    imupic32mcj.h
00003  * Author:  nsasp
00004  *
00005  * Created on August 31, 2022, 6:40 PM
00006  */
00007
00008 #ifndef IMUPIC32MCJ_H
00009 #define IMUPIC32MCJ_H
00010
00011 #ifdef __cplusplus
00012 extern "C" {
00013 #endif
00014
00015 #include <stddef.h>                // Defines NULL
00016 #include <stdbool.h>              // Defines true
00017 #include <stdlib.h>               // Defines EXIT_FAILURE
00018 #include <math.h>
00019 #include "definitions.h"          // SYS function prototypes
00020 #include "imu.h"
00021 #include "timers.h"
00022
00023 #define SDEBUG_psd // parse sensor data
00024
00025
00026     /*
00027     * Board hardware configurations
00028     * and hardware specific functions
00029     */
00030
00031     /*
00032     * what type of Ethernet TCP adapter are we using
00033     */
00034     //#define USR_TCP
00035 #define WAVESHARE
00036
00037     /*
00038     * what IMU chip are we using
00039     */
00040     //#define SCA3300 // this includes the SCL3300 device, SPI2 32-bit transfers
00041     //#define BMA400 // Bosch devices, SPI2 8-bit transfers
00042 #define BNO086 // SPI2 8-bit transfers 3MHz mode 3
00043
00044
00045 #ifdef BNO086
00046 #undef BMA400
00047 #undef SCA3300
00048 #endif
00049
00050 #ifdef SCA3300
00051 #define SPI2_32BIT
00052 #undef BMA400
00053 #endif
00054
00055

```

```

00056 #ifdef BMA400 // vibration sensor used
00057 #define BMA490L // this device is not longer used but functions are needed
00058 #undef SCA3300
00059 #endif
00060
00061 #define B_ID_0 0xFF0108
00062 #define B_ID_1 0x10534743
00063
00064 #define IMU_CS 0
00065 #define LCD_CS 0
00066
00067 #define BMA_LOG_TIMEOUT 150
00068 #define BNO_LOG_TIMEOUT 100
00069 #define SCA_LOG_TIMEOUT 10 // 8
00070
00071 // #define SHOW_LOG
00072 #define SHOW_LCD
00073 #define SHOW_VG
00074 #define FFT_MIX false
00075
00076 #define FBUFFER_SIZE 256
00077 #define max_buf FBUFFER_SIZE-1
00078 #define RBUFFER_SIZE FBUFFER_SIZE
00079
00080 #ifndef LED_GREEN_On
00081 #define LED_GREEN_On() LED_GREEN_Set()
00082 #define LED_GREEN_Off() LED_GREEN_Clear()
00083 #endif
00084 #ifndef LED_RED_On
00085 #define LED_RED_On() LED_RED_Set()
00086 #define LED_RED_Off() LED_RED_Clear()
00087 #endif
00088
00089 /* hardware dependant define */
00090 #define BLANK_COUNTER TMR4
00091
00092 #define CFD_RETRAN 0b01
00093 #define CFD_RETRAN_MODE 1
00094
00095 /*
00096  * setup external interrupt #2 for IMU BMA4x0 data update interrupt trigger output
00097  */
00098 void init_imu_int(const imu_cmd_t *);
00099 /*
00100  * user callback function per BMA490L data interrupt
00101  */
00102 void update_imu_int1(uint32_t, uintptr_t);
00103
00104 /*
00105  * setup external interrupt #2 for IMU BNO08X data update interrupt trigger output
00106  */
00107 void init_imu_int_bno(const imu_cmd_t *);
00108 /*
00109  * user callback function per BNO08X data interrupt
00110  */
00111 void update_imu_int_bno(uint32_t, uintptr_t);
00112 void delay_us(uint32_t);
00113 /*
00114  * start core-timer
00115  * ms tick-timer
00116  * set cpu serial ID numbers
00117  */
00118 void start_tick(void);
00119 /*
00120  * configure the SPI port bit size for data transfers
00121  */
00122 uint8_t set_imu_bits(void);
00123 void canfd_set_filter(uint32_t, uint32_t);
00124 void clear_can_errors(void);
00125 void set_can_retran(void);
00126 extern uint32_t board_serial_id, cpu_serial_id;
00127 extern volatile uint32_t bno08x_int_count;
00128
00129 #ifdef __cplusplus
00130 }
00131 #endif
00132
00133 #endif /* IMUPIC32MCJ_H */
00134

```

4.53 pid.h

```

00001 /*

```



```

00002  * File:    pid.h
00003  * Author:  root
00004  *
00005  * Created on July 19, 2020, 3:04 PM
00006  */
00007
00008 #ifndef PID_H
00009 #define PID_H
00010
00011 #ifdef __cplusplus
00012 extern "C" {
00013 #endif
00014
00015 #include "imupic32mcj.h"
00016
00017     typedef struct _SPid {
00018         double dState; // Last position input
00019         double iState; // Integrator state
00020         double iMax, iMin; // Maximum and minimum allowable integrator state
00021         double iGain, // integral gain
00022         pGain, // proportional gain
00023         dGain; // derivative gain
00024     } SPid;
00025
00026     double UpdatePI(volatile SPid * const, const double);
00027     void ResetPI(volatile SPid * const);
00028
00029 #ifdef __cplusplus
00030 }
00031 #endif
00032
00033 #endif /* PID_H */
00034

```

4.54 quat.h

```

00001 /*
00002  * File:    quat.h
00003  * Author:  root
00004  *
00005  * Created on April 28, 2023, 6:54 PM
00006  */
00007
00008 #ifndef QUAT_H
00009 #define QUAT_H
00010
00011 #ifdef __cplusplus
00012 extern "C" {
00013 #endif
00014
00015 // Copyright (C) 2019 Martin Weigel <mail@MartinWeigel.com>
00016 //
00017 // https://github.com/MartinWeigel/Quaternion
00018 // Permission to use, copy, modify, and/or distribute this software for any
00019 // purpose with or without fee is hereby granted, provided that the above
00020 // copyright notice and this permission notice appear in all copies.
00021 //
00022 // THE SOFTWARE IS PROVIDED "AS IS" AND THE AUTHOR DISCLAIMS ALL WARRANTIES
00023 // WITH REGARD TO THIS SOFTWARE INCLUDING ALL IMPLIED WARRANTIES OF
00024 // MERCHANTABILITY AND FITNESS. IN NO EVENT SHALL THE AUTHOR BE LIABLE FOR
00025 // ANY SPECIAL, DIRECT, INDIRECT, OR CONSEQUENTIAL DAMAGES OR ANY DAMAGES
00026 // WHATSOEVER RESULTING FROM LOSS OF USE, DATA OR PROFITS, WHETHER IN AN
00027 // ACTION OF CONTRACT, NEGLIGENCE OR OTHER TORTIOUS ACTION, ARISING OUT OF
00028 // OR IN CONNECTION WITH THE USE OR PERFORMANCE OF THIS SOFTWARE.
00029
00030 #pragma once
00031 #include <stdbool.h>
00032 #include <stdint.h>
00033 #include <stdio.h>
00034 #include <math.h>
00035
00036 #define QUATERNION_EPS (1e-4)
00037
00038     typedef struct Quaternion {
00039         double w;
00040         double v[3];
00041     } Quaternion;
00042
00043     void Quaternion_set(double w, double v1, double v2, double v3, Quaternion* output);
00044
00045     void Quaternion_setIdentity(Quaternion* q);
00046
00047     void Quaternion_copy(Quaternion* q, Quaternion* output);
00048

```

```

00068
00072 bool Quaternion_equal(Quaternion* q1, Quaternion* q2);
00073
00077 void Quaternion_fprint(FILE* file, Quaternion* q);
00078
00086 void Quaternion_fromAxisAngle(double axis[3], double angle, Quaternion* output);
00087
00095 double Quaternion_toAxisAngle(Quaternion* q, double output[3]);
00096
00102 void Quaternion_fromEulerZYX(double eulerZYX[3], Quaternion* output);
00103
00109 void Quaternion_toEulerZYX(Quaternion* q, double output[3]);
00110
00116 void Quaternion_fromXRotation(double angle, Quaternion* output);
00117
00123 void Quaternion_fromYRotation(double angle, Quaternion* output);
00124
00130 void Quaternion_fromZRotation(double angle, Quaternion* output);
00131
00136 double Quaternion_norm(Quaternion* q);
00137
00141 void Quaternion_normalize(Quaternion* q, Quaternion* output);
00142
00146 void Quaternion_conjugate(Quaternion* q, Quaternion* output);
00147
00155 void Quaternion_multiply(Quaternion* q1, Quaternion* q2, Quaternion* output);
00156
00160 void Quaternion_rotate(Quaternion* q, double v[3], double output[3]);
00161
00168 void Quaternion_slerp(Quaternion* q1, Quaternion* q2, double t, Quaternion* output);
00169
00170
00171 #ifdef __cplusplus
00172 }
00173 #endif
00174
00175 #endif /* QUAT_H */
00176

```

4.55 remote_cmd.h

```

00001 /*
00002  * File:    remote_cmd.h
00003  * Author:  root
00004  *
00005  * Created on March 2, 2023, 10:11 AM
00006  */
00007
00008 #ifndef REMOTE_CMD_H
00009 #define REMOTE_CMD_H
00010
00011 #ifdef __cplusplus
00012 extern "C" {
00013 #endif
00014 #include "imupic32mcj.h"
00015 #include "cmd_scanner.h"
00016 #include "host.h"
00017
00018 #define REMOTE_DRIVER "V1.001"
00019 #define REMOTE_ALIAS "REMOTE CMD"
00020
00021 /*
00022  * command execute functions
00023  * prototypes
00024  */
00025 void fh_show_idle(void *a_data);
00026 void fh_stop_trigger(void *a_data);
00027 void fh_start_trigger(void *a_data);
00028 void fh_start_safe(void *a_data);
00029 void fh_start_lock(void *a_data);
00030 void fh_start_unlock(void *a_data);
00031 void fh_start_warn(void *a_data);
00032 void remote_cmd_decode(imu_host_t *);
00033
00034 extern void fh_start_AT(void *a_data);
00035
00036 extern char response_buffer[RBUFFER_SIZE];
00037 extern char cmd_buffer[FBUFFER_SIZE];
00038 extern imu_host_t host0;
00039 extern t_cmd g_cmds[];
00040 extern char hbuffer[FBUFFER_SIZE];
00041 extern imu_cmd_t imu0;
00042

```

```

00043 #ifdef __cplusplus
00044 }
00045 #endif
00046
00047 #endif /* REMOTE_CMD_H */
00048

```

4.56 sca3300.h

```

00001 /*
00002  * File:   sca3300.h
00003  * Author: nsasp
00004  *
00005  * Created on August 31, 2022, 6:28 PM
00006  * SPI mode 0, 8MHz, 32-bit transfers
00007  */
00008
00009 #ifndef SCA3300_H
00010 #define SCA3300_H
00011
00012 #ifdef __cplusplus
00013 extern "C" {
00014 #endif
00015 /*
00016  * using code from Linux IIO driver for sca3300
00017  * https://elixir.bootlin.com/linux/latest/source/drivers/iio/accel/sca3300.c
00018  */
00019 #include <stddef.h> // Defines NULL
00020 #include <stdbool.h> // Defines true
00021 #include <stdlib.h> // Defines EXIT_FAILURE
00022 #include <math.h>
00023 #include "definitions.h" // SYS function prototypes
00024 #include "imupic32mcj.h"
00025 #include "imu.h"
00026 #include "timers.h"
00027 #include "canfd.h"
00028
00029 #define SCA3300_DRIVER "V1.006"
00030
00031 #define SCA3300_ALIAS "SCx3300"
00032
00033 /* Device commands and response */
00034 #define SCA3300_REG_WHOAMI 0x10
00035 #define SCA3300_WHOAMI_ID 0x51
00036 #define SCA3300_WHOAMI_ID_SCL 0xC1
00037 #define SCA3300_WHOAMI_32B 0x40000091
00038 #define SCA3300_SWRESET_32B 0xB4002098
00039 #define SCA3300_RS_32B 0x180000E5
00040 #define SCA3300_ACC_X_32B 0x040000F7
00041 #define SCA3300_ACC_Y_32B 0x080000FD
00042 #define SCA3300_ACC_Z_32B 0x0C0000FB
00043 #define SCA3300_TEMP_32B 0x140000EF
00044 #define SCA3300_MODE1 0xB400001F
00045 #define SCA3300_MODE2 0xB4000102
00046 #define SCA3300_MODE3 0xB4000225
00047 #define SCA3300_MODE4 0xB4000338
00048 #define SCL3300_MODE1 0xB400001F
00049 #define SCL3300_MODE2 0xB4000102
00050 #define SCL3300_MODE3 0xB4000225
00051 #define SCL3300_MODE4 0xB4000338
00052 #define SCL3300_ANGLE 0xB0001F6F
00053 #define SCL3300_E_FLAG1 0x1C0000E3
00054 #define SCL3300_E_FLAG2 0x200000C1
00055 #define SCL3300_ANG_X_32B 0x240000C7
00056 #define SCL3300_ANG_Y_32B 0x280000CD
00057 #define SCL3300_ANG_Z_32B 0x2C0000CB
00058 #define SCA3300_BANK0 0xFC000073
00059 #define SCA3300_BANK1 0xFC00016E
00060 #define SCA3300_SERIAL1 0x640000A7
00061 #define SCA3300_SERIAL2 0x680000AD
00062
00063 #define SCA3300_CHIP_ID_DELAY 10000 // ID command repeat delays in usec
00064 #define SCA3300_CHIP_SWR_DELAY 10000 // chip software reset delay
00065 #define SCA3300_CHIP_MODE_DELAY 100000 // chip G mode setup delay
00066 #define SCA3300_CHIP_CS_DELAY 11 // CS high min duration between toggles
00067 #define SCA3300_CHIP_BYTES_PER_SPI 4 // 32-bit transfers, 4 bytes
00068
00069 /*
00070  * function pointer templates
00071  */
00072 void sca3300_set_spimode(void *);
00073 bool sca3300_getid(void *);

```

```

00074     bool sca3300_getserial(void *);
00075     bool sca3300_getdata(void *);
00076     void sca3300_version(void);
00077
00078     enum sca3300_crc_check {
00079         SCA3300_REC = 0, // first item in the 32-bit receiver buffer
00080         SCA3300_TRM = 0, // first item in the 32-bit transmit buffer
00081     };
00082
00083     uint8_t CalculateCRC(uint32_t);
00084     bool sca3300_imu_transfer(imu_cmd_t *, uint32_t);
00085     bool sca3300_check_crc(imu_cmd_t *, uint8_t);
00086
00087 #ifdef __cplusplus
00088 }
00089 #endif
00090
00091 #endif /* SCA3300_H */
00092

```

4.57 timers.h

```

00001 #ifndef TIMERS_H
00002 #define TIMERS_H
00003
00004 #include <stdint.h>
00005 #include <stdbool.h>
00006
00007 //Software timers - use these to refer to timers rather than
00008 //integers. Add more as needed.
00009 //NOTE: 32767 ms max delay
00010
00011 enum APP_TIMERS {
00012     TMR_INTERNAL = 0, //Used in timers.c - do not remove or use elsewhere
00013     TMR_IMU,
00014     TMR_LOG,
00015     TMR_CS,
00016     TMR_HOST,
00017     TMR_REPLY,
00018     TMR_BNO_INT,
00019     TMR_BNO_TO,
00020     TMR_IMU_DATA,
00021     //
00022     //(Add timers here as needed)
00023     //
00024     TMR_COUNT //number of timers - always keep at end of enum!
00025 };
00026
00027 void StartTimer(const uint8_t timer, const uint16_t count);
00028 bool TimerDone(const uint8_t timer);
00029 void WaitMs(const uint16_t numMilliseconds);
00030 void timer_ms_tick(uint32_t, uintptr_t);
00031
00032 void delay_ms_tick(uint32_t);
00033
00034 #endif //TIMERS_H

```

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